
Computational Discrete Global Geometric Structures

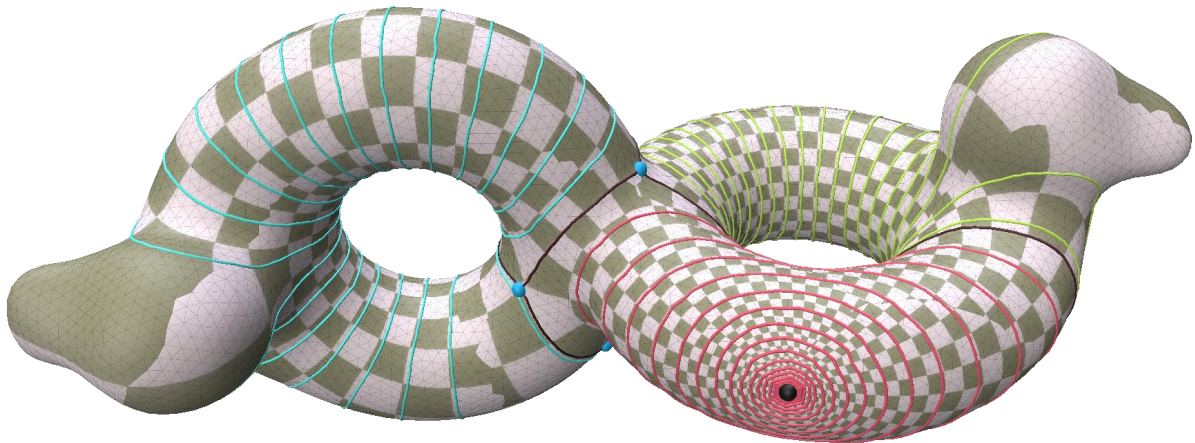
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Abstract : This paper proposes a brand-new research direction, field, content, and tool: computational Discrete Global Geometric Structures (CDGGS). It serves as an innovative interdisciplinary and transdisciplinary research direction that integrates computer graphics, differential geometry and topology theory, numerical computation, mechanical applications, physical topological materials, and other disciplines. Computational Discrete Global Geometric Structure focuses on three key areas: 1) developing and implemented algorithms for constructing diverse global geometric structures on discrete meshes to catch their "global properties"; 2) visualizing global geometric structures; 3) exploring the applications of these implemented algorithms across various technological domains. The paper puts forward a key perspective: Many pivotal works in the history of mathematics and physics have essentially centered on global geometric structures.

Key Words : Super-structured Quadrilateral Mesh, Surface Mapping Class, Moduli Space, Geometric Topology, Global Geometric Structure.



(Figure 1: A harmonic foliation with a pole and its corresponding meromorphic quadratic differential, generated by Hui Zhao with the software Geometric .)

1 Overview

Based on three decades of accumulated work experience, practical insights in the field of algorithm design for geometric topology applications on meshes, as well as reflections on the future development of theories and algorithm design in this field, Zhao Hui proposed the concept of "Computational Global Discrete Geometric Structures (CGDGS)" in 2022. Concurrently, he clearly defined its innovative research direction, scope, content, and tools. The innovations of this concept have been extensively discussed and systematically elaborated on a WeChat official account in previous works. This paper condenses the scattered explanations and clarifications from recent years, and provides a detailed organization and summary of the various innovations of CGDGS from multiple perspectives, laying a solid foundation for in-depth research and development in subsequent studies.

Global geometric structures refer to geometric structures of fiber bundles defined on various manifolds, and they are more refined than topological structures. The author has defined the concept of Computational Global Discrete Geometric Structures and proposed this research direction. The research scope of this field focuses on three core areas:

- (1) Developing and implemented algorithms to capture the global properties of various global geometric structures;
- (2) Visualizing various global geometric structures using computer graphics technologies;
- (3) Applying the aforementioned algorithms to diverse industrial technology fields.



(Figure 2: A harmonic foliation, generated by Hui Zhao with the software Geometric .)

Centered on algorithm research, "Computational Global Discrete Geometric Structures" primarily conducts algorithm design based on cutting-edge differential geometry and topology theories — especially theories of global geometry. Its main objective is not to conduct theoretical research on differential geometry and topology or develop new theories, but rather to investigate computer algorithms corresponding to existing theories and concepts. However, during the process of algorithm research, it may conversely yield new insights and understandings into geometric topology theories.

The cutting-edge differential geometry and topology theories are vast and profound, while the research content of Computational Global Discrete Geometric Structures is clearly defined. Its focus lies in studying the theories of global geometric structures and designing algorithms for various types of global geometric structures—such as harmonic foliations, meromorphic quadratic differentials, and other global geometric structures with inherent global properties. Mathematical concepts and theories are interdependent and intertwined; through the research of global geometric structures, it is possible to essentially cover all concepts in differential geometry and topology.

At present, the algorithms under research mainly focus on 2-dimensional manifolds (surfaces) and various global geometric structures on their tangent bundles. In the future, it is expected to expand to other manifolds and fiber bundles.

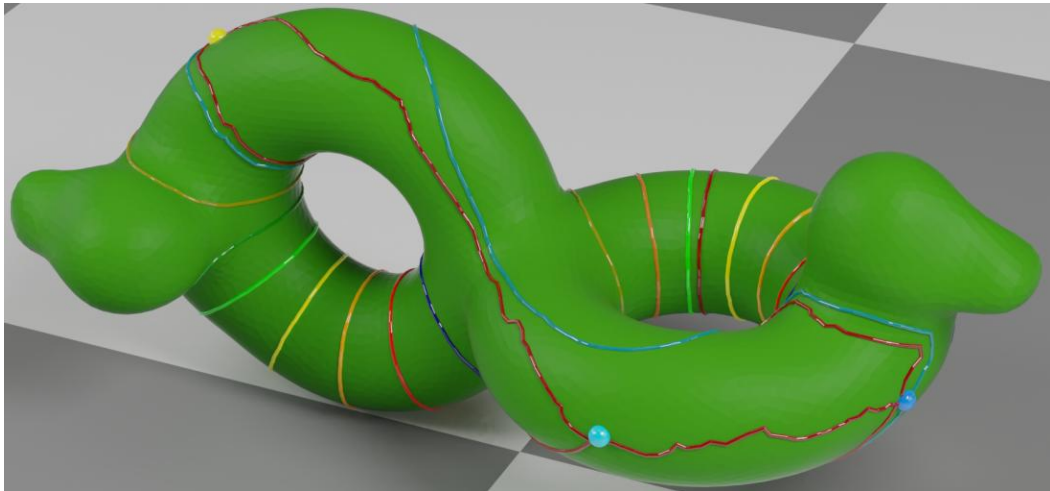
Unlike abstract mathematics, whose driving force stems from the invention of new structures and the proof of new theories, the research on Computational Global Discrete Geometric Structures (CGDGS) is driven by potential technological applications—specifically, solving technical challenges that cannot be addressed by other mathematical computing tools. That said, CGDGS research shares a commonality with abstract mathematics research: curiosity-driven exploration also serves as an important source of motivation.

2 Global Geometric Structures

The research on computational discrete global geometric structures has transcended the conceptual stage and now finds tangible applications in industrial technology. These applications have demonstrated immense technological potential, addressing problems that remain intractable with other mathematical theories and tools—such as the use of super-structured quadrilateral meshes in industrial software. This paper provides an overview of these applications.

The proposal of the research direction on computational discrete global geometric structures is rooted in years of algorithmic research conducted by numerous scholars both domestically and internationally, evolving as a refined new field of study. We believe this direction holds tremendous potential to drive and enable a host of entirely new technologies.

The development of Euclidean plane geometry spans millennia. Following the advent of calculus, local differential geometry emerged—a discipline that applies calculus to the study of curves and surfaces in three-dimensional space. The founder of global geometry was the preeminent mathematician Shiing-Shen Chern. Around the 1930s, prior to Chern's groundbreaking work, research on local differential geometry had already advanced considerably, leading many scholars to believe that the field had little remaining room for exploration. It was not until the 1940s that Chern revolutionized the field by pioneering the new domain of "global geometry" through pivotal achievements such as his proof of the Gauss-Bonnet theorem and the construction of Chern classes. Its core lies in establishing a connection between local geometric information and global topological properties, marking a breakthrough advancement over the traditional, locally focused classical differential geometry.



(Figure 3: A harmonic foliation and the Ribbon graph, generated by Geometric .)

In the decades since, and continuing to the present day, leading mathematicians—including Fields Medal laureates such as Shing-Tung Yau, Bill Thurston, Curtis McMullen, Maxim Kontsevich, Maryam Mirzakhani, Anton Zorich, Gang Tian, Alex Eskin, Scott A. Wolpert, Howard Masur, Benson Farb, Alex Wright, Vincent Delecroix, Jayadev S. Athrey, and Carlos Matheus—have (dedicated themselves to) advancing research in this area. As a result, cutting-edge geometric and topological theories related to global geometry have gradually matured. Notable contributions within this body of work include: Professor Shing-Tung Yau's use of geometric analysis to prove the existence of Calabi-Yau global geometric structures; and Bill Thurston's landmark achievements in the fields of foliations and the classification of 3-manifolds.

Today, global geometry remains a dynamically evolving research area, with the mathematical community developing new tools and methods to investigate existing global geometric structures and explore/discover new ones. However, unlike established mathematical theories such as calculus, linear algebra, and local differential geometry—all of which matured centuries ago and have been widely adopted in industrial technology — these cutting-edge geometric and topological theories of global geometric structures have only evolved in recent decades. Consequently, their dissemination into engineering and technological fields is still in its infancy. Against this backdrop, the innovation of the "computational discrete global geometric structures" research direction outlined in this paper lies in leveraging algorithm design as its engine, facilitating the translation of theoretical research findings on global geometric structures into practical technological applications.

3 Global Geometric Structures and Topology

The concepts of geometry and topology intersect and integrate with each other. Due to its relatively intuitive nature, the concept of topology has gained wide recognition in fields such as physics, computer science, mechanics, and mechanical engineering — particularly topology concepts with visual representations, such as the genus of a surface. Many topological concepts have already found relatively extensive technical applications: for instance, in mechanics, the study of topological optimization is a field with numerous industrial and product-related applications; in physics, there are topological materials, among other examples.

Compared to the popularization of topological concepts, the concept of global geometric structures is less known in the field of engineering and technology.

Global geometric structures connect the properties of local differential geometry with large-scale topological properties. Global geometry can be regarded as a refinement of topology: topological structures are relatively "coarse" and cannot accurately distinguish between cases that are topologically identical but geometrically different. Global geometric structures are constrained by topology; in many cases, direct research on topology is difficult, so topology can be studied indirectly by investigating geometry.

Global geometric structures define more detailed mathematical concepts and theorems based on topology, thereby enabling the differentiation of more distinct geometric structures. In this way, they provide more mathematical theoretical tools than topology for algorithm research.



(Figure 4: Some topologists.)

Currently, topology is relatively more mature than global geometric structure research in terms of theory, and topology also has a longer history of research and development. Consequently, in terms of technology diffusion, topology has found relatively more applications. For example, in the research on topological algorithms and applications, there are fields like topological analysis, which have been widely applied to biological data and other types of data. Therefore, with the maturity of algorithm research in

topology, it is an inevitable trend in historical development to further propose algorithm research on global geometric structures.

A currently important research direction in physics is topological materials, which heavily relies on topological theory. However, if global geometric structures can be incorporated into this research, we anticipate that by relying on theoretical guidance to design more innovative physical experiments — especially by using "algorithms" as a bridge to connect these global geometric structures — more experimental results can be obtained.

It is evident that although the diffusion of global geometric structures from theory to technology is still in its infancy, it will possess greater potential to drive the development of applications. Topology emerged first, followed by global geometric structures; moreover, the scholars engaged in research on geometry and topology overlap to a great extent. For specific information about some representative mathematicians who have made pioneering contributions to topology, please refer to the figure below.

4 The Definition of Global Geometric Structures

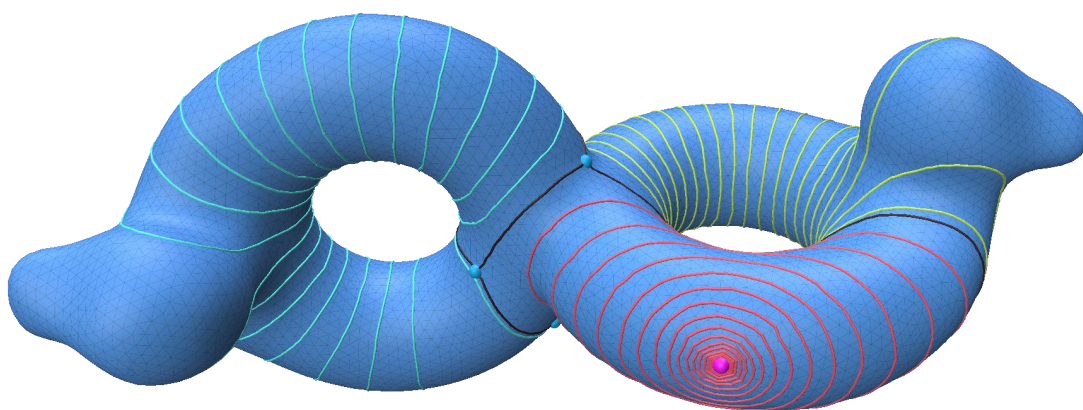
Global Geometric Structures refer to geometric structures defined on manifolds and various fiber bundles—these fiber bundles can be tangent bundles or other types of bundles.

Mathematicians have discovered and developed hundreds of distinct global geometric structures. Examples include vector fields on surfaces, differential 1-forms, holomorphic 1-forms, holomorphic 2-forms, meromorphic 1-forms, and harmonic foliations.

One of the core research directions in this field is to determine: whether a specific global geometric structure exists on a given manifold or its fiber bundles, and what properties such a structure possesses. Taking mathematicians like Shing-Tung Yau and Gang Tian as examples, they primarily use partial differential equations (PDEs) and geometric analysis methods to investigate the existence of various global geometric structures, such as Calabi-Yau manifolds.

Mathematical definitions require quantification via formulas and equations. Global geometric structures are characterized by the fact that they cannot be defined by a single formula or equation within a single coordinate system. Their key defining features are as follows:

- (1) A global geometric structure is defined on the entire manifold.
- (2) Its concept and definition cannot be described by a single equation or formula.
- (3) It must be expressed jointly through different formulas and equations under multiple coordinate systems.
- (4) These equations, which belong to different coordinate systems, must satisfy specific transformation rules.
- (5) The core information of the global structure is inherently contained in the transformation relationships of these equations.
- (6) Each distinct global geometric structure has its unique equation form and transformation rules.



(Figure 5: A harmonic foliation with a pole, generated by Geometric.)

Therefore, the study of global geometric structures involves developing various tools, methods, and

techniques to analyze the transformations within these formulas and equations, thereby deriving various properties related to the global nature of these geometric structures on manifolds and fiber bundles. For instance, geometric analysis employs tools from partial differential equations and analysis to study global geometric structures.

Historically, the mathematical community has primarily studied global geometric structures using abstract mathematical tools. The innovation of the "computational discrete global geometric structures" we propose lies in introducing "algorithms" as a tool to investigate various global geometric structures.

Unlike global geometric structures, the structures in local differential geometry can be represented by formulas or equations within a single coordinate system. Examples include Gaussian curvature, mean curvature, and maximum/minimum principal curvatures.

原始模型	Gaussian 曲率	平均曲率	最大主曲率	最小主曲率
				
				
				
				

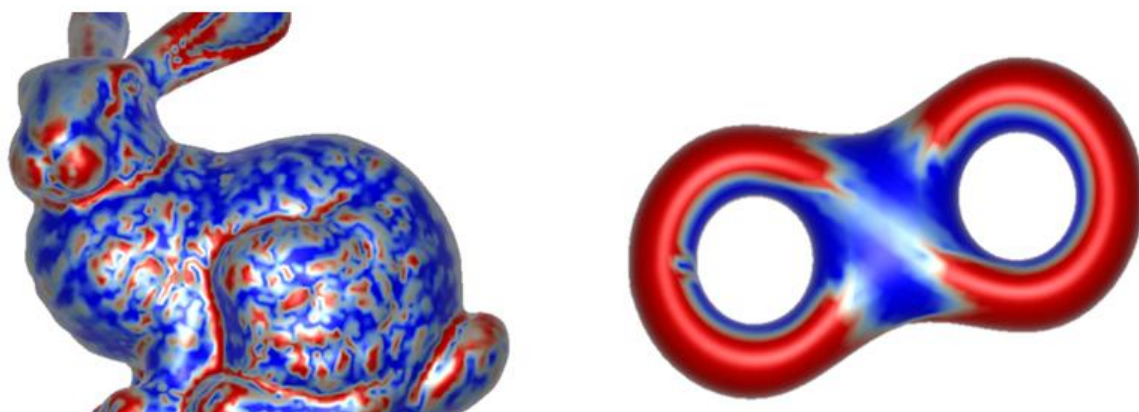
(Figure 6: Several kinds of curvatures, curvature numerical value are colored.)

The theory of local differential geometric structures has a long development history, dating back to the era of Gauss. It had already reached relative maturity over a century ago, which has facilitated its

well-advanced extension to industrial and technological applications. This theory has helped solve numerous technical challenges, and a wealth of robust algorithms have been developed to compute specific numerical values of various local geometric structures (as shown in the figure).

The numerical calculation of these specific local geometric structures also requires first discretizing surfaces and manifolds, followed by designing corresponding computational methods to approximate the formulas defined on smooth surfaces.

Computer graphics comprises four core modules: modeling, rendering, animation, and interaction. Local differential geometry has been widely applied in the modeling module of computer graphics—particularly in the research of mesh processing within computer graphics, which extensively employs theories and concepts from local differential geometry (as illustrated in the figure below). For more detailed demonstrations, please refer to the mesh processing code platform meshDGP.



(Figure 7: Gaussian curvature and mean curvature, curvature numerical value are colored.)



(Figure 8: some mathematicians of global geometry.)

A major focus of mathematicians' research on various global geometric structures lies in proving existence—that is, verifying whether a specific type of global geometric structure exists. Even if a global

geometric structure can be expressed by equations in a local coordinate system, it may not be valid globally across the entire fiber bundle. After proving the existence of such a structure, mathematicians then proceed to investigate its various properties and characteristics.

Numerous mathematicians have made significant contributions to the research and development of global geometry, pioneering many new fields. For a list of some outstanding mathematicians associated with global geometry research, please refer to the figure.

The primary focus of computational discrete global geometric structures is to study existing global geometric structures whose existence has already been proven, and to design algorithms for them. However, during the process of algorithm design, new types of global geometric structures may also be discovered.

5 Global Geometric Structures and Computational

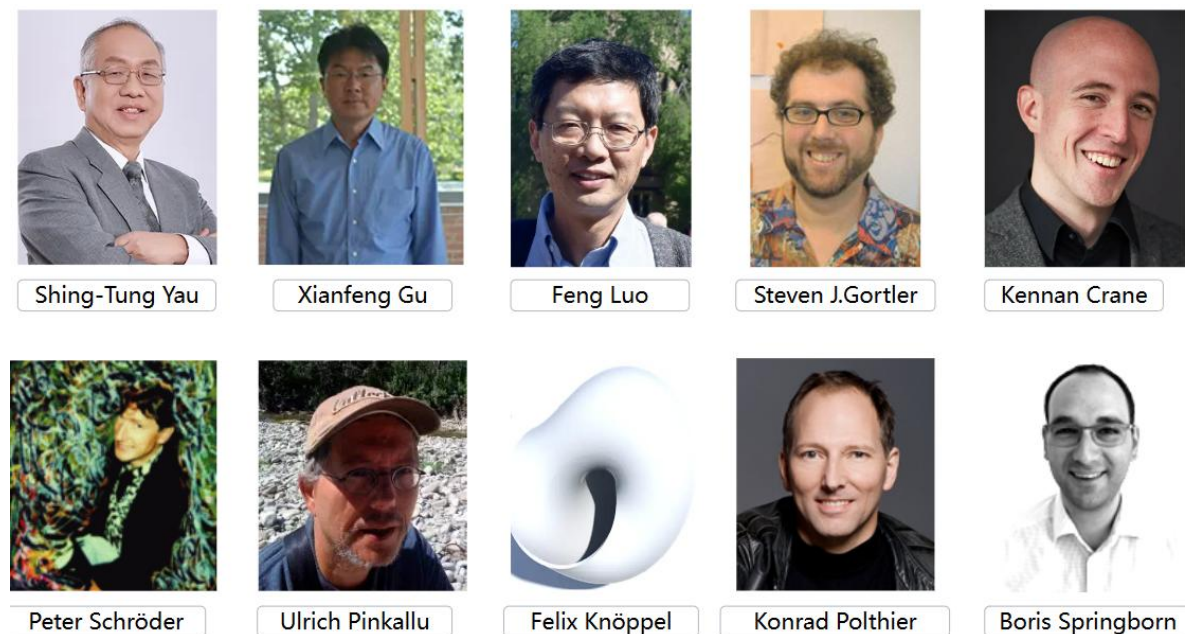
Mathematical research in the field of global geometry has gradually advanced since the 1950s. Consequently, the diffusion of such mathematical theories from the mathematical community to the field of engineering and technology is still an ongoing historical process, and most experts in engineering and technology are not yet sufficiently familiar with or proficient in these theories.

To apply relevant theories to engineering and technology, it is necessary to build on the mathematical community's proofs of existence and enable the computation of these global geometric structures within the engineering community. Only when computational results are obtained can these structures be further applied to various engineering and technical scenarios.

The development of local differential geometric structures previously followed a similar path: progressing from abstract theory to computable implementation, and then to industrial and technological applications. The key difference lies in the fact that the mathematical theory of local geometric structures matured much earlier. After over a century of development, it has already yielded a large number of mature computational methods. During our research, we have gradually refined the insight that new algorithms and computational tools should be developed from the perspective of "global geometric structures."

Around the 2000s, some computer scientists, driven by research needs in mesh processing, began collaborating with differential geometers and ventured into the field of global geometric structure computation. These research efforts originally stemmed from the mesh research direction in computer graphics. Over more than two decades of development, some global geometric structures have gradually transitioned from abstract mathematical theories to implementable numerical computation methods, and have further found applications in the industrial sector.

As shown in the figure, a selection of scientists who have contributed in various ways to the computability of global geometric structures is presented.



(Figure 9: Some mathematicians and computer scientists of computational global discrete geometric structures.)

The origin of the computable development of global geometric structures is driven and motivated by specific applications in computer graphics, rather than emerging from abstract mathematics for the sake of computation itself.

For instance, in film visual effects and video games, texturing 3D animated characters requires first flattening the character's 3D mesh onto a 2D plane—only then can 2D images be mapped onto it. This process is known as mesh parameterization flattening.

To advance research on mesh parameterization flattening, Professor Xianfeng Gu (Harvard University), Shing-Tung Yau, and Steven J. Gortler conducted studies around the 2000s on the computable theory, algorithm design, and code implementation of differential 1-forms—a type of global geometric structure.

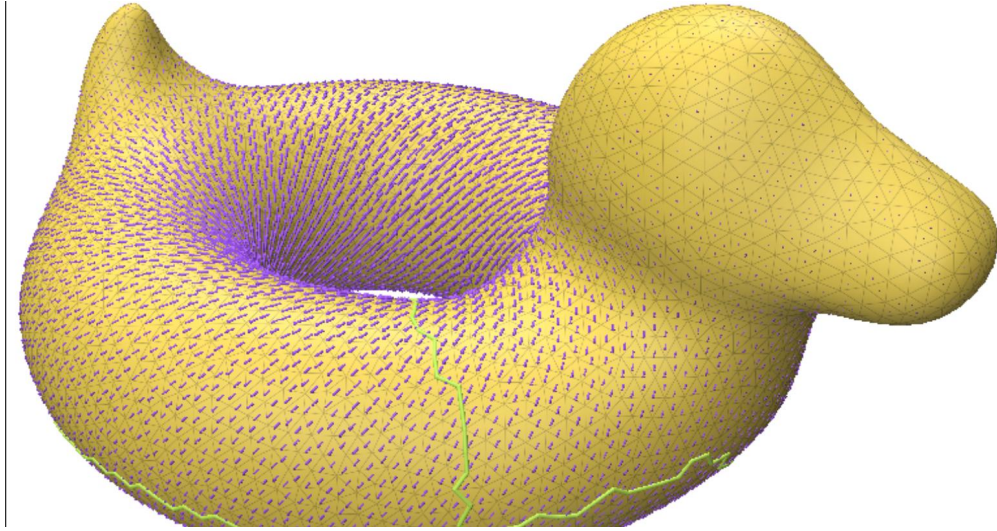
Prior to this work, parameterization flattening based on local differential geometry theory could only handle disk-shaped meshes (i.e., topologically simple meshes with no "holes"). By leveraging the theory of global geometric structures, this limitation was overcome, enabling parameterization flattening for high-genus meshes (i.e., meshes with multiple "holes" or complex topological structures).



(Figure 10: Differential 1-form, generated by software Geometric.)

Around the 2010s, Professor Keenan Crane from Carnegie Mellon University (CMU) conducted research on another type of global geometric structure—vector fields—focusing on their computable theory, algorithm design, and code implementation.

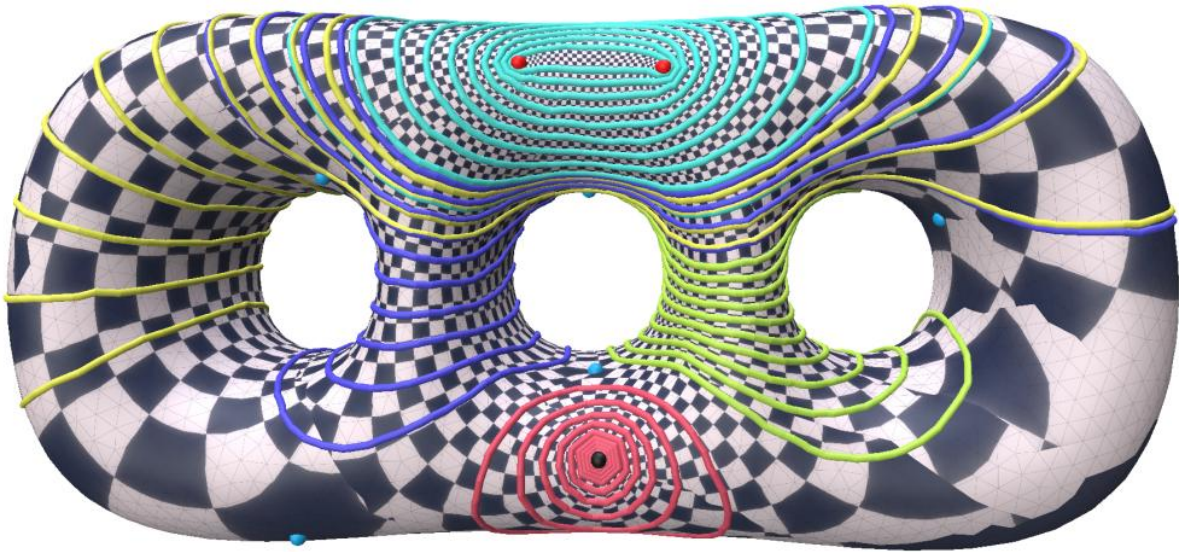
A vector field is defined as assigning a directional vector to each point on a surface; the collection of vectors across all points on the entire surface forms a complete vector field.



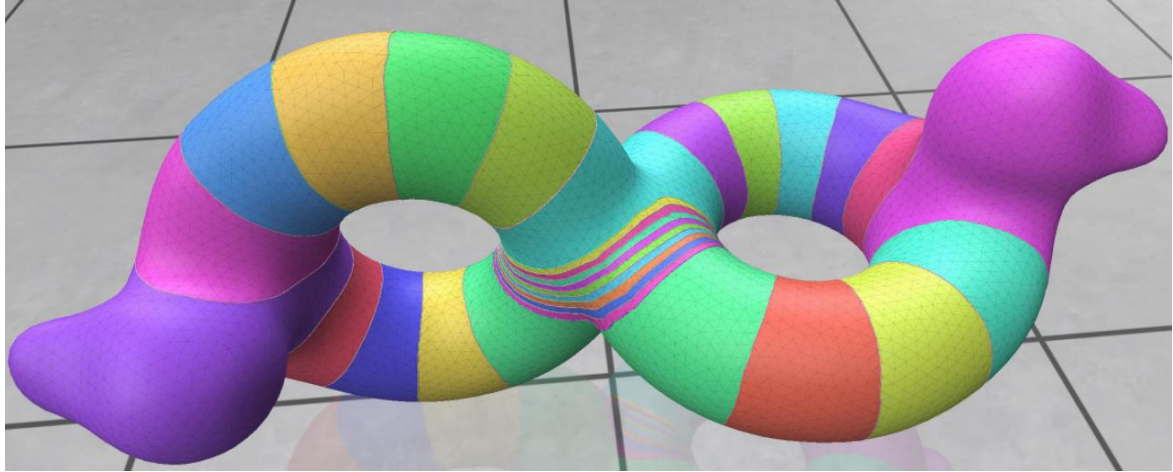
(Figure 11: Vector field, generated by software Geometric.)

For example, around 2019, Zhao Hui conducted research on discretization algorithms for several other types of global geometric structures, including harmonic foliations, holomorphic quadratic differentials, harmonic foliations with poles, and meromorphic quadratic differentials.

The research on the computability of these global geometric structures was also driven by applications in parameterized texturing on high-genus meshes within the field of computer graphics.



(Figure 12: A harmonic foliation with a pole and its meromorphic quadratic differential, generated by Geometric.)



(Figure 12: A non-harmonic foliation without poles, generated by Geometric.)

6 Research Viewpionts

A single mathematical concept can be approached and understood from numerous distinct angles. Each perspective may lead to the development of new theories and mathematical structures that are difficult to achieve through other viewpoints.

For instance, Fields Medal laureate William P. Thurston, in his paper *On Proof and Progress in Mathematics*, discussed seven different perspectives for thinking about and understanding the derivative—a fundamental concept in calculus.

- (1) Infinitesimal: the ratio of the infinitesimal change in the value of a function to the infinitesimal change in a function.
- (2) Symbolic: the derivative of x^n is nx^{n-1} , the derivative of $\sin(x)$ is $\cos(x)$, the derivative of $f \circ g$ is $f' \circ g * g'$, *etc.*
- (3) Logical: $f'(x) = d$ if and only if for every ϵ there is a δ such that when $0 < |\Delta x| < \delta$,

$$\left| \frac{f(x + \Delta x) - f(x)}{\Delta x} - d \right| < \delta.$$

- (4) Geometric: the derivative is the slope of a line tangent to the graph of the function, if the graph has a tangent.
- (5) Rate: the instantaneous speed of $f(t)$, when t is time.
- (6) Approximation: The derivative of a function is the best linear approximation to the function near a point.
- (7) Microscopic: The derivative of a function is the limit of what you get by looking at it under a microscope of higher and higher power.

Regarding the aforementioned algorithms studied on meshes, our innovative proposition is to center research on global geometric structures and conduct algorithm design from the perspective of global geometric structures. However, when scholars investigated these specific algorithms at the time, they approached them from other perspectives, including:

The Discrete Exterior Calculus (DEC) perspective, represented by Professor Keenan Crane from Carnegie Mellon University (CMU);

The Computational Conformal Geometry perspective, represented by Professor Xianfeng Gu from Stony Brook University (New York);

The Finite Element Exterior Calculus (FEEC) perspective, proposed by Professor Douglas N. Arnold from the University of Minnesota (USA);

The more mature Topology Analysis perspective.

Each of the aforementioned perspectives has its unique strengths in solving specific technical problems that require its distinct framework. Nevertheless, the content they study essentially belongs to a certain type of "global geometric structure."

Our proposed new perspective is to conduct algorithm design research based on the lens of "global geometric structures". This approach enables successful algorithm design for a broader range of global geometric structures, which can then be applied to industrial technologies.

Similar to the seven distinct perspectives on the derivative of a function, many problems can only be solved more conveniently through the thinking perspective of "global geometric structures". In other words,

a new perspective must be capable of achieving results that existing perspectives cannot. For example, algorithms that transform non-harmonic foliations into harmonic ones are difficult to develop using the aforementioned perspectives.

Refining a brand-new perspective for algorithm design is one of the key innovations in our proposal of "computational discrete global geometric structures."

7 Proposing a Brand-New Research Field

In 2022, Zhao Hui proposed the research direction, content, domain, tools, framework, and methodologies for "computational discrete Global Geometric Structures". He described and elaborated on this concept through channels such as WeChat official accounts; this paper summarizes this work in a comprehensive and detailed format.

Nomenclature Breakdown: "Computable" - "Discrete" - "Global" - "Geometric Structures"

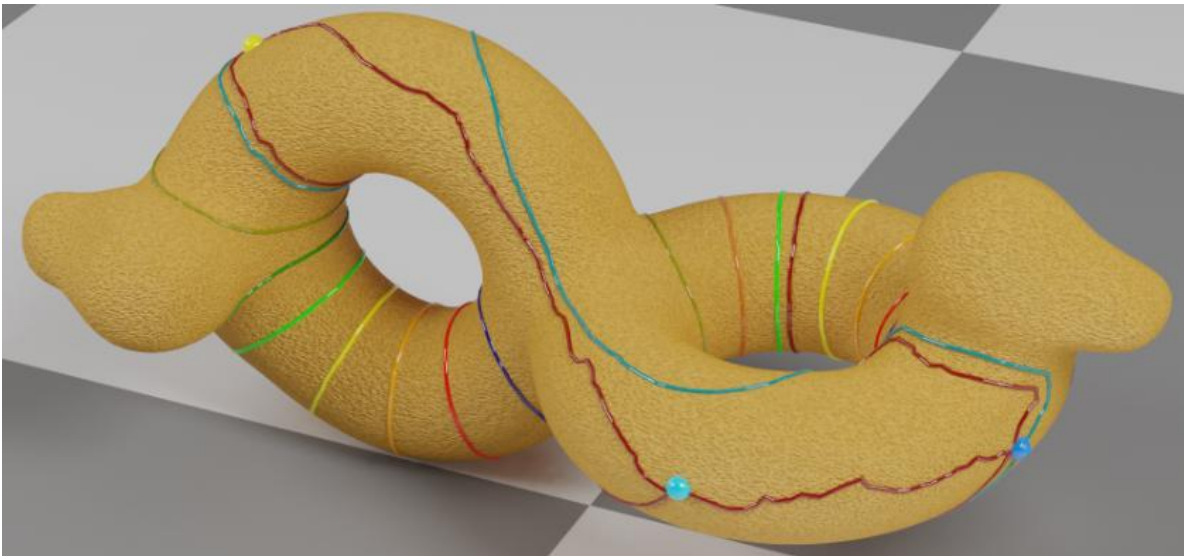
The term "computational discrete Global Geometric Structures" is segmented and defined as follows, with each component reflecting a core attribute of the research field:

"Discrete": When computer scientists compute global geometric structures, their primary approach involves discretizing smooth surfaces into meshes and then designing algorithms for computation on these meshes. This discretization process is the origin of the term "Discrete" in the nomenclature.

"Global": This component emphasizes the core focus of this new research direction, distinguishing it from the more familiar field of local differential geometry (which centers on local geometric properties rather than global characteristics).

"Geometric Structures": This specifies that the research object is "geometry" rather than "topology," clarifying the disciplinary boundary of the field.

"Computable": Here, "computable" refers to "algorithms" rather than simple formula-based calculations. It highlights the field's emphasis on developing systematic algorithmic frameworks to enable practical computation of global geometric structures.



(Figure 14: A harmonic foliation and the Ribbon graph, generated by Geometric .)

For most scholars in science and engineering, the concept of global geometry is generally unfamiliar and thus difficult to grasp. The key and obvious differences between local geometric structures and global geometric structures are as follows:

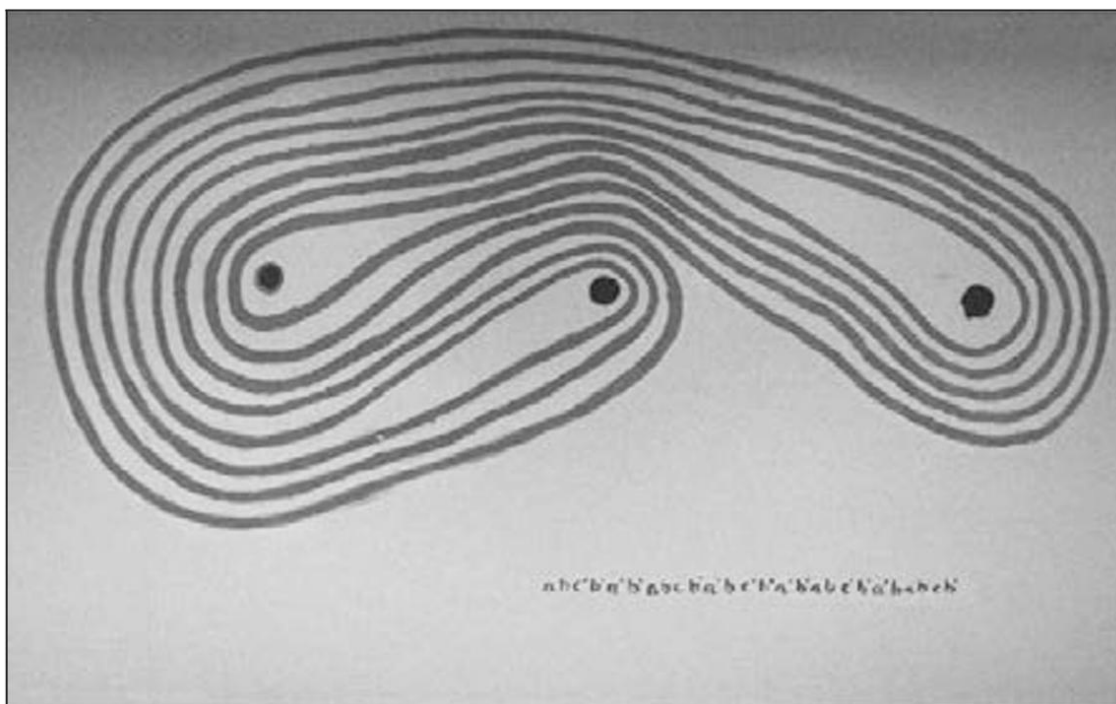
If the topology of an entire manifold changes (e.g., a "hole" is punctured in a surface) while the geometric shape of other regions remains unchanged, the local geometric values (such as Gaussian curvature) of those unchanged regions will not alter.

However, global geometric structures (e.g., harmonic foliations) will undergo changes. This is because global geometric structures cannot be defined at individual local points—they are defined on the entire manifold.

Although a global geometric structure can be expressed as local point-wise values under a specific coordinate system, if the manifold changes, these local values will shift even if the local shape of the point itself remains unchanged. This is a "ripple effect" where changes to distant regions of the manifold impact local values.

For example, if a small hole is punctured in a surface, global geometric structures such as differential 1-forms, vector fields, harmonic foliations, and holomorphic quadratic differentials will be completely altered.

To date, the motivation for researching the computability of various global geometric structures has originated from mesh processing applications in the field of computer graphics. However, with the proposal of our research direction on "computational discrete global geometric structures"—which clarifies innovations and analytical perspectives—we anticipate that future research will extend beyond computer graphics and find applications in other interdisciplinary fields, such as topological materials and biological genomics.



(Figure 15: A foliation, drawn by Thurston William and Dennis Sullivan.)

Zhao Hui's thinking on "computational discrete global geometric structures" evolved gradually, rooted in his algorithmic research on a specific type of global geometric structure. Initially, he did not recognize the need to frame his work within the "global geometric structure" framework. It was only during the algorithm design process that he gradually familiarized himself with the mathematical properties of that specific global geometric structure, mastered its global attributes, and eventually realized that many other geometric structure algorithms also belong to the category of global geometric

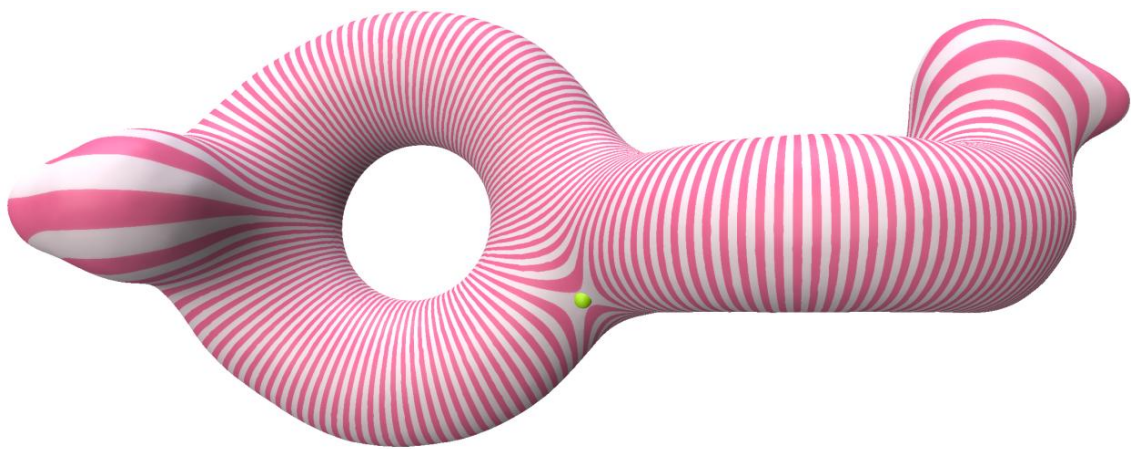
structures—and thus should be studied under the unified framework of "global geometric structures."

The global geometric structure that provided crucial inspiration for the establishment of "computational discrete global geometric structures" is harmonic foliations.

Foliations are a type of global geometric structure. The most renowned mathematician to study foliations from an abstract perspective is Fields Medal laureate Bill Thurston. As shown in the figure, in the 1970s (when Thurston was a doctoral student), he and Dennis Sullivan hand-drew diagrams of foliations on the corridor walls of the Mathematics Department at the University of California, Berkeley. These diagrams were preserved for decades, inspiring generations of students and faculty to pursue research in geometric topology.

In the 1970s, Bill Thurston conducted extensive and in-depth research on foliations, achieving remarkable progress in the field. Many scholars believed that Thurston had "concluded" the research on foliations—meaning he had addressed the core questions and advanced the field to such an extent that they chose not to enter this research area afterward.

Foliations can exist on manifolds of various dimensions. The simplest form of a foliation is a set of parallel lines on a plane. On a 2D surface, a foliation also consists of a set of curves. In a certain geometric sense, these curves can be regarded as "parallel" to one another—analogueous to the set of parallel lines on a plane. Due to this property, 2D foliations have intuitive visual representations, as shown in the figure.



(Figure 16: A foliation, generated by Geometric.)

Building on the work of Professor Steven J. Gortler (his collaborative advisor during a visit to the Harvard University School of Engineering), Dylan Thurston (son of Bill Thurston), and students supervised by Gortler—work that began many years ago—the author conducted in-depth algorithmic research on foliations of 2D surfaces. After years of effort, the author designed a convergence-guaranteed algorithm for generating harmonic foliations.

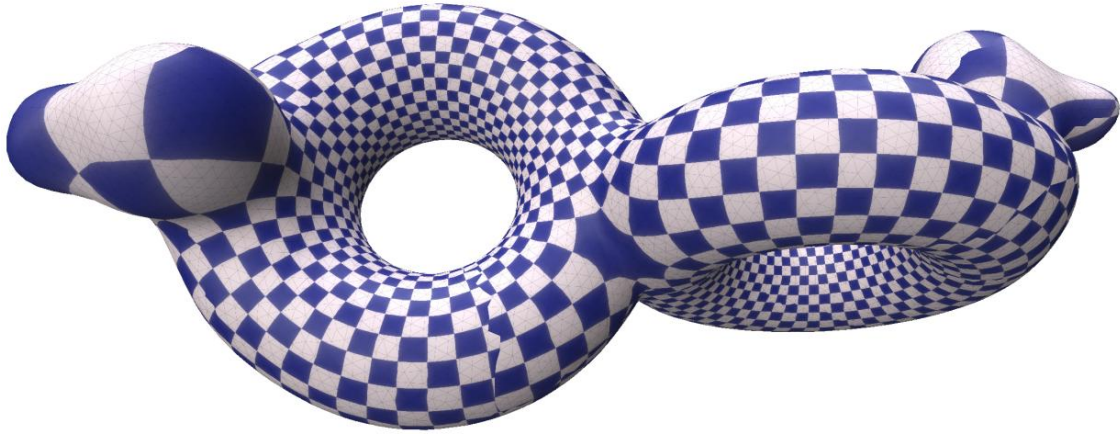
Algorithm design serves as a tool for understanding global geometric structures. Through the algorithmic process of transforming non-harmonic foliations into harmonic ones, and using harmonic foliations as a concrete case study, the author gradually gained insight (from an algorithmic perspective) into the global properties of a broader class of global geometric structures and their differences from local geometric structures.

Without such a concrete example or the hands-on process of researching algorithms, it would be extremely difficult to grasp the core essence of underlying global geometric structures from the perspective of abstract concepts alone.

The research field of "computational discrete global geometric structures" was established to solve problems that are difficult to address from other perspectives. For instance:

Differential 1-forms can be viewed as a subset of harmonic foliations, while harmonic foliations represent an extension of differential 1-forms.

Algorithms for differential 1-forms can be designed from the perspectives of conformal geometry or exterior calculus; however, it is very challenging to extend the thinking behind these two perspectives to the algorithmic design of foliations.



(Figure 17: A holomorphic quadratic differential, generated by Geometric.)

For a harmonic foliation consisting of a set of horizontal lines, rotating it by 90 degrees yields a harmonic foliation of vertical lines. Combining these two foliations results in another related global geometric structure: holomorphic quadratic differentials, as shown in the figure.

The successful algorithmic design research on harmonic foliations (a type of global geometric structure) enabled the mastery of algorithmic principles for other global geometric structures related to foliations—such as holomorphic quadratic differentials.

This developmental approach specifically adopts the "global geometry" perspective to analyze problems. While holomorphic quadratic differentials can also be studied from perspectives like conformal geometry, geometric analysis, or partial differential equations (PDEs), algorithm design tends to be relatively more straightforward when viewed through the lens of global geometric structures.

Prior to this, although algorithms for "global geometric structures" (e.g., vector fields, differential 1-forms, and frame fields) already existed, scholars primarily conducted research from diverse perspectives and frameworks—such as computational conformal geometry and discrete exterior calculus.

Based on practical experience and judgments regarding the future development of the field of differential geometric topology computation, we argue that all these studies can be unified under the new framework we propose: "computational discrete Global Geometric Structures".

The core guiding principle of this framework is "the computability of global geometric structures". It focuses on algorithmic research for various types of "global geometric structures" and systematically conducts the design of discretization algorithms.

Thus, drawing on Zhao Hui's research insights and practical experience in the fields of harmonic foliations and super-structured quadrilateral meshes, combined with a review and analysis of relevant research by other scholars, this paper proposes and establishes the new research direction of "computational discrete Global Geometric Structures"—clearly defining its research domain, core content, and methodological system.

The core of this research lies in conducting theoretical studies, algorithm design, code implementation, and industrial application exploration for various "global geometric structures" developed by mathematicians—with a particular focus on global geometric structures on 2D surfaces.



(Figure 18: A meromorphic quadratic differential, generated by Geometric.)

8 The Significance of Constructing Computable Global Geometric Structures

While the core of computational discrete Global Geometric Structures lies in algorithm research, the inherent properties of global geometric structures themselves drive numerous groundbreaking innovations. This is primarily because most current technologies are built on the study of local mathematical structures, whereas research on global mathematics has not yet been popularized in technical fields. Consequently, any algorithm related to global geometric structures is poised to drive progress and development across various technical domains.

The evolutionary trajectory—Local Differential Geometry → Topology → Global Geometry → Global Geometric Structures → computational discrete Global Geometric Structures—clearly demonstrates how abstract mathematics gradually advances toward industrial applications. This path aligns more closely with the inherent logic of scientific and technological development.

Currently, global geometric structures are in the historical process of evolving from abstract theory to computable implementation. The algorithmic research of computational discrete global geometric structures is not a mere "simple calculation" of abstract global geometric structures. Instead, it first requires discretizing the manifold on which these structures depend, followed by algorithm design on the discrete manifold. The core objective is to preserve the global properties of the global geometric structures (as defined on smooth manifolds) during discretization, rather than merely achieving local approximation.

In the process of researching algorithms for various global geometric structures, new geometric objects and research tools are emerging—this is not merely a simple calculation of existing mathematical theories. In fact, new tools and methods developed based on the computability of certain global geometric structures have successfully solved many long-standing challenges in industrial technology. For example, the index counting technique proposed by Professor Steven J. Gortler (Harvard University) during his algorithmic research on differential 1-forms (a type of global geometric structure) was later applied to the field of rigidity, addressing numerous technical problems in that domain.

The research direction of "computational discrete Global Geometric Structures" established by the author encompasses four key dimensions: theoretical research, algorithm design, code implementation, and industrial application. Its application scenarios extend beyond the field of CAD-CAE-CAM industrial software to support research in areas such as topological materials in physics. Currently, many topological materials and quantum phenomena are, in essence, physical realizations of specific global geometric structures.

The academic community has generally recognized the profound role of abstract mathematics in physical experiments. However, physicists face three practical dilemmas:

Difficulty in allocating additional time and energy to learn and master cutting-edge geometric topology theories;

Even if they master these theories, the challenge of accurately identifying the specific mathematical tools needed from a vast body of theoretical knowledge;

The need to explore concrete pathways to apply these theories to their own physical research.

All three steps require physicists to invest enormous time and effort, and cannot be resolved solely by relying on external resources.

To address this, the approach proposed in this paper is: first, conduct algorithm design and

computational implementation for the global geometric structures involved in these physical fields; then, use these results to guide the practical execution of physical experiments. By leveraging algorithms as a bridge, a direct connection is established between abstract geometric theories and physical experiments—thus resolving the three aforementioned investment dilemmas faced by physicists.

Among the various global geometric structures studied by the mathematical community:

Some can be artificially constructed using computer algorithms;

Others are naturally generated through physical processes in nature. For instance, physical particles such as electrons and photons are global geometric structures constructed by nature—nature itself can be viewed as a "computer."

Nevertheless, a large number of global geometric structures remain confined to the realm of abstract theory: their existence has been proven abstractly, but they lack both corresponding physical entities and algorithmic construction. With technological advancement, it is expected that future innovations in physical materials or algorithmic breakthroughs will enable the construction of these structures.

The paper proposes the following approach: for natural global geometric structures not yet "constructed by nature," we can first use algorithms to study their properties and potential construction methods. Based on the results of this algorithmic research, we then design corresponding physical experiments, ultimately realizing the construction of various new global geometric structures using physical materials. This process transforms more global geometric structures from abstract concepts into real-world existences.

Just as the application of natural global geometric structures (such as electrons and photons) has driven the development of technological civilization over the past two centuries, artificially constructed physical materials based on global geometric structures will undoubtedly propel even greater progress in technological civilization in the future.

This represents the core contribution and ultimate goal of the research direction, methods, domains, and tools of "computational discrete Global Geometric Structures" proposed in this paper.

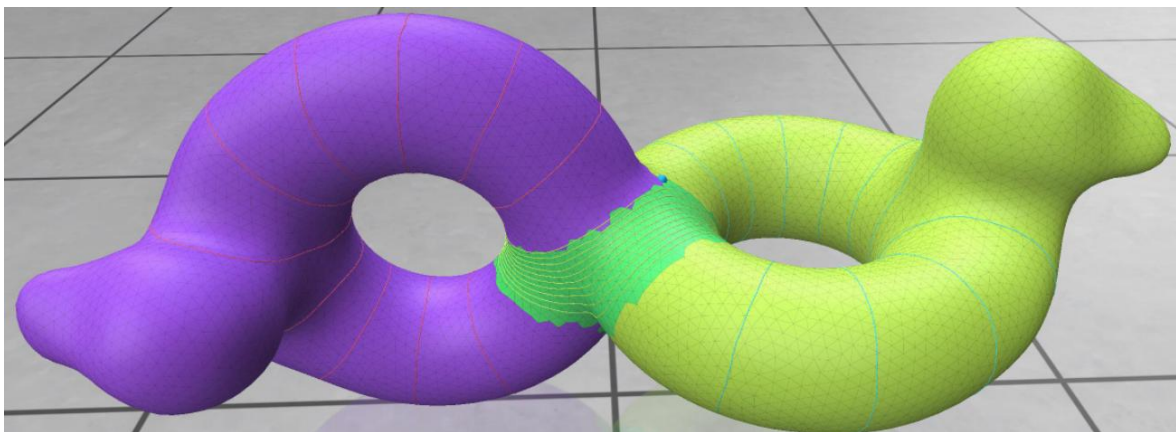
9 Zhao Hui' s Work on computational discrete Global Geometric Structures

All innovations proposed in this paper are based on the author' s three decades of extensive practical research in algorithm design, code implementation, and visualization. They have been gradually refined through reflection and long-term accumulation, rather than being developed from abstract concepts to other abstract concepts. The research work that provided inspiration for the establishment of "computational discrete Global Geometric Structures" mainly includes the following:

- (1) Non-harmonic foliations;
- (2) Harmonic foliations;
- (3) Ribbon-Graph;
- (4) Optimization from non-harmonic to harmonic states;
- (5) Foliations with poles;
- (6) Holomorphic quadratic differentials;
- (7) Meromorphic quadratic differentials;
- (8) Harmonic foliations on surfaces with boundaries;
- (9) Super-structured quadrilateral meshes.

Through the algorithm design, code implementation, and visualization observation of these global geometric structures, combined with cross-referencing differential geometric topology theories and consulting relevant research methods of other scholars, a complete framework of ideas has been gradually summarized and formed.

This paper only provides a brief overview of the relevant intuitive descriptions and algorithmic experimental results for the mathematical concepts and theories involved. For more detailed and specific mathematical concepts, please refer to relevant textbooks.



(Figure 19: A non-harmonic foliation, generated by Geometric.)

(1) Non-harmonic Foliations

Foliations can be either harmonic or non-harmonic. A harmonic foliation refers to a set of "parallel lines" that are the smoothest under the current metric, while a non-harmonic foliation consists of

non-smooth curves. As shown in the figure below, the curves of the non-harmonic foliation are relatively dense in the middle of the mesh model and sparse in other areas; in contrast, the curves of a harmonic foliation would be evenly distributed across the surface.

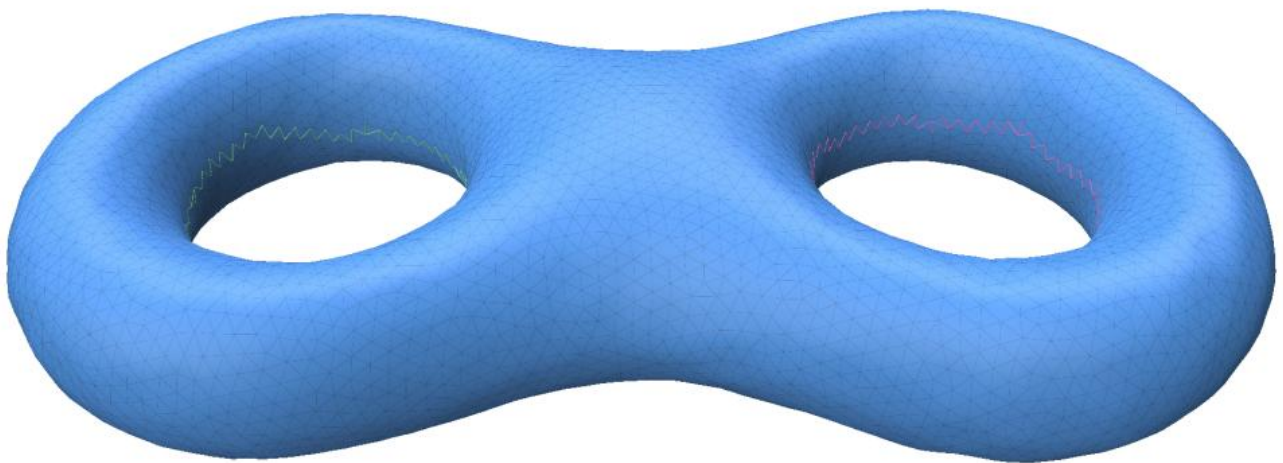
A non-harmonic foliation cannot be represented as a harmonic function at each local point on the surface—i.e., it does not satisfy the conditions of a harmonic function, hence the name "non-harmonic foliation." However, it is topologically equivalent to a certain harmonic foliation.

A non-harmonic foliation may also fail to cover the entire surface and only exist in a specific part of the surface. As shown in the figure, the two annuli of different colors represent a foliation—this foliation is not only non-harmonic, but also has local values of 0 in the regions outside the annuli; only the regions on the annuli have non-zero values.

However, through optimization, the regions with non-zero values can be gradually expanded to the regions with zero values. Nevertheless, these foliations are all equivalent in the sense of Whitehead Classes.

This phenomenon reflects the characteristic of foliations as a global geometric structure: even though the values in most regions are 0, the topology of the entire foliation is already determined by the foliation in the initial region.

This characteristic cannot be studied from other perspectives such as discrete exterior calculus. Therefore, this is the reason why the author proposes that "algorithm design" should be approached from the perspective of "global geometric structures"—thereby enabling the development of more algorithms and applications related to global geometric structures.



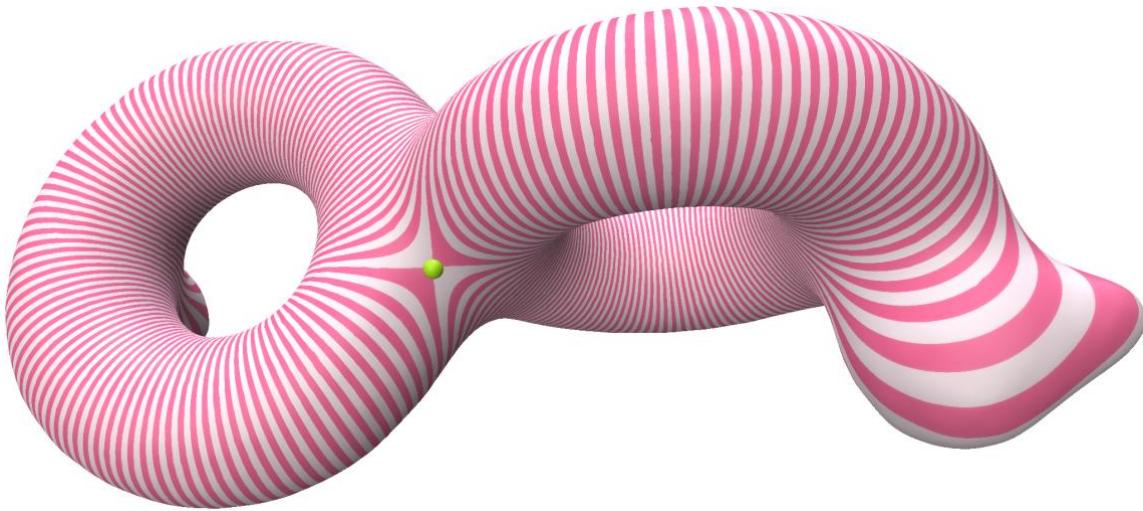
(Figure 20: A non-harmonic foliation, generated by Geometric.)

(2) Harmonic Foliation

Each point of a harmonic foliation can be represented by a local harmonic function, but this requires multiple harmonic functions expressed in multiple coordinate systems as well as the transformations between them. A harmonic foliation cannot be represented by a single harmonic function on the entire surface. This is the global property of a global geometric structure. As shown in the figure, the curves on

the harmonic foliation are evenly distributed, achieving the smoothest result under the metric of the surface. If the genus of the surface is not 1, zeros (i.e., points where the values of several independent local functions are 0) will appear when the foliation covers the entire surface. The value 0 of these zeros does not change with the transformation of coordinate systems and remains 0 in any coordinate system, so they are special points. However, the number of zeros of a foliation on each surface is finite and conforms to a specific formula.

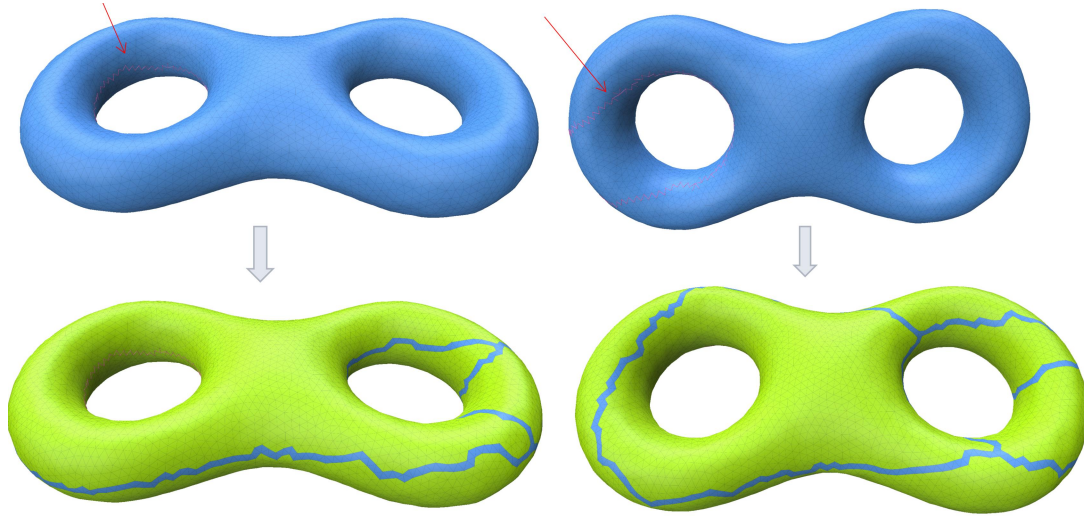
The small spheres in the figure represent the zeros of the harmonic foliation. On a surface with genus g , a harmonic foliation has $4g - 4$ zeros of degree 1. Three curves meet at a zero of degree 1. Two zeros of degree 1 can merge into a zero of degree 2, where four curves meet—just like the zero shown in this figure. For a non-harmonic foliation that covers the entire surface, there are also $4g - 4$ zeros.



(Figure 21: A harmonic foliation, generated by Geometric.)

(3) Ribbon-Graph

A Ribbon-Graph is a special type of graph on a surface. Each annulus of a non-harmonic foliation can generate several Ribbon-Graphs, as shown in the figure. The red arrows point to the foliation, and the white arrows point to the generated Ribbon-Graphs. Ribbon-Graphs have many important applications in the research of various geometric theories; here, they are used in the optimization process of foliations from non-harmonic to harmonic states.



(Figure 22: A foliation and a ribbon graph, generated by Geometric.)

(4) Non-harmonic to harmonic foliation

A non-harmonic foliation can be optimized into a harmonic foliation through an "algorithm". As shown in the figure, the non-harmonic foliation at the top-left (a blue annulus) is gradually optimized into the harmonic foliation at the bottom-right via various intermediate non-harmonic foliations. Within the same Whitehead class, there can be many non-harmonic foliations, but only one harmonic foliation.

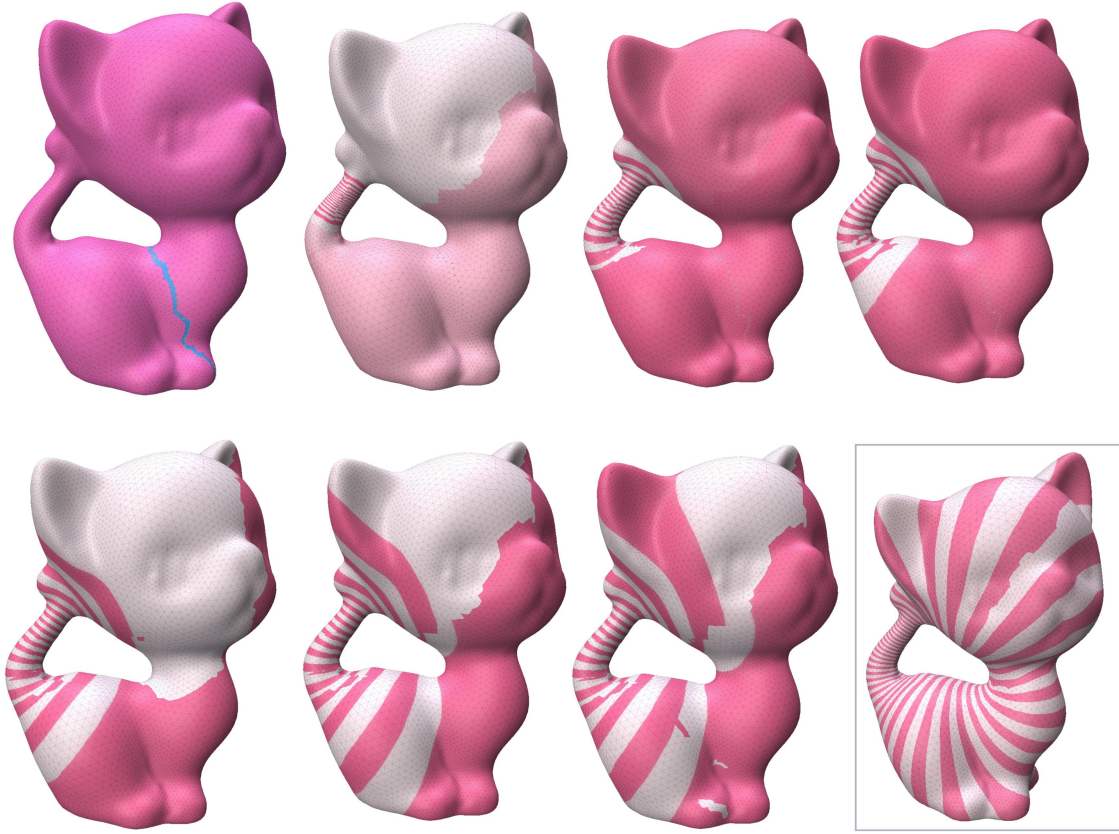
This optimization process does not exist in the field of mathematics and is difficult to achieve from the perspective of abstract mathematical tools. However, from the perspective of algorithmic tools, it is relatively natural to conceive of this method. This optimization process has inspired the idea proposed in this paper—using "algorithms" as tools to explore "global geometric structures"—which constitutes the core of computational discrete Global Geometric Structures.

During the optimization of a non-harmonic foliation, the support of Ribbon-Graphs is required; otherwise, a harmonic result cannot be obtained. The optimization process will get trapped in a local optimum, resulting in another foliation that belongs to a different Whitehead class and contains additional poles.

The design of the algorithm requires a large number of technical tools, such as computer graphics rendering. For example, the stripes in the figure are rendered and presented using texture mapping technology in computer graphics.

The "harmonicity" of a harmonic foliation is defined with respect to the metric of the surface. If two surfaces have the same topology but different metrics, the shape of the harmonic foliation on each surface will also differ.

A surface can have an infinite number of Whitehead equivalence classes. If the initial foliation does not belong to a certain Whitehead class, the resulting harmonic foliation will belong to another Whitehead class.



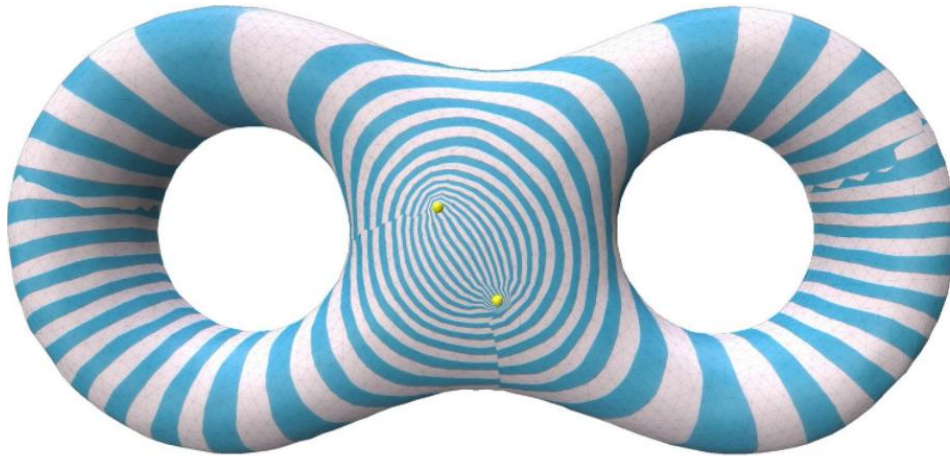
(Figure 23: A non-harmonic foliation to a harmonic one, generated by Geometric.)

(5) Harmonic foliation with poles

Harmonic foliations can also have poles, where the degree of a pole can be 1 (as shown in the figure below), 2, or other values. For a surface with genus g that contains poles, the harmonic foliation on it must satisfy the following equation:

$$(\text{Number of poles}) \times (\text{Degree of poles}) + (\text{Number of zeros}) \times (\text{Degree of zeros}) = 4g - 4$$

A pole can be regarded as an infinitely small hole cut out of the surface. During the optimization process of transforming a foliation from non-harmonic to harmonic, the positions of the poles remain unchanged, while the positions of the zeros [keep moving until they reach their optimal positions.](#)



(Figure 24: A harmonic foliation with two poles of degree one, generated by Geometric.)

In his classic 1994 paper *On Proof and Progress in Mathematics*, William P. Thurston went beyond traditional formalized mathematics and conducted in-depth discussions on the essence of mathematical research, the diversity of mathematical understanding, and the socialization process of mathematical knowledge. Thurston pointed out that mathematical understanding is not a single logical process, but a multi-dimensional and individualized cognitive activity. He listed seven different ways to understand the mathematical concept of "derivative":

- (1) Infinitesimals (early idea in calculus);
- (2) Symbolic computation (algebraic rules);
- (3) Logical definition (ε - δ language);
- (4) Geometric intuition (slope of the tangent line);
- (5) Physical rate (instantaneous velocity);
- (6) Linear approximation (optimal local approximation);
- (7) Microscopic limit (limit behavior under high-magnification).

These ways of understanding are not mutually exclusive, but complementary and interdependent. The work of mathematicians lies not only in constructing rigorous definitions, but also in reconciling different perspectives to enable mathematical concepts to be effectively applied in many fields such as physics, engineering, and computer science. As Thurston emphasized, one of the core challenges of mathematics education is to guide students to internalize these diverse ways of thinking, rather than merely staying at the mechanical memorization of formulas and proofs.

Different understandings mentioned above can lead to the development of different new mathematical structures. For example, the concept of exterior calculus can be built on the basis of the fourth understanding; however, it is difficult to develop the concept of exterior calculus from perspectives such as the third one. In other words, one of the purposes of having different perspectives on the same mathematical concept is to go beyond the concept itself and establish new developments in different directions.



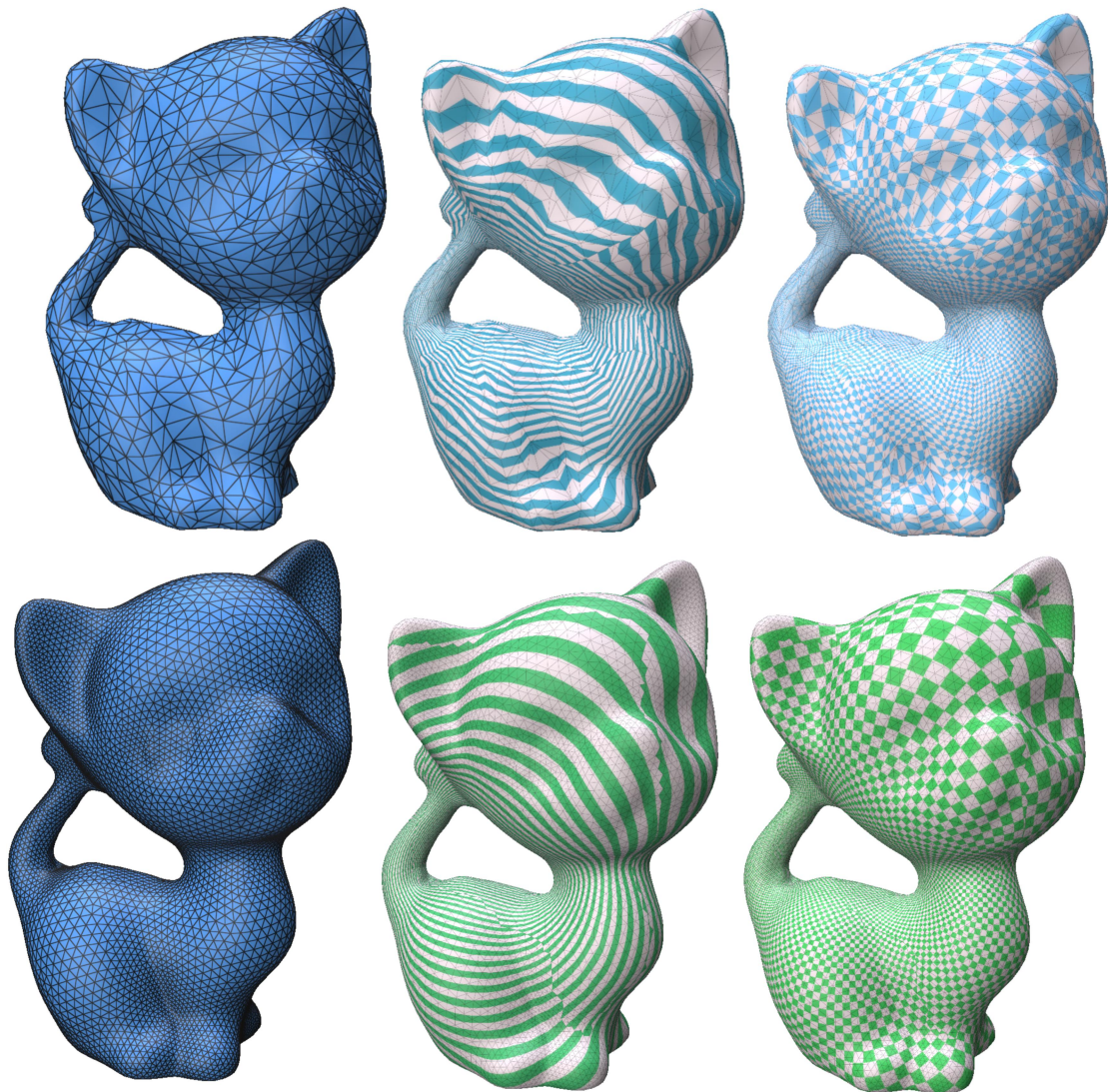
(Figure 25: A harmonic foliation with three poles of degree 2, generated by Geometric.)

The approach proposed in this paper — understanding "global geometric structures" from the perspective of "algorithms" — is an innovative way to comprehend global geometric structures. It helps develop new mathematical structures that cannot be obtained from other perspectives such as discrete exterior calculus, finite element exterior calculus, and conformal geometry. For instance, the transformation process from non-harmonic to harmonic states is only very natural and clear when viewed from the "algorithmic" perspective; moreover, new structures and new concepts can be developed in the course of the "algorithm."

The proposal of the research direction of computational discrete Global Geometric Structures in this paper is based on the gradual formation of a concrete framework of ideas through the observation of these specific algorithms and experiments in practice. For example, through the aforementioned research on "algorithms" for capturing harmonic foliations with poles, the author gradually realized that in addition to the seven different understandings of mathematical concepts familiar to mathematicians proposed by Thurston, the "algorithms" in computer science also provide an additional dimension for understanding mathematical concepts. The most crucial point is to abandon the formalized way of memorization and transform concepts into a process and act of understanding.

The proposal of the research direction of computational discrete Global Geometric Structures did not follow the sequence of first mastering differential geometry, then learning global geometry, and finally researching algorithms. Instead, it started with "algorithms" from the very beginning. During the process of researching some specific algorithms that were not known to belong to global geometric structures at that

time, the author gradually condensed various phenomena into a coherent thread, and finally formed the concept and framework of studying "global geometric structures" using "algorithms" as tools.



(Figure 26: A harmonic foliation and its holomorphic quadratic differential.)

(6) Holomorphic Quadratic Differentials

A holomorphic quadratic differential can be regarded as composed of two mutually perpendicular harmonic foliations, as shown in the figure. The algorithm for generating it first creates a set of horizontal harmonic foliations, and then generates the corresponding vertical harmonic foliations.

The poles and zeros of a holomorphic quadratic differential are consistent with those of the harmonic foliations—specifically, the zeros and poles of the horizontal and vertical harmonic foliations are located at the same positions.

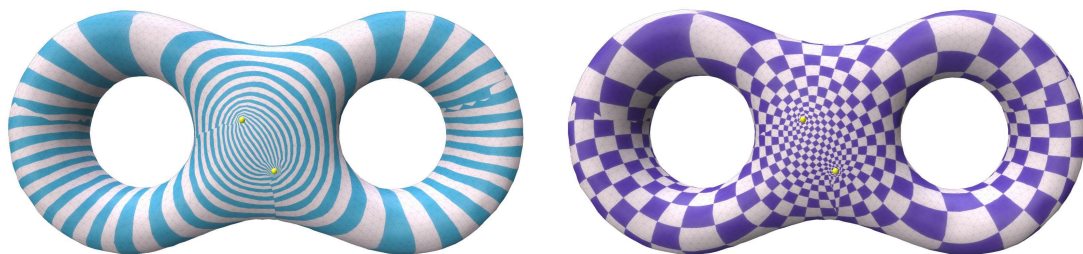
Here, the "holomorphy" of the holomorphic quadratic differential needs to satisfy the definition of conformality, and it can also be analyzed from the perspective of conformality. However, the conformal

perspective is a local one, making it difficult to understand global properties. Moreover, it is challenging to capture global constraints when designing algorithms from the conformal perspective.

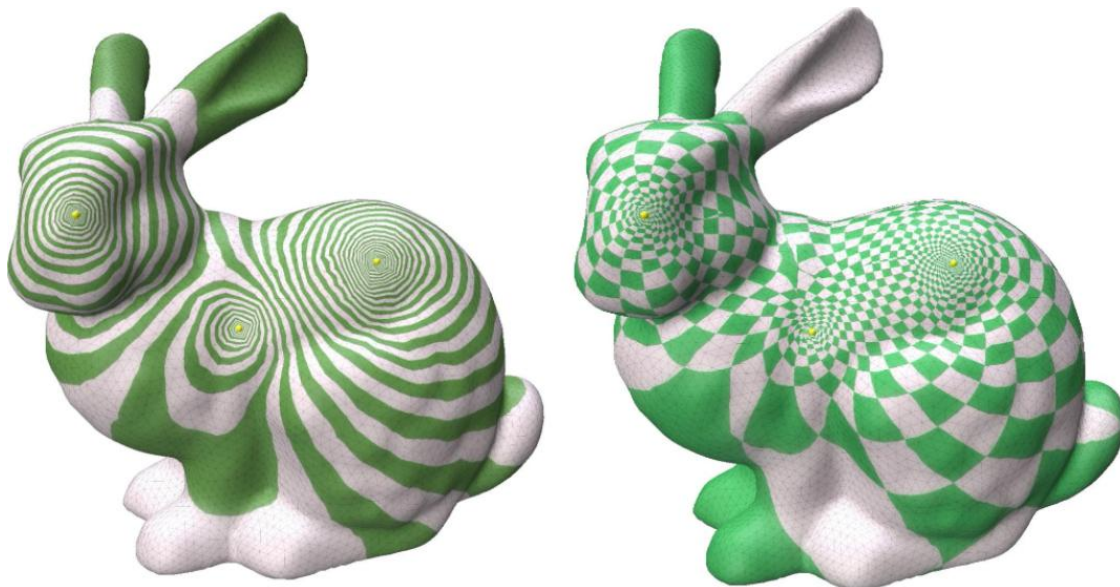
These insights reflect the author's thinking process during the development of the research direction of "computational discrete Global Geometric Structures." Gradually, the author established this research direction, which adopts the perspective of studying "global geometric structures" and uses "algorithms" as the core tool.

(7) Meomorphic Quadratic Differentials

A holomorphic quadratic differential has only zeros, not poles. If a harmonic foliation has poles, the corresponding structure is a meromorphic quadratic differential.



(Figure 26: A harmonic foliation and its meromorphic quadratic differential, generated by Geometric.)

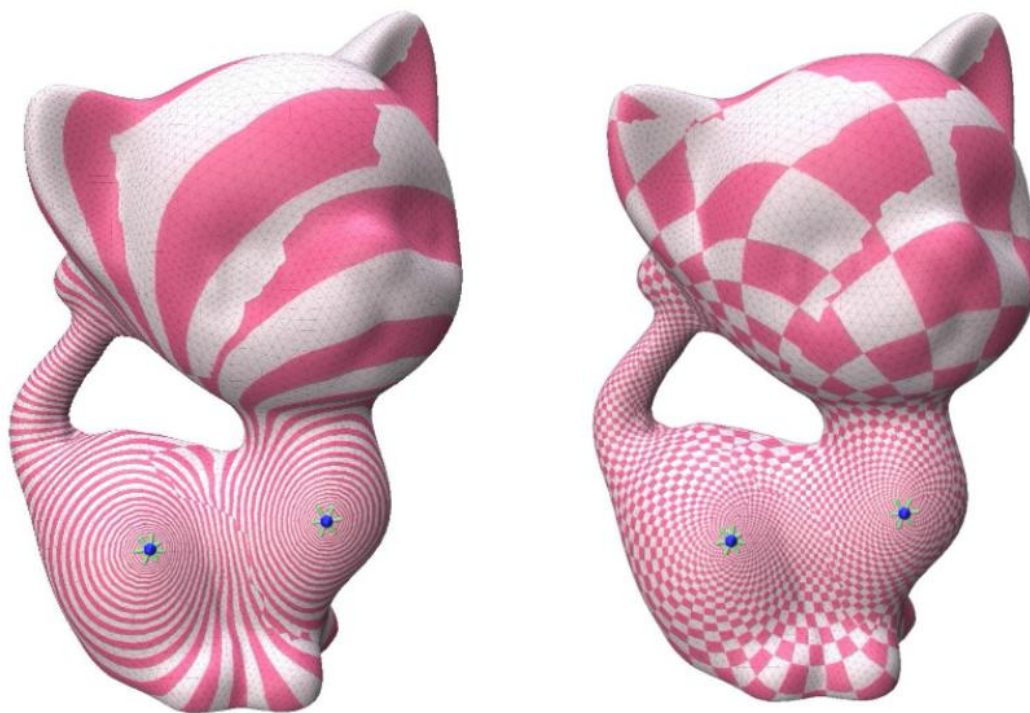


(Figure 27: A harmonic foliation and its meromorphic quadratic differential.)

During the research on foliations, the author was initially unaware that foliations are global geometric structures, nor had the concept of "global geometric structures" been established at the outset. The establishment of the concept of global geometric structures was derived from the practical research on "algorithms," rather than being obtained by memorizing a definition from a mathematics textbook first.

Instead, when researching "algorithms" related to foliations, the author gradually discovered certain

phenomena that could not be computed through approximation using local formulas or methods. Centering on "algorithms," the author progressively reflected on relevant mathematical concepts and theories. Eventually, from a thinking perspective, the author realized that a unified framework could be formed by adopting the perspective of "global geometric structures" and using "algorithms" as tools — and this framework could lead to the development of entirely new research fields.



(Figure 28: A harmonic foliation and its meromorphic quadratic differential, generated by Geometric.)

(8) Harmonic foliations with Boundaries

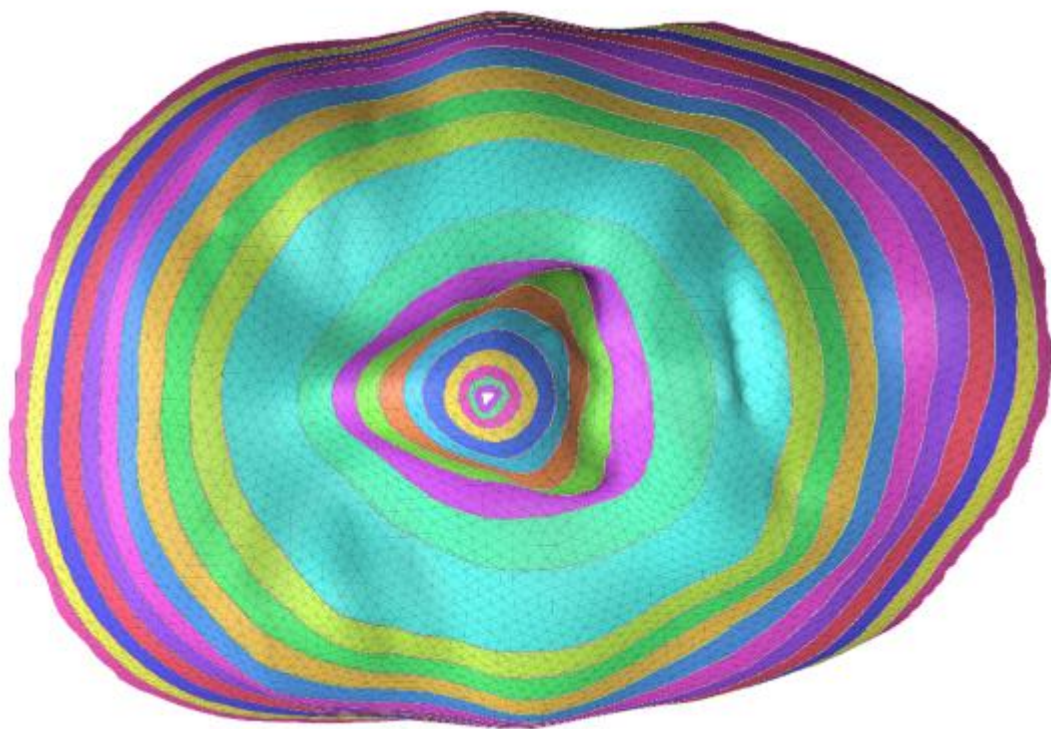
The foliations and holomorphic quadratic differentials discussed earlier all exist on closed surfaces; however, foliations can also exist on surfaces with boundaries. A surface with boundaries is different from a surface with certain infinitesimal features. The harmonic foliations with poles mentioned previously have infinitesimal holes (Punctures) at the pole positions on the surface, whereas the holes of a surface with boundaries are not infinitesimal—they have a specific perimeter. Therefore, it is impossible to treat the boundaries as poles, which means corresponding meromorphic quadratic differentials (and similar structures) cannot be obtained.

As a result, the two (surfaces with poles and surfaces with boundaries) differ not only in mathematical concepts but also in the process of algorithm design. Although the algorithms follow a roughly similar framework, detailed analysis must be re-conducted. When expressed in mathematical formulas, both harmonic foliations with poles and harmonic foliations on surfaces with boundaries take the form of harmonic functions in local coordinate systems; thus, it is difficult to capture their global properties from this perspective. From an algorithmic perspective, however, since direct replication of existing methods is not feasible and algorithm design must be re-considered, a different viewpoint is provided to examine the essence of foliations—and gradually, their global geometric characteristics are revealed. The process of

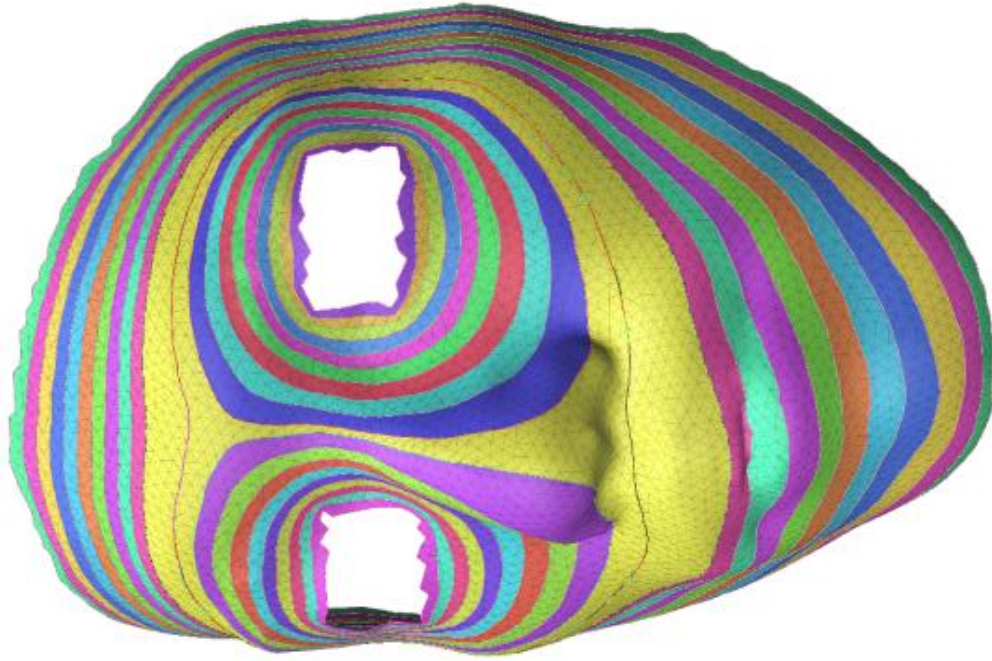
designing these detailed algorithms and conducting repeated experiments gradually inspired the author of this paper to establish the research framework for the computability of global geometric structures.

In the case of surfaces with boundaries, the annuli at the boundaries need to be treated as non-harmonic foliations for optimization, and finally, harmonic foliations are obtained. Alternatively, a line can be drawn between the boundaries to serve as the initial foliation.

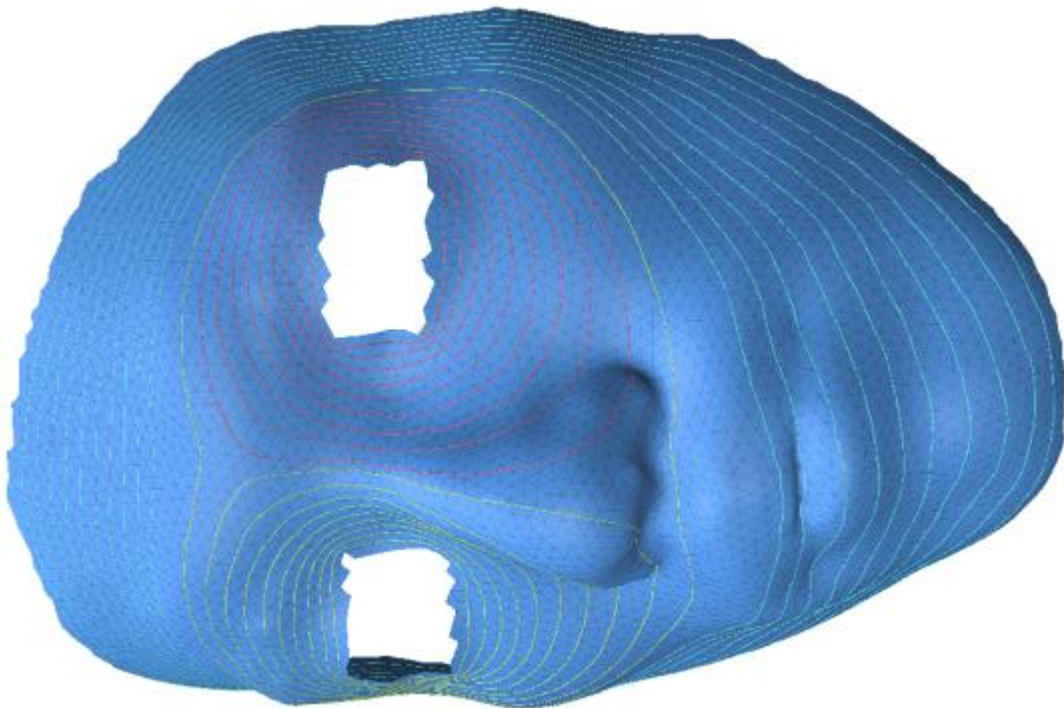
It is also possible to sew two identical surfaces with boundaries along their boundaries to form a complete surface without boundaries; the resulting foliation will be consistent on both original surfaces. The sewn surface is a surface of higher genus, but since the two original surfaces are identical, it appears visually as a single surface in three-dimensional space. During algorithm design, however, computations can be performed on the sewn surface — only by reclassifying the original boundary annuli as non-boundary annuli in the new surface, and the results obtained will remain the same.



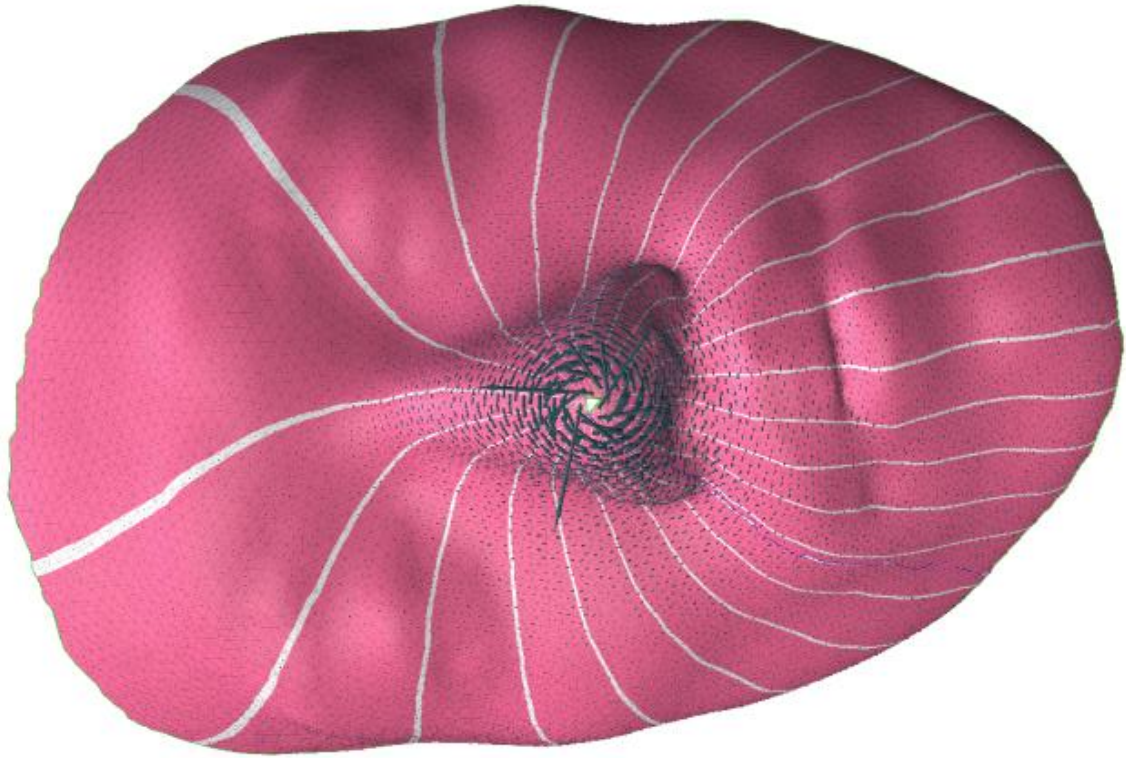
(Figure 29: A harmonic foliation on a surface with boundaries, generated by Geometric.)



(Figure 30: A harmonic foliation on a surface with boundaries, generated by Geometric.)



(Figure 31: A harmonic foliation on a surface with boundaries, generated by Geometric.)

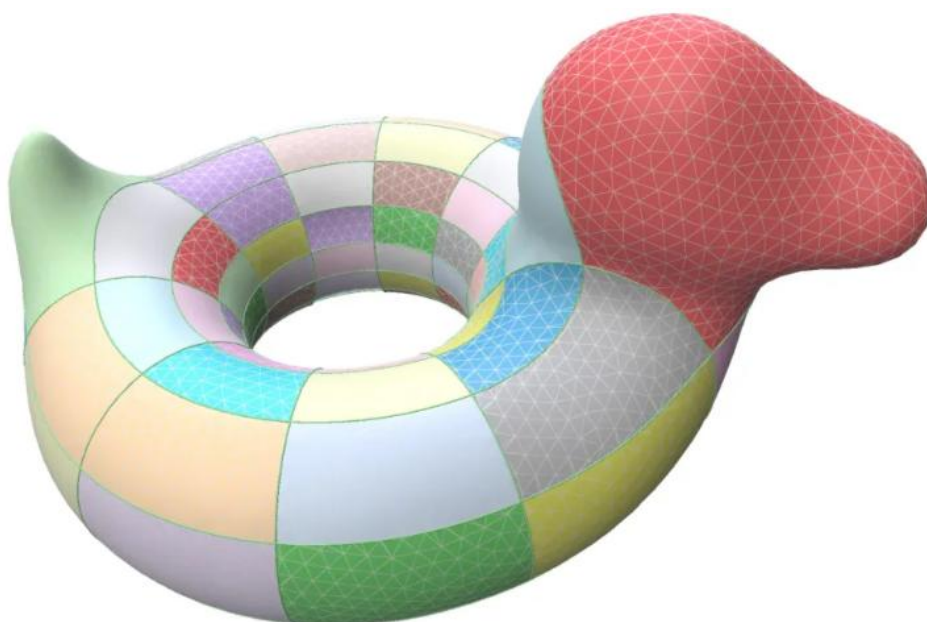


(Figure 31: A harmonic foliation on a surface with boundaries, generated by Geometric.)

(9) Super Structured Quad Mesh

The concept of "super-structured quadrilateral meshes" has no corresponding counterpart in the field of mathematics; it is an entirely new concept proposed by the author of this paper. The proposal of this concept is primarily aimed at solving certain technical challenges in industrial software.

A super-structured quadrilateral mesh refers to a quadrilateral mesh where "the overall arrangement and structure of quadrilaterals are controllable and designable" — this is a definition oriented toward algorithm development.

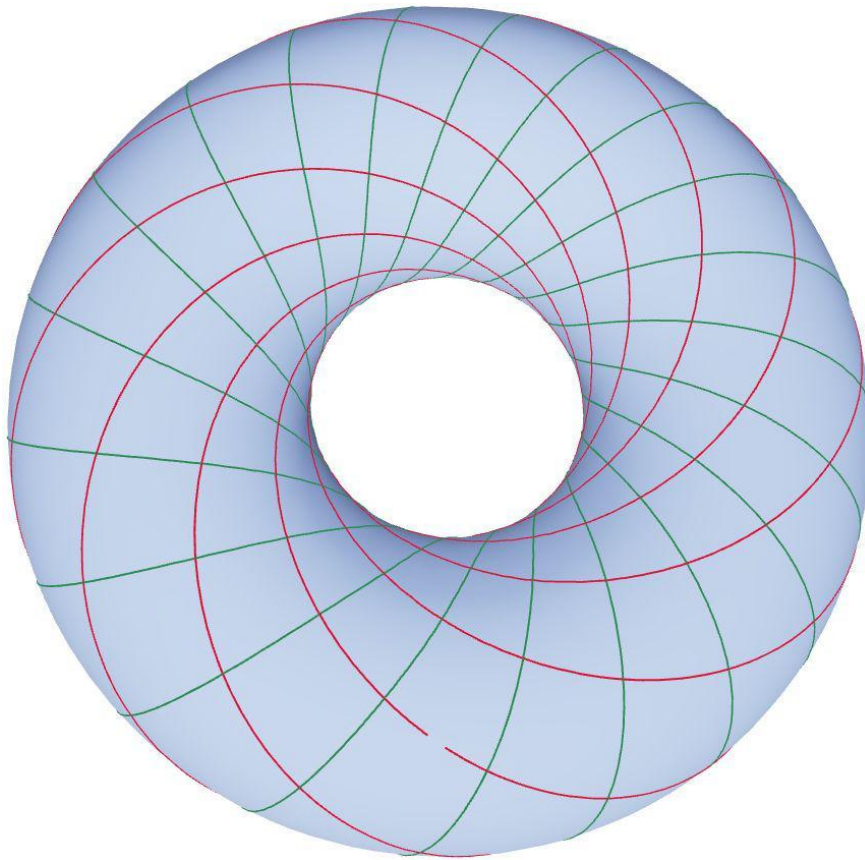


(Figure 32: Square Tiling, generated by Geometric.)

This concept encapsulates several concepts and theories related to global geometric structures into a single concept. To realize this concept, it is necessary to design algorithms for a number of mathematics concepts associated with global geometric structures and ultimately integrate them into a unified whole—unlike the previous focus on foliations, which targeted a single type of global geometric structure. For example, the development of super-structured quadrilateral meshes requires algorithm design for, and the integration and synthesis of, the following geometric topology theories:

- (1) Moduli spaces;
- (2) Teichmüller spaces;
- (3) Dynamical systems on surfaces;
- (4) Ribbon-Graphs (Figure 25);
- (5) Harmonic foliations (Figure 26);
- (6) Holomorphic quadratic differentials;
- (7) Meromorphic quadratic differentials;
- (8) Thurston norm;
- (9) Translate surfaces;
- (10) Half-translation surfaces;
- (11) Flat surfaces;
- (12) Riemann surfaces;
- (13) Square-tiled surfaces of surfaces;
- (14) Masur-Veech volume;

-
- (15) Meanders;
 - (16) Billiards;
 - (17) Interval exchange transformations;
 - (18) Teichmüller flow;
 - (19) Strata of abelian and quadratic differentials;
 - (20) Mapping classes of surfaces;
 - (21) Research on surfaces and 3-manifolds by the Thurston school, among others;
 - (22) ...



(Figure 33: Super structured quad mesh, generated by Geometric.)

In most cases, one can directly start from the abstract concept of a global geometric structure and then directly develop the corresponding algorithm design. However, the example of super-structured quadrilateral meshes demonstrates that viewing problems from the "algorithmic" perspective can bring insights that are not easily visible from the abstract perspective. Yet this perspective does not emerge out of thin air; instead, it relies on existing global geometric structures to derive new ideas.

From this, it can be seen that the "computational discrete Global Geometric Structures" proposed in this paper is not merely about targeting a single geometric structure and then finding a one-to-one "algorithm" to capture the global mathematics of that structure. Rather, it requires developing a comprehensive algorithm based on a variety of "global geometric structures." There is no simple,

mechanical one-to-one correspondence between computational discrete Global Geometric Structures and global geometric structures themselves. This is precisely where the significance of this brand-new research direction lies.

10 Global Geometric Structures and Physical Construction

Another source of inspiration for the research direction of "computational discrete Global Geometric Structures" proposed in this paper stems from reflections on the research methods of physicists. Based on the author's perspective on global geometric structures, established through algorithm research, it is argued that many important works in physics — such as those on quantum theory and topological materials — also investigate global geometric structures using "physical experiments," as well as various physics-related methods that involve data analysis from physical experiments and mathematical theoretical derivation. Examples include:

(1) Although Newton's research focused on local (rather than "global geometric") structures, Newtonian mechanics indirectly gave rise to quantum mechanics, which later accounts for global geometric structures.

(2) Max Planck studied global geometric structures by mathematically analyzing physical experimental data on black bodies and deriving mathematical formulas — this marked the germination of quantum theory.

(3) Pioneers of quantum mechanics such as Werner Heisenberg and Niels Bohr derived equations and conclusions from observations of physical experiments, rather than through abstract mathematical approaches or by building abstractions upon other abstractions.

(4) Paul Dirac identified inconsistencies in existing formulas from an abstract-to-abstract perspective, then developed new formulas that describe global geometric structures. The new formulas in question are Dirac's equations. Chen-Ning Yang commented on this approach to discovering new global geometric structures, stating it was like "prose by autumn waters, untainted by dust," and praised Dirac as someone whose "innate intellect brings forth myriads of phenomena, and whose integrity transcends ordinary norms." Yang's praise primarily stemmed from the fact that this approach did not rely on experimental data — unlike Planck, who analyzed data, Dirac analyzed formulas. While Dirac worked from abstraction to abstraction and from formula to formula, the initial formulas he used corresponded to specific physical quantities. Consequently, the global geometric structures he derived could directly reveal new physical phenomena.

(5) Physicists such as André-Marie Ampère and Michael Faraday derived certain local equations through observations of electromagnetic physical experiments; these local equations represent the local expressions of global geometric structures.

(6) James Clerk Maxwell developed Maxwell's equations by synthesizing the equations of scientists including Charles-Augustin de Coulomb, Hans Christian Ørsted, Ampère, and Faraday. Although Maxwell did not conduct direct physical experiments himself, he engaged in further reflection based on the results of previous physical experiments. The equations he derived also serve as local expressions (under a specific coordinate system) of a complete global geometric structure.

(7) In physical experiments on electrons, photons, and elementary particles conducted by physicists, these physical particles are infinitely fascinating precisely because they are all global geometric structures. Specifically, these particles exist on fiber bundles where the entire universe acts as the manifold; thus, physical experiments capture global properties through local observations. The equations derived from local observations (under a single coordinate system) inherently contain the properties of global geometric structures.

(8) Physicists such as Professor Klaus von Klitzing conducted experiments on various quantum Hall effects; through these physical experiments, they obtained local formulas for global geometric structures.

(9) Physicists like David Thouless analyzed the formulas of the quantum Hall effect and provided explanations of global geometric structures.

(10) Professor Chen-Ning Yang's work on Yang-Mills equations also involved deriving global geometric structures through abstract reasoning from existing equations—however, the existing equations he built upon were themselves grounded in physical experiments.

(11) Contemporary physicists such as Qikun Xue have developed topological materials; these efforts essentially involve constructing specific global geometric structures directly through physical experiments.

(12) Professor Hong Ding, recipient of the 2025 Future Science Prize in Physical Sciences, also conducted work that involved constructing physical materials related to topological global geometric structures through physical experiments.

There are numerous other groundbreaking works by prominent physicists—whether experimental or theoretical physicists. From the perspective of this paper, all these works revolve around various global geometric structures.

This is because research related to global geometric structures—whether in mathematics, algorithms, or physics—cannot be mechanically achieved through solving local equations (as is possible with local equations themselves). Instead, all such research requires creative development to capture "global properties." For local phenomena described by local equations, mechanical calculations of the equations may be feasible, but this approach rarely leads to new innovations.

Although global geometric structures possess holistic characteristics, their definitions ultimately still rely on local equations for implementation. However, if a local equation is arbitrarily constructed under a certain coordinate system, it will often fail to hold globally across the manifold and thus lack research value.

To derive meaningful equations for global geometric structures, one approach (commonly used by mathematicians) involves developing new structures based on logical reasoning within existing mathematical frameworks. For example, mathematicians propose mathematical conjectures such as the Poincaré Conjecture and the Calabi-Yau Conjecture. Most of these conjectures are built on existing research and have a high probability of being valid—this is what gives them research value.

Another important approach originates from observations of physical experiments. Conclusions from physical experiments correspond to objective realities already constructed by nature; this means the corresponding equations inherently possess a foundation for validity, even if they have not yet undergone systematic mathematical research and analysis to establish their rigor. Therefore, such global geometric structures derived from physical phenomena have already been "validated" by nature in terms of their existence. Even if they have not yet been proven through rigorous mathematical methods by mathematicians, they are undoubtedly of great value for mathematical research. Typical examples include Dirac's equations proposed by Dirac and the equations of general relativity established by Albert Einstein.

Physics research relies on physical experiments, and conducting experiments requires substantial external resources and conditions (such as experimental equipment), which are often difficult to obtain easily. In contrast, for thousands of years—since Euclid—mathematicians have used "abstract thinking" as their core tool, freeing themselves from the constraints of external physical conditions and enabling them

to advance unburdened. This way of thinking, unshackled by reality, has driven humanity to create numerous achievements in mathematical civilization that are only limited by the "boundaries of thought"—some of these mathematical structures may not yet have manifested in nature.

While mathematicians can deduce new mathematical structures from existing frameworks starting from abstract concepts, it is not easy to identify new structures that are truly valid—especially in the case of global geometric structures. If an equation is constructed arbitrarily, it will most likely fail to hold globally across fiber bundles and thus lack research value. In contrast, physicists' research directly begins with objective results already realized by nature; therefore, any new structure discovered in physics is inherently of great mathematical research value, as its existence has been confirmed through physical experiments.

Once equations and mathematical structures derived from physical experiments are thoroughly studied by mathematicians, they can break free from the constraints of external conditions associated with physical experiments. For instance, some physical experiments are difficult to conduct due to massive investment requirements, but mathematicians can conduct comprehensive research on the relevant physical equations at almost no cost, derive "abstract" conclusions, and in turn feed these conclusions back to guide physical experiments.

Even if, hypothetically, physical experiments had no cost constraints and conclusions could be obtained directly through massive experiments, it would still be necessary to conduct systematic research on these conclusions using "abstract" mathematical tools. Only in this way can all conclusions be ensured to be consistent, rigorous, logical, and extensible—rather than remaining a simple accumulation of fragmented results.

This paper argues: "Using abstract thinking as a tool to break free from the constraints of external material conditions, and exploring new geometric structures worthy of research from the results of physical experiments already constructed by nature—this precisely embodies the power of mathematics itself and the profound connection between mathematics and physics."

Currently, some global geometric structures studied in abstract mathematics do not have corresponding counterparts in physical materials or physical phenomena found in nature. However, it may be possible to artificially construct entirely new physical materials and physical phenomena that correspond to certain abstract global geometric structures. There are two approaches to this end:

The first approach involves conducting physical experiments, accidentally discovering new physical materials, and finally investigating which global geometric structure they correspond to.

The second approach involves designing physical experiments for a specific global geometric structure and obtaining new physical materials under theoretical guidance.

Currently, the first approach is more common, but it is highly accidental and unpredictable (examples include the discovery of the Hall effect). The second approach, however, requires mastery of mathematical knowledge related to global geometric structures as well as knowledge of physical experimental materials to enable planned design. Nevertheless, it is generally difficult to acquire sufficient knowledge in both fields within a feasible timeframe and with available effort.

Although theories and concepts of global geometric structures have long been widely applied in physics, physical experiments and theoretical physics research have not explicitly taken "global geometric structures" as a guiding ideology. By adding a "computable" bridge between abstract mathematics and

physics applications, this paper anticipates establishing a rapid channel connecting the abstract concepts of global geometric structures to their physical applications. This is because the "algorithms" of computational discrete Global Geometric Structures consist of step-by-step procedures—i.e., constructive proofs—which naturally align and correspond one-to-one with the constructive nature inherent in physical experiments.

11 Algorithm vs. Computation

In the brand-new research field of "computational discrete Global Geometric Structures" proposed by us, the core of "computable" lies in "algorithms," rather than simple "formula application" or "theory application"—that is, it is not about inputting a formula and then performing numerical calculations on it. For many local geometric structures, it is indeed feasible to complete calculations by inputting formulas and using computers for approximate approximation; however, "global geometric structures" are different. Although local formulas exist for them, their global nature needs to be constructed through transformations between multiple formulas, and a single local formula or equation cannot fully define a global geometric structure. Therefore, the approach of simply inputting formulas or equations and having computers perform direct calculations fails to yield results with "global characteristics."

A separate section is dedicated to emphasizing this point because teachers and students in science and engineering are familiar with the "calculation" of mathematical formulas—most of these formulas can be directly computed by computers using approximation methods (such as Taylor expansion with the first few terms truncated) to obtain results. For example, experts in disciplines like mechanics, mechanical engineering, and mathematics have less exposure to "algorithms" compared to computer scientists. As a result, they are more accustomed to the approach of "inputting formulas or equations and performing direct calculations via paper-and-pencil derivation or software," and have limited experience with calculation methods centered on "algorithms."

This situation arises because most current technologies in many science and engineering fields involve local mathematical structures, which can all be solved through such "calculation" methods—and this has further solidified the perceptions of experts in these fields.

Nevertheless, technologies that can be solved by relying solely on local calculations are already relatively mature. The core breakthrough of the "computational discrete Global Geometric Structures" proposed in this paper can only be achieved by using "algorithms" as tools.

This point is particularly prominent in practice: many experts in related fields consider "formulas and equations" to be the focus of research, with subsequent calculations being mere afterthoughts. However, this paper argues that the key to research lies in "algorithms," and the preceding formulas and equations all serve the algorithms. In short, it is necessary to design "algorithms" to capture the global properties of global geometric structures, rather than relying on local formulas within global geometric structures for calculations. The latter will lead to the loss of global properties due to calculation errors, ultimately failing to truly obtain global geometric structures—and thus contributing nothing to the computable framework of global geometric structures proposed in this paper.

Currently, the inherent perception of relying on local calculations is relatively widespread, stemming from the fact that the application of global geometric structures in industrial fields has been relatively limited in the past. As the research field of "computational discrete Global Geometric Structures" proposed in this paper becomes more widespread, the essential distinction between "algorithms" and "computation" will undoubtedly gain broader recognition.

Solving a scientific problem requires dozens of interconnected steps, which can be roughly simplified as follows: Abstract Geometric Theory → Applied Geometry → Algorithm Design → Code Implementation → Engineering Technology Application. These steps often span multiple disciplines, and each discipline differs in its values, research objectives, behavioral logic, and psychological motivations.

The field of computational discrete Global Geometric Structures falls under the category of applied mathematics, with a clear focus on "algorithms."

The academic community has always had a vague boundary between abstract mathematics and applied mathematics, which to a certain extent restricts the development of applied mathematics. In response, this paper proposes a clear criterion for distinction: "Research that can yield specific numerical values under current computer hardware, function libraries, and software conditions belongs to applied mathematics; research that cannot yield specific numerical values under existing practical computing conditions belongs to abstract mathematics."

Take Shor's algorithm in quantum computing as an example: since there are no practically usable quantum computers at present, this algorithm remains at the theoretical level and belongs to the category of abstract mathematics. However, if quantum computing hardware is realized in the future, enabling it to output specific numerical values, Shor's algorithm will be transformed into a part of applied mathematics. Beyond hardware, the development of function libraries and software also affects this boundary: Thirty years ago, due to the lack of robust linear system function libraries and optimization problem solvers, many algorithms that now rely on these tools could only be studied theoretically at that time and could not yield specific results, thus belonging to abstract mathematics. With technological progress, when these algorithms can be actually solved, they are transformed into applied mathematics. Another example is Turing machine theory: when Alan Turing first proposed it, it belonged to abstract mathematics; it was only after computer hardware was invented and could perform specific calculations that it was classified into applied mathematics.

It is evident that using "whether specific numerical values can be calculated under current conditions" as a criterion for distinction can provide clear guidance for the development of applied mathematics. The core of applied mathematics lies in "application," and its value must be reflected through immediate service to industrial technology practice. If specific numerical values cannot be obtained under current conditions, it will be difficult to truly implement applications, and thus it can hardly be called "applied mathematics." This criterion for distinction is expected to provide beneficial support for the development of applied mathematics.

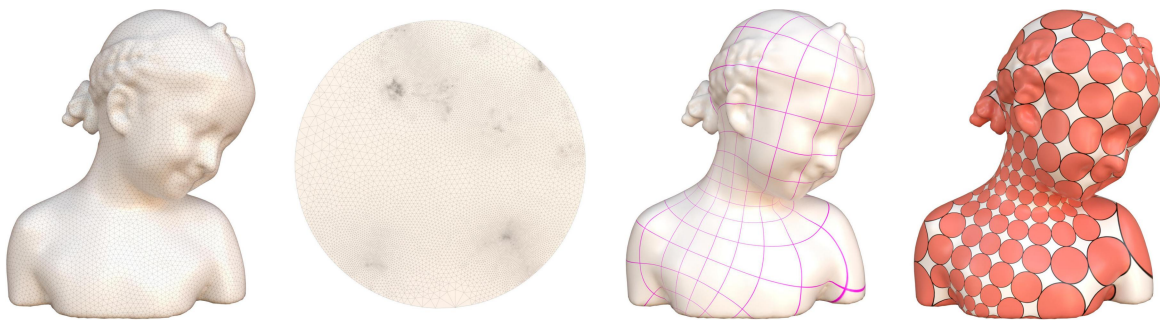
The core reason why this paper discusses the distinction between abstract mathematics and applied mathematics is that the brand-new research field of computational discrete Global Geometric Structures belongs to applied mathematics, not the field of abstract science. However, breakthroughs in this research require close collaboration between different disciplines in the upstream and downstream. Due to differences in research objectives and value orientations, various disciplines are prone to cognitive confusion. Such confusion will increase collaboration costs and may even lead to research failure. Therefore, establishing a clear and feasible criterion for distinction will provide crucial support for the development of research on computational discrete Global Geometric Structures.

The clear distinction between "algorithms" and "computation" can not only help mathematicians distinguish between the familiar "computation" and the computer scientists' tool of "algorithms" but also, conversely, help mathematicians think about corresponding concepts from the perspective of algorithms. Thurston proposed seven different perspectives for understanding functions, but all seven are from the perspective of abstract mathematical tools. This paper suggests that thinking and understanding can also be carried out from the perspective of "algorithms"—especially for "global geometric structures."

In summary: Due to the limited precision of computer computation, all computations require finite

calculations to approximate accurate numerical values. For global geometric structures, it is also necessary to express them as a local equation under a local coordinate system, but their global properties are embedded in the transformations between equations under different coordinate systems. For local geometric structures and mathematical structures, it is sufficient to perform numerical approximation calculations on the local equation. However, for global geometric structures, what needs to be approximated is not the locally expressed equation, but the indirect information embedded in the transformations between equations. In this case, the mindset of local computational approximation cannot be used to solve the problem; instead, a "algorithmic" mindset is required to capture "global information."

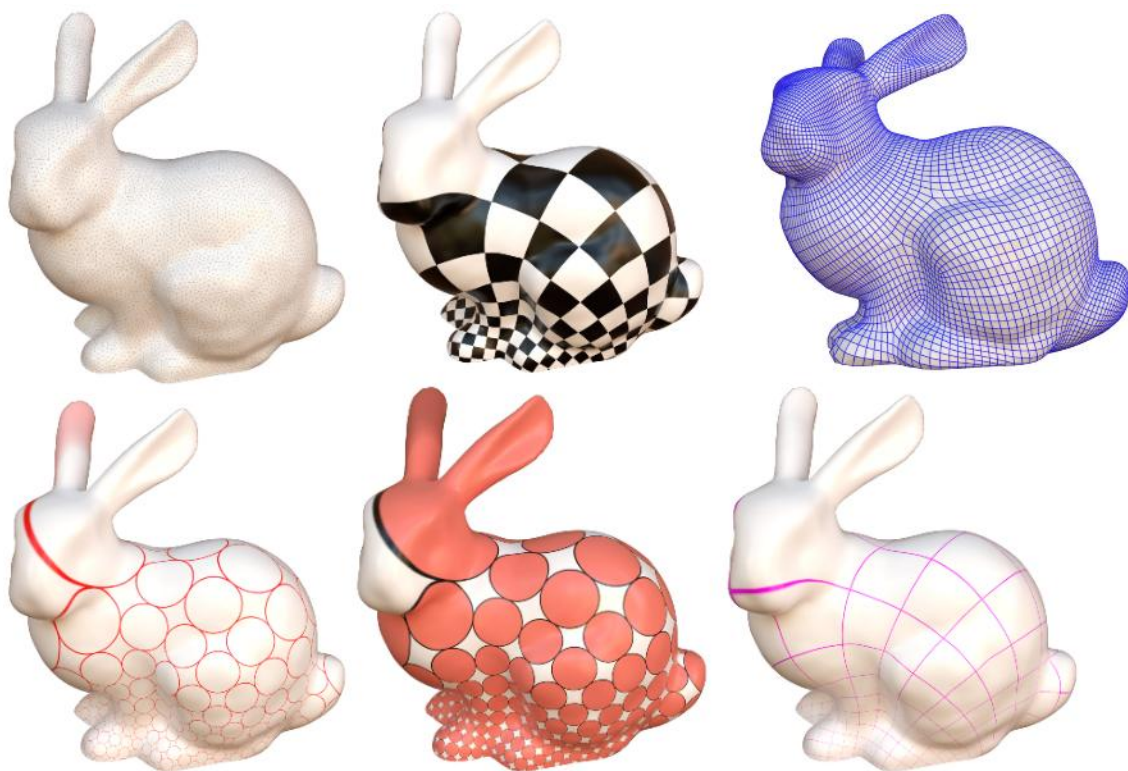
These are all practical conclusions gradually drawn by the author of this paper during the research on various algorithms for global geometric structures, and they also form the foundation for proposing computational discrete Global Geometric Structures. This process is similar to deriving new insights from physical experiments, rather than obtaining them through logical deduction relying on abstract tools. The development of science and technology relies on various tools: previously, "abstract tools" drove tremendous leaps in mathematics; with the development of computer science, "algorithms" — as a new tool — will also bring a brand-new perspective. Moreover, compared to physical experiments (which require substantial external conditions), "algorithms" only require a computer — needing barely more resources than "abstract" tools, while being capable of handling the required external conditions.



(Figure 34: Parameterization by Discrete Calabi flow, generated by Geometric.)

12 Visualization of Global Geometric Structures

In the research on computational discrete Global Geometric Structures, this paper proposes that visualizing manifolds, fiber bundles, and global geometric structures through computer graphics rendering technology is a "necessary condition" and "essential tool" for the research—not a dispensable accessory that merely adds extra value to research results. For example, as shown in the figure below, various rendering presentations of a rabbit-shaped surface allow for intuitive visual observation of the corresponding geometric structures. Compared with the "paper, pencil, and blackboard" commonly used in abstract mathematics, "visualization" is an additional tool proposed in this paper, which can deepen the understanding and popularization of relevant mathematical concepts and theories.



(Figure 35: Several renderings of a bunny mesh, generated by Geometric.)

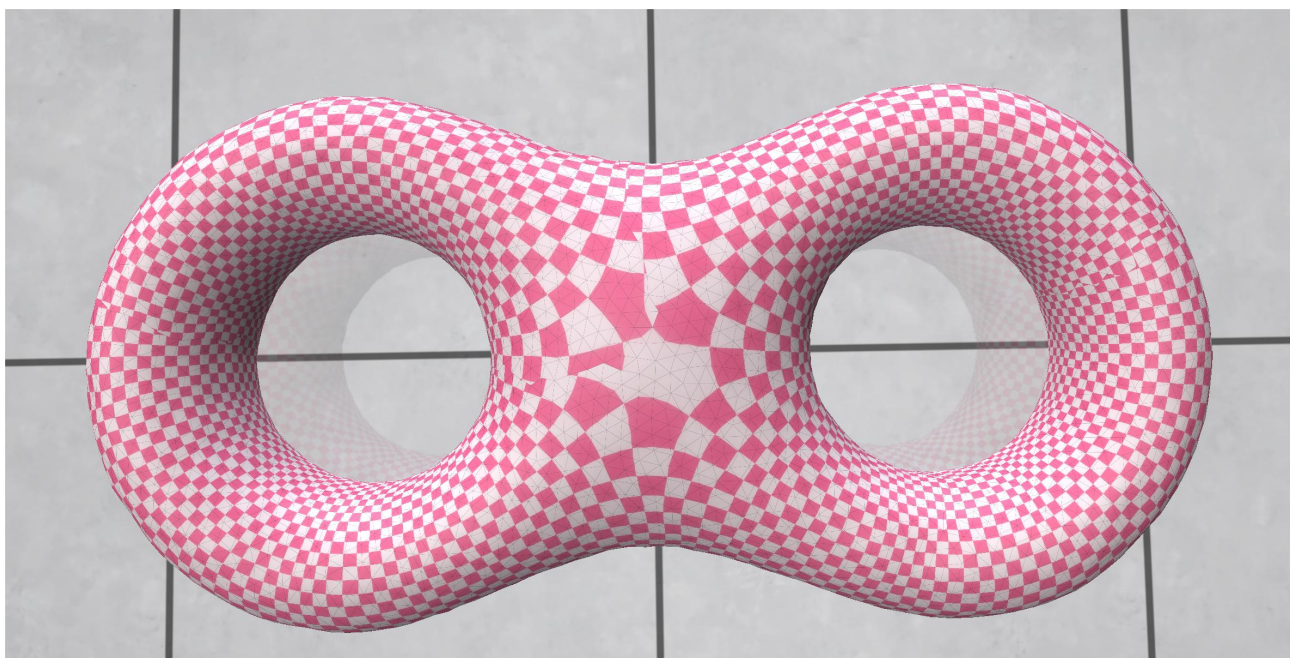
Research on algorithms and theories requires a variety of tools—such as mathematical tools from differential geometric topology, programming tools like function libraries and solvers, and hardware tools including computers. These are all universally recognized as essential research tools.

Manifolds (especially 2D surfaces) studied in the field of computational discrete Global Geometric Structures can often be embedded in Euclidean space, possessing an inherent attribute of 3D visual representation. This paper explicitly proposes that visualization of 3D models and their various elements using computer graphics rendering technology should be regarded as an essential tool for research. "Visualization" is also part of the research itself, rather than an ornamental byproduct added after the successful completion of algorithm and theoretical research. For instance, the various rendering techniques applied to the rabbit-shaped surface shown in the figure are themselves important and critical research content in the field of "computational discrete Global Geometric Structures."

The exploration of algorithms for any given global geometric structure does not happen overnight; each type of structure requires entirely new ideas and tools for algorithm design. Therefore, before a mature algorithm is developed, approximate "visualization" technology can be used to generate visual representations. Although such displays may be globally inconsistent and fail to satisfy the global properties of the structure, presenting results calculated based on local formulas and equations can still roughly reflect the characteristics and patterns of the global geometric structure at the visual level — thereby providing inspiration for research. In other words, local formulas of global geometric structures can be utilized: while they cannot capture global properties, they can produce partial visual effects, which in turn motivate the gradual development of a complete "algorithm."

The representation of holomorphic quadratic differentials shown in the figure is a typical example of such a "visualization" result, rather than an accurate expression that strictly satisfies the global properties of holomorphic quadratic differentials. The reason is as follows: although an algorithm exists for the horizontal harmonic foliations (a component of holomorphic quadratic differentials) to ensure their global properties, the component obtained by rotating these foliations by 90 degrees remains an approximate result. When combined, the two components only form a visual approximation, not an approximate solution that captures the overall mathematics of holomorphic quadratic differentials in the mathematical sense.

Even so, research on the "visualization" of holomorphic quadratic differentials still plays an important supporting role in understanding their abstract geometric concepts. Therefore, "visualization algorithm design" should also be treated as a key research content in this field.



(Figure 36: A holomorphic quadratic differential, generated by Geometric.)

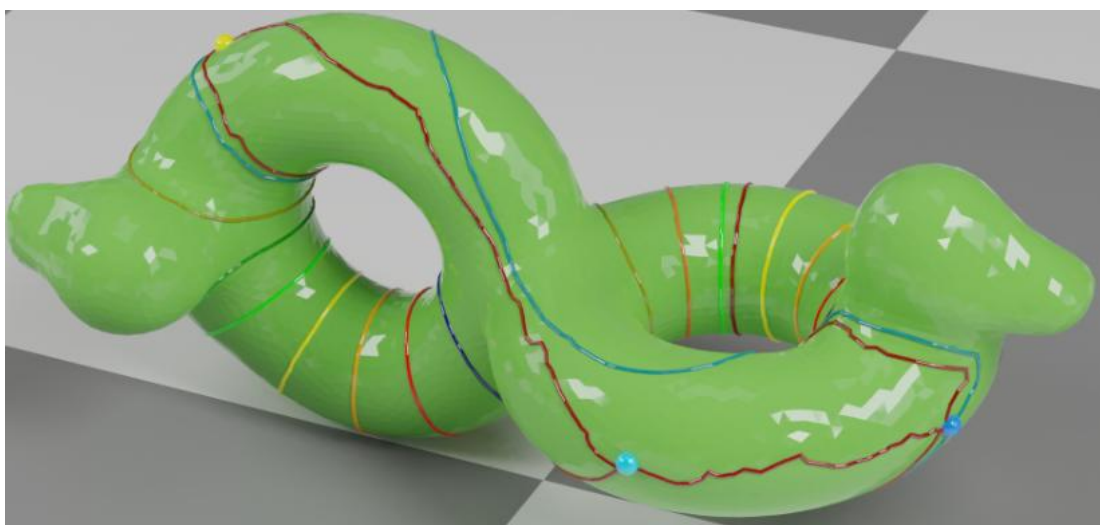
The "visualization" tool for computational discrete Global Geometric Structures proposed in this paper requires in-depth understanding of various rendering technology theories and algorithms in computer graphics, as well as proficient mastery of coding — similar to how mathematical research requires

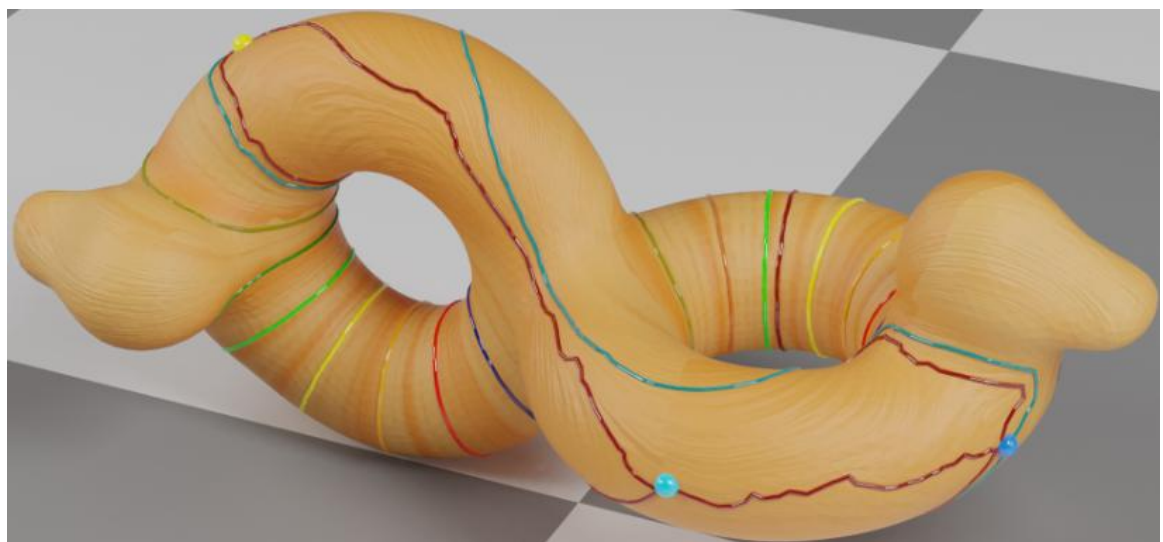
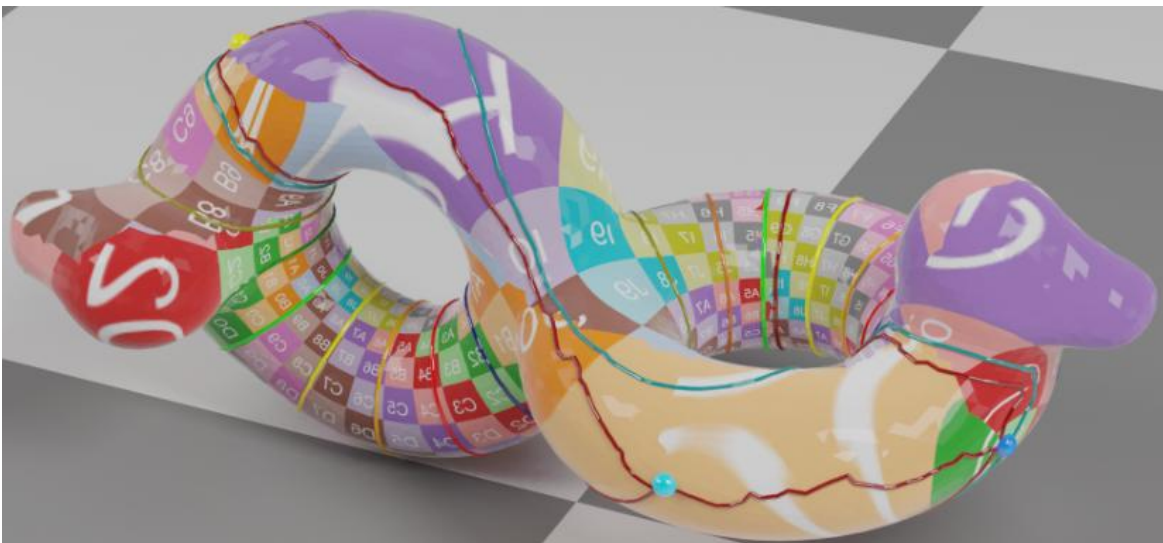
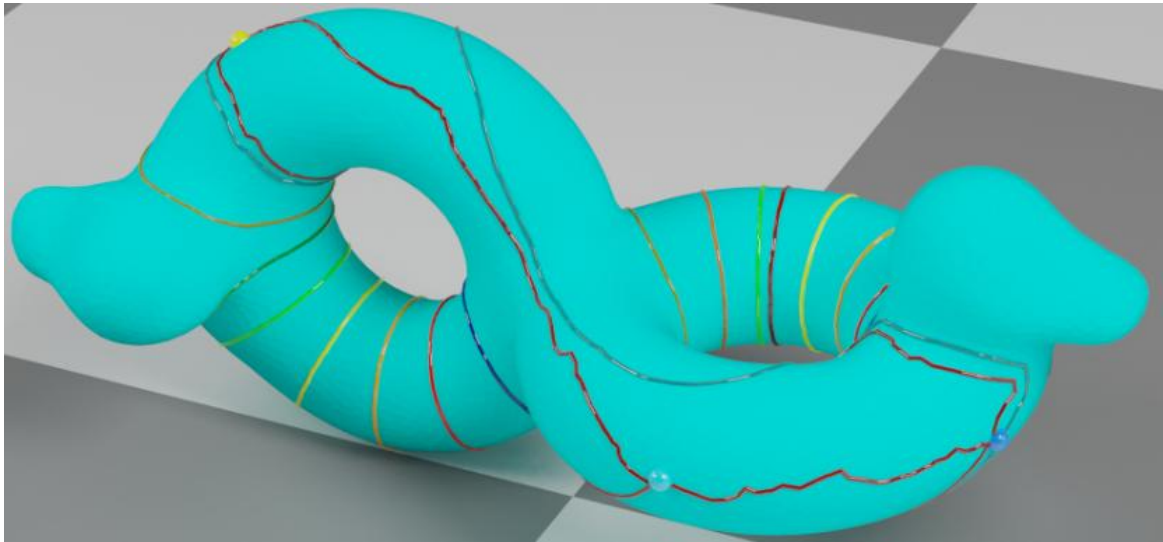
proficiency in writing on a blackboard.

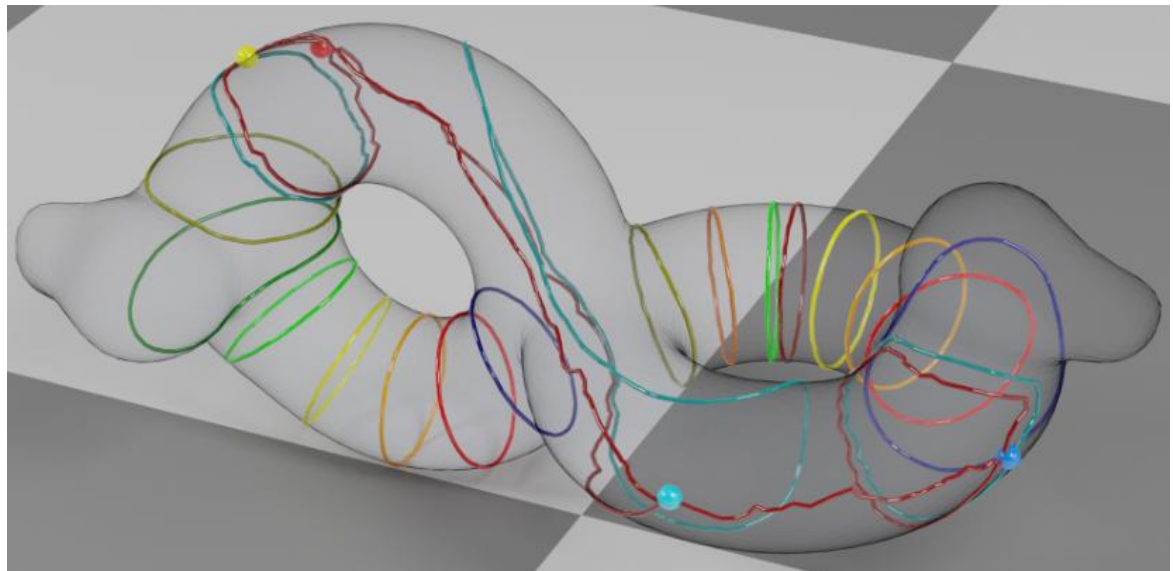
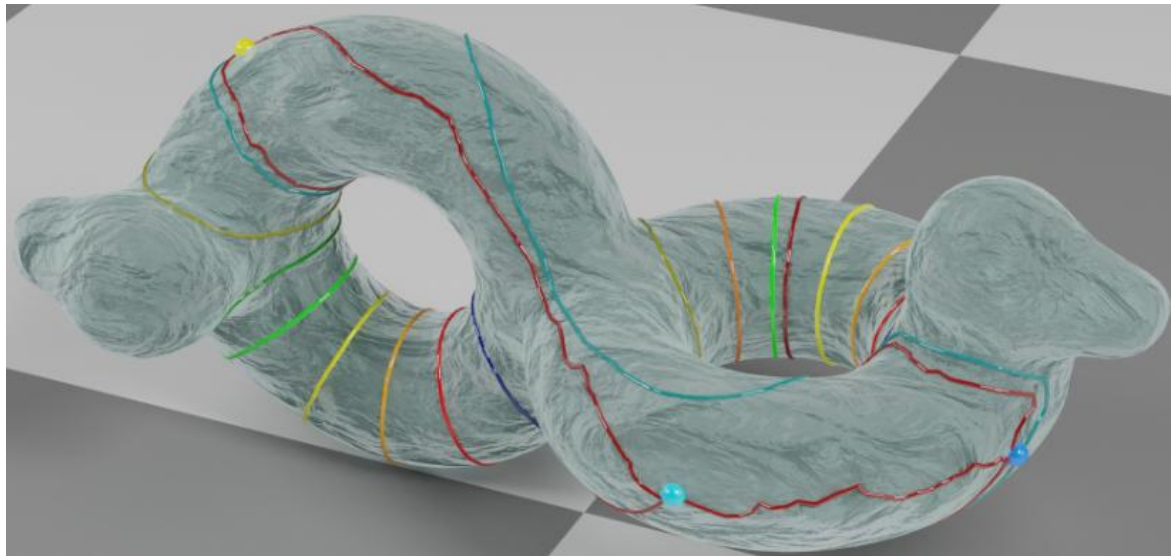
The "visualization" tool holds significance in the following aspects:

- (1) Presenting the results of algorithms for global geometric structures;
- (2) Serving as a visual tool to display various elements of geometric structures during algorithm research;
- (3) Assisting in code writing and debugging through the presented visual tools;
- (4) Obtaining visual displays of pseudo-global geometric structures via visualization using local formulas;
- (5) Aiding in the understanding and popularization of relevant global geometric structure concepts through visual means.

In the research on computational discrete Global Geometric Structures, the author of this paper has conducted a large number of visualization experiments, which have further facilitated the experimentation of many algorithms. For example, thousands of different renderings have been generated for the same result of a harmonic foliation, with some of these illustrations shown below. It can be seen from this that the research direction of computational discrete Global Geometric Structures relies heavily on rendering and visualization—this also represents an innovation in the research philosophy and research tools of this paper.







13 Existing algorithms for global geometric structures

Over the past two to three decades, through collaboration between computer scientists and differential geometric topologists, algorithms have been successfully designed for many global geometric structures. Although the scholars at that time did not aim to conduct research on global geometric structures, nor did they view the algorithms from the perspective of global geometric structures, these algorithms can actually be categorized under the broader framework of the research direction of computational discrete Global Geometric Structures. Furthermore, this paper argues that from the perspective of this broader framework — with a clear focus on studying "global geometric structures" — the implementation of algorithms for more global geometric structures will undoubtedly be advanced at a faster pace. An innovative research direction does not emerge out of thin air; it is only based on the research practices of numerous previous scholars that a relatively comprehensive new framework of ideas can be gradually refined.

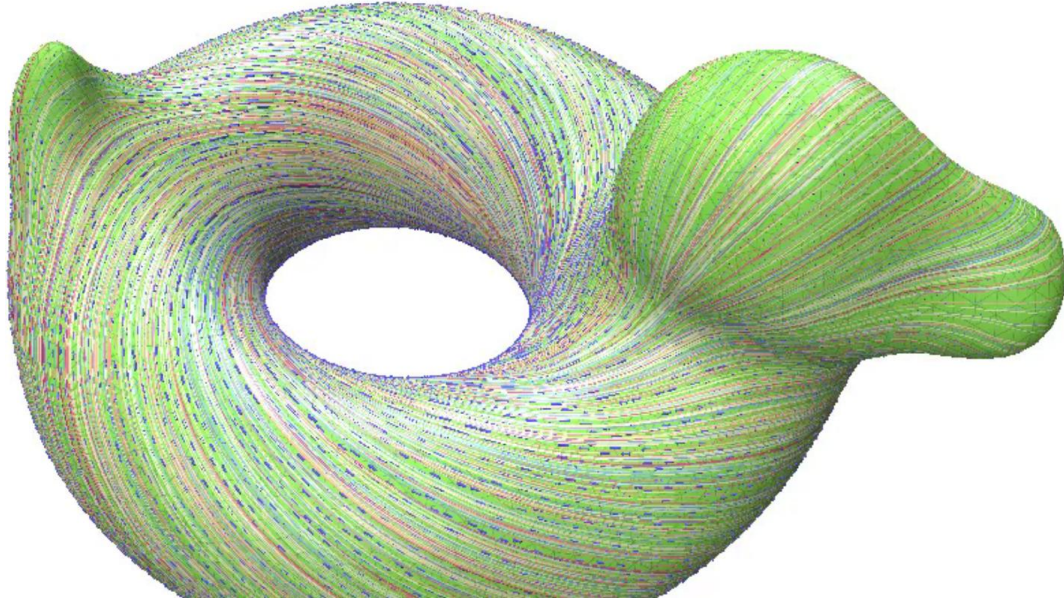
This paper provides an overview of several global geometric structures for which algorithms have already been implemented; for detailed specifics, please refer to the corresponding literature on each algorithm.

(1) Differential 1-forms

In the book *The Geometry of Physics: An Introduction* (1997, First Edition), Theodore Frankel proposed a theoretical method for discrete differential 1-forms. In the article "Global Conformal Surface Parameterization" published in 2003, Xianfeng Gu and Shing-Tung Yau designed an algorithm for computing harmonic one-forms and successfully calculated differential 1-forms.

Rotating a differential 1-form by 90 degrees yields a perpendicular dual differential 1-form; the combination of these two forms results in a holomorphic 1-form. In the same article, Xianfeng Gu and Shing-Tung Yau also designed an algorithm to compute the visualization of the corresponding holomorphic 1-form. The reason for this is that the zero positions of the dual differential 1-form differ from those of the original differential 1-form; thus, this algorithm does not capture the global properties of the holomorphic 1-form. However, the visualization results are still important for the research of holomorphic 1-forms, as they help to intuitively understand this mathematical concept from a visual perspective.

Differential 1-forms on discrete meshes play a crucial role and have since been widely applied in many mesh processing algorithms. This includes extending the computational methods and application scope of traditional finite elements in the field of finite element exterior calculus.



(Figure 37: Harmonic differential 1-form, generated by software Geometric.)

(2) Index Counting

In the article "Discrete One-Forms on Meshes and Applications to 3D Mesh Parameterization," Steven J. Gortler, Craig Gotsman, and Dylan Thurston studied discretized differential 1-forms on meshes. They extended discrete harmonic differential 1-forms and obtained a global geometric structure called "index counting," which has no corresponding concept on smooth surfaces.

By virtue of this brand-new global geometric structure of index counting, the article also presented a novel proof for harmonic mappings of disk-shaped meshes. This proof can be further extended to non-disk-shaped meshes by leveraging the global properties of the index-counting global geometric structure, thereby solving problems that could not be addressed using other mathematical tools previously. Discrete 1-forms have also found application in polycube generation, as shown in the figure.

This example also demonstrates that adopting different solutions to the same problem can lead to new mathematical tools, and these new tools may in turn solve other problems. Particularly for algorithms related to global geometric structures, the research on each type of global geometric structure algorithm can yield numerous new computational tools and methods. This not only facilitates the solution of the current problem but also holds potential contributions to other problems—and this is the reason why this paper proposes studying algorithms for "global geometric structures."

This case shows that the research on algorithms for discrete global geometric structures does not involve simply copying or establishing a one-to-one correspondence with global geometric structures on smooth surfaces. Instead, it takes global geometric structures on smooth surfaces as a foundation, and in the process of research, entirely new discrete global geometric structures can be discovered.

When Steven J. Gortler and his collaborators conducted this research, they did not have the concept of studying "global geometric structures" in mind. A large portion of their article focused on applications such as harmonic mappings for parameterization.

This paper proposes adopting the concept and framework of "global geometric structures" to examine

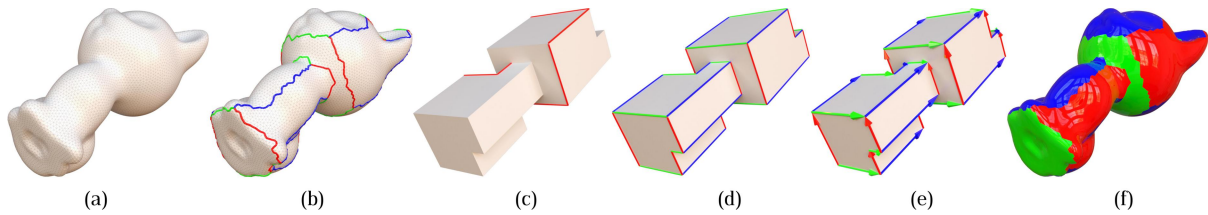
all these algorithms and research efforts with commonalities. This approach can clarify the research direction, unify many research contents that seem unrelated and independent on the surface, and thus enable the systematic advancement of more research of this type.

The new global geometric structure of index counting has also been applied in many subsequent key mesh applications. In the research on discrete foliations, the index counting for differential 1-forms was extended to support index counting for foliations, thereby laying the foundation for algorithms of discrete foliations.

If a mesh has a genus of 1, since there are no zeros, the differential 1-form and its dual differential 1-form can form a discrete holomorphic 1-form; in other words, the genus-1 mesh can be flattened. However, for meshes with higher genus, current algorithms fail to achieve successful flattening due to inconsistent zero positions.

According to the research direction of "super-structured quadrilateral meshes" proposed by the author of this paper, it is expected that through a brand-new approach, the zero positions of the dual differential 1-form and the given differential 1-form on high-genus meshes can be aligned. This would allow the algorithm to capture the global properties of holomorphic 1-forms, and this work is still in progress.

The proposal of the research direction of "computational discrete Global Geometric Structures" draws part of its inspiration from the detailed analysis of the achievements and limitations of the aforementioned related algorithm research efforts, as well as the reasons behind these limitations.

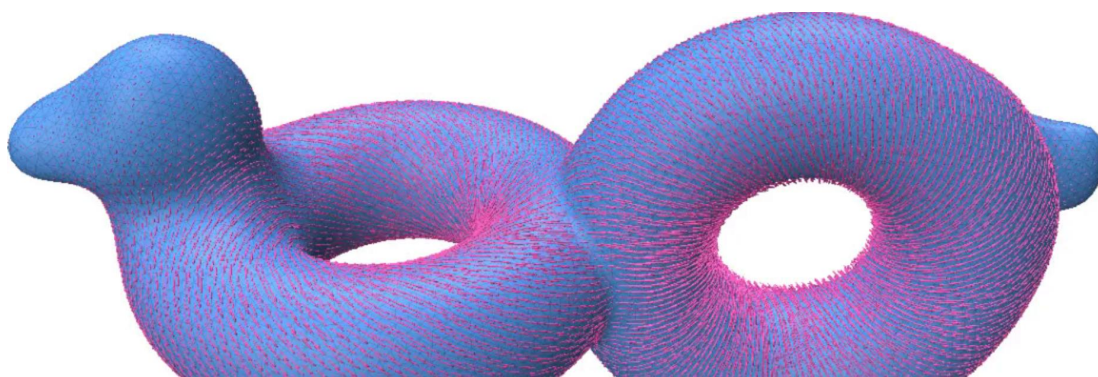


(Figure 38: one form for polycube generation.)

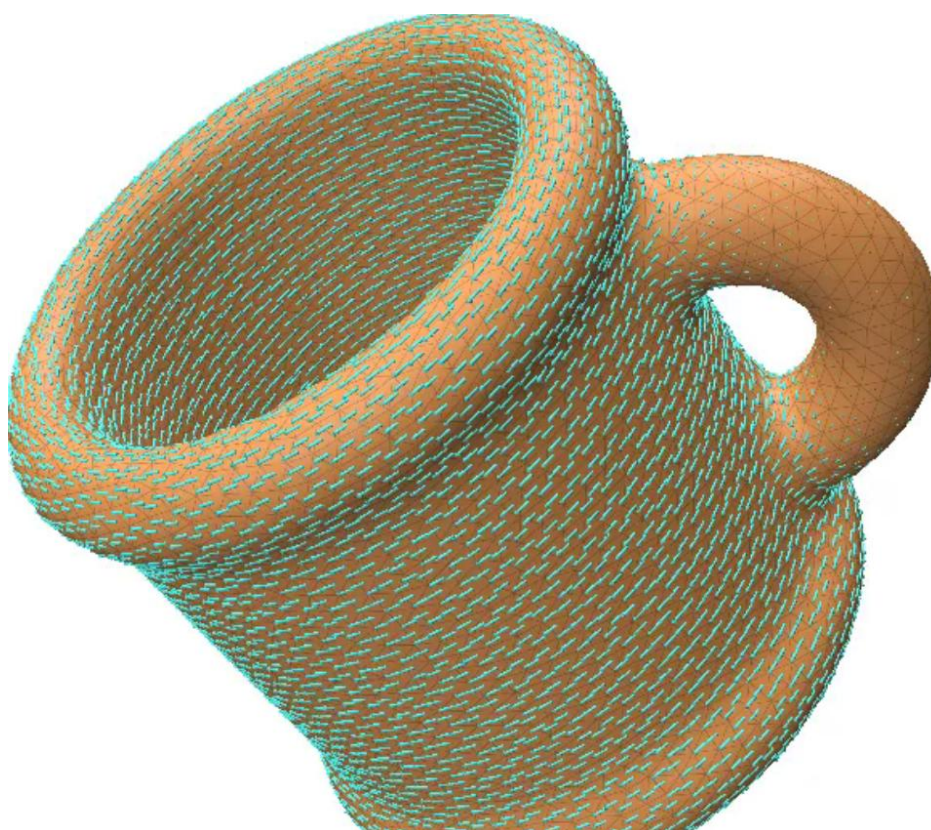
(3) Vector Filed

In the article "Trivial Connections on Discrete Surfaces" published in 2010, Keenan Crane, Mathieu Desbrun, and Peter Schröder successfully discretized the differential geometric concept of "connection." Furthermore, relying on this discretized connection, they designed an algorithm to compute vector fields on meshes.

A discrete vector field can be regarded as the dual of a differential 1-form. However, harmonic differential 1-forms are defined with respect to the metric on the surface, whereas vector fields utilize connections and thus do not depend on the metric. By extending vector fields—for instance, equipping each point with four mutually perpendicular vectors—a frame field is obtained.



(Figure 39: Vector field, generated by software Geometric.)



(Figure 40: Vector field, generated by software Geometric.)

(4) Spin Structure

In the article "Shape from Metric," Albert Chern, Felix Knöppel, Ulrich Pinkall, and Peter Schröder designed an algorithm to realize the algorithmic design of spin structures on 2D surface meshes. For a mesh with genus g , there are (2^g) discrete spin structure equivalence classes.

For example, on a torus, there are 2 spin structure equivalence classes — and the article also successfully implemented these discrete spin structure equivalence classes. The design of the spin structure algorithm relies on the proposal of discrete connections in vector field algorithms. This shows that many discrete differential geometric concepts are intertwined with one another, gradually laying the foundation for the algorithmic design of more subsequent differential geometric topology concepts in a relay-like manner. Algorithms such as discrete Ricci flow, Yamabe flow, and Calabi flow can generate numerous

metrics for a mesh that do not require embedding in Euclidean space.

If a mesh only has metric information (i.e., only edge length information) and no coordinate information in 3D Euclidean space, how can the embedding or immersion of this mesh be obtained? This problem was successfully solved through spin structures on 2D surface meshes.

Motivation

There are also other global geometric structures that have been successfully made computable by other computer scientists in the field of computer graphics research—for instance, conformal structures obtained through edge swapping.

There is no unified template to copy for the discretization algorithm of each type of global geometric structure; however, every such algorithm requires a motivation for research. For example, the motivations behind the algorithms for the aforementioned global geometric structures all stem from applications in computer graphics: harmonic differential 1-forms and index counting originated from the application of mesh parameterization and texturing; vector fields and frame fields originated from the generation of quadrilateral meshes from triangular meshes; and the algorithm design for spin structures originated from the application of embedding and immersing metric-equipped meshes in 3D space.

The concepts and theories behind each type of global geometric structure are extremely extensive—each global geometric structure constitutes a mathematical system in its own right. It is difficult to persist in learning it purely from a mathematical perspective.

Therefore, to research the algorithm design of a global geometric structure, one must first identify a research motivation. Beyond applications in computer graphics, current areas such as topological materials in physics also offer significant research motivations; however, research in this regard still awaits future collaboration between computer scientists, differential geometers, and physicists to explore.

Currently, algorithmic research on global geometric structures focuses on meshes corresponding to 2D surfaces—this is because meshes are one of the primary research objects in computer graphics. Research on discretization algorithms in 3D and other dimensions requires identifying corresponding applications to continuously provide research momentum. This is because such research is not abstract mathematical research; it involves engineering work such as code implementation. Without a sustained driving force, it is difficult to carry out engineering work, and thus research on pure algorithms without code implementation cannot be sustained.

The proposal of "computational discrete Global Geometric Structures" is based on the analysis of these algorithms starting from the code level, rather than beginning with a mathematical concept of "global geometric structures." The process of establishing this field is reversed: starting from programs to code, then to algorithms, and finally to geometric theories. Gradually, a specific framework of thought—arguing that algorithmic research on "global geometric structures" should be carried out systematically—was refined.

It is worth noting that many computer scientists in this field have educational backgrounds and experience in mathematics such as differential geometric topology, or they collaborate with differential geometers. They need to be well-versed in both mesh-based algorithms and code in computer graphics, as well as the theories developed in modern differential geometry.

14 Relevant Academic and Other Activities

To advance the research on computational discrete Global Geometric Structures, a great deal of relevant activities have been carried out, and the significance of these activities is well-aligned with the insights of Fields Medal laureate William P. Thurston.

In his paper *On Proof and Progress in Mathematics*, William P. Thurston pointed out:

(1) The development of mathematics cannot proceed without effective communication. The visualization proposed in this paper as an essential tool is one such example.

(2) The mathematical community places excessive emphasis on publishing cutting-edge results, while neglecting the dissemination of fundamental cognitive frameworks. The various relevant exchange activities introduced in this paper serve as an illustration of efforts to address this.

(3) Mathematicians should focus more on "how to convey mathematical ideas to beginners with no prior background" rather than merely presenting technical proofs to peers. The national touring art exhibition on computational discrete Global Geometric Structures is a case in point.

(4) Mathematicians need to enable non-specialists to understand their core ideas. The numerous rendered images of foliations included in this paper exemplify this.

(5) Mathematical language must not only describe problems with precision but also adapt to the cognitive needs of learners from different backgrounds. For instance, understanding concepts by drawing comparisons from the perspective of "algorithms" (as proposed in this paper) meets this requirement.

(6) More efforts should be invested in popularizing mathematical thinking, rather than solely pursuing the latest theoretical breakthroughs. The use of a WeChat subscription account to popularize relevant knowledge (as adopted in this research) is an example of this approach.

(7) The acceptance of many mathematical theorems does not rely solely on written proofs; instead, it is gradually established through verbal communication, citations, and collective recognition within the academic community.

(8) Recognized theorems may lack complete proofs or even contain errors, but the mathematical community maintains their reliability through continuous discussion and revision.

(9) Mathematical understanding does not exist solely in the text of papers; it is more deeply embedded in the researchers' cognitive networks and social interaction processes.

(10) Thurston's views challenge the traditional notion that "mathematical proofs are absolute truth."

(11) Mathematical knowledge is a dynamic, social cognitive construct, rather than a static product of logic.

(12) The true progress of mathematics is reflected not only in the accumulation of the number of theorems but also in the improvement of humanity's depth of understanding of mathematics and the efficiency of its dissemination.

(13) The core mission of mathematicians is not only to prove theorems but also to advance the development of human thinking.

(14) Mathematics is not merely a formal system but a complex combination of human cognitive activities and social practice. Its intrinsic value needs to be re-examined from dimensions such as the

diversity of thinking, the effectiveness of communication, and the socialization of knowledge.

Thurston's paper illustrates that the work of mathematicians extends far beyond the scope of formal logic; they are shapers and pioneers of human thinking. The "computational discrete Global Geometric Structures" we propose supplements this perspective from the angle of algorithms.

The various forms of work we have conducted to advance computational discrete Global Geometric Structures are as follows.

14.1 Organizing Academic Seminars

In the spring of 2022, at the Yanqi Lake Institute of Applied Mathematics in Beijing, the author of this paper, together with Professor Shing-Tung Yau and Professor Zhengwei Liu, co-organized a seminar titled "[Tsinghua University - Yanqi Lake Institute of Applied Mathematics Seminar on computational discrete Global Geometric Structures, Topological Materials, and Quantum Computers](#)". By inviting scholars in the fields of physical materials, computer algorithms, and differential geometric topology to conduct relevant discussions and research exchanges, this seminar aims to promote the understanding of global geometric structures with algorithms as the core, as well as their applications in physics.

BIMSA-Tsinghua可计算离散整体几何结构、拓扑材料和量子计算机讨论班

This seminar talks about the algorithms which compute global geometry structures, and furthermore discusses their potential applications in topological materials and quantum computer. In differential geometry, local geometric properties work together to produce global structures, such as harmonic one-form, foliation, holomorphic quadratic differentials, conformal structures, and so on. The local properties, global geometry structures, and their relationship have been fully studied in mathematical community in last century. However, their applications are still not fully explored. Around 2000, with the seminar work of Gu-Yau algorithm, computer graphics researchers began to design algorithms to compute global geometry structures on surfaces to solve many difficult mesh modeling problems. The experience and achieved advances in computation of global geometry structures could bridge pure math and their applications in topological materials and quantum computer. In this seminar, we initiate this ambitious goal of exploring research possibility among computational global geometry structures, topological materials and quantum computer.

组织者

刘正伟, 丘成桐, 赵辉

日期

2022年03月01日至05月04日

历史讲座

A generalization of Tutte's harmonic parameterization

演讲者: Edward Chien

时间: 2022年04月29日 09:30 至 10:30

线上: Zoom 388 528 9728 密码: BIMSA

构筑拓扑量子物态世界——从量子反常霍尔系统到拓扑超导体

演讲者: 何珂

时间: 2022年04月22日 14:00 至 15:00

地点: JCY-1 线上: Zoom 427 154 2002 密码: BIMSA

Discrete Conformal Geometry of Surfaces

演讲者: Feng Luo

时间: 2022年04月15日 09:00 至 10:00

线上: Zoom 388 528 9728 密码: BIMSA

Rigidity and deformation of discrete conformal structures on polyhedral surfaces

演讲者: 徐旭

时间: 2022年04月08日 14:00 至 15:30

线上: Tencent 494 085 7642 密码: 2022

Mesh Generation Using Computational Conformal Geometry

演讲者: 顾险峰

时间: 2022年04月01日 09:30 至 10:30

线上: Zoom 388 528 9728 密码: BIMSA

拓扑磁结构及其电操控

演讲者: 杜海峰

时间: 2022年03月25日 14:00 至 15:15

地点: Tsinghua-Ningzhai-W11 线上: Tencent 482 3868 6453 密码: 2022

Quantum transport phenomena in topological materials

演讲者: 王亚愚

时间: 2022年03月18日 14:00 至 15:30

地点: JCY-3

The "spectral topology" and skin effect in non-Hermitian band theory

演讲者: 方辰

时间: 2022年03月10日 14:00 至 15:15

地点: JCY-1

Iron-based superconductors as a new Majorana playground

演讲者: 丁洪

时间: 2022年03月01日 14:00 至 16:00

地点: 中会议室一 线上: Tencent 336 497 186 密码: 2022

14.2 Offering Courses

The author of this paper, in collaboration with the Yanqi Lake Institute of Applied Mathematics, offered the course "[computational discrete Global Geometric Structures](#)," which covers the fundamental content of relevant algorithms.

This was the first time the course was offered. It integrates computer graphics, mesh processing, differential geometry, visualization rendering, and other disciplines, and is based on the author's original meshDGP code framework.

The course attracted the participation of teachers and students from many related majors across the country. Notably, the availability of code for comparative learning allows participants to access the course without entry barriers; in addition, the visual graphics enable them to gain a preliminary intuitive understanding even without grasping the mathematical concepts.

可计算离散整体几何结构

This course introduces the algorithms and philosophy of Computational Discrete Global Geometric Structures. There are six components in this course: 1) the mathematical theory of global geometric structures on smooth manifolds; 2) their discrete counterparts; 3) the algorithms to compute them; 4) the programming and coding techniques for the implementations of the algorithms; 5) the visualization techniques to demonstrate these geometry structures; 6) besides computer graphics, the possible discussion of their potential applications in other interdisciplinary fields, and so on. Computational discrete global geometric structures (CDGGS) is pioneered by Prof. Shing-Tung Yau and Prof. Xianfeng Gu with the classic Gu-Yau algorithm to compute discrete harmonic one-form around 2000. After twenty years' research by mathematicians and computer scientists, currently, CDGGS can handle many other discrete global geometric structures, such as holomorphic one-form, foliation, holomorphic quadratic differentials, conformal structures, and so on, and play an important role in the application of geometry modeling and meshing.

讲师

赵辉

日期

2022年09月14日至12月07日

网站

<https://www.bimsa.cn/newsinfo/749654.html>

修课要求

Differential Geometry, Linear Algebra, Topology

听众

Undergraduate, Graduate

视频公开

公开

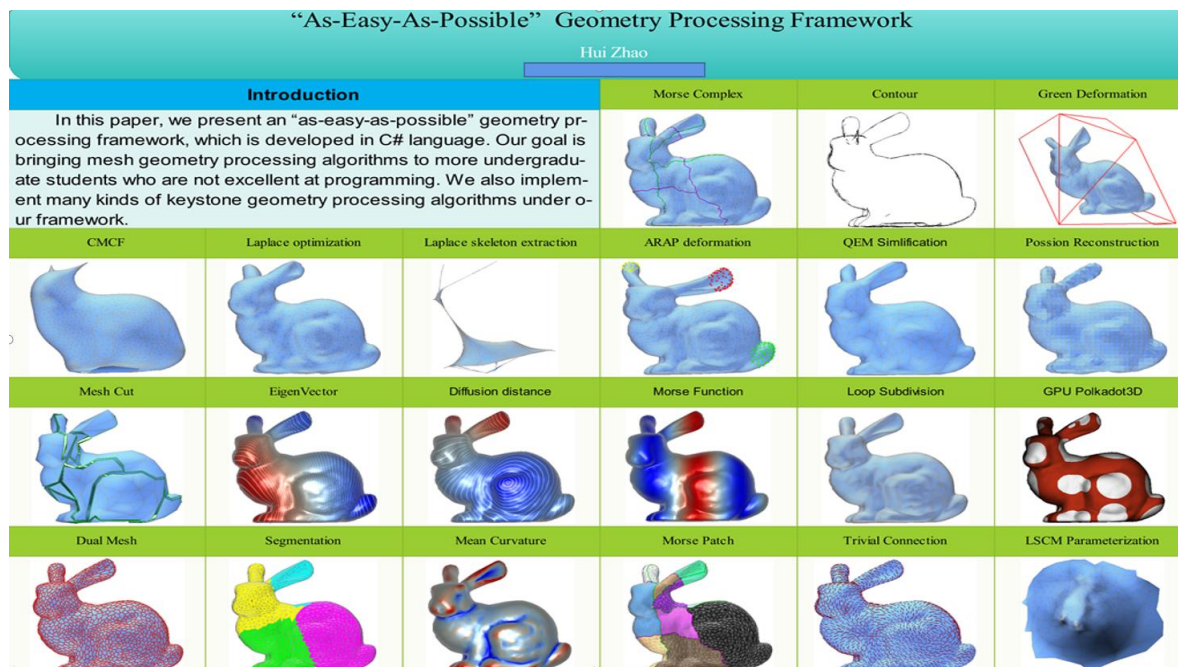
笔记公开

14.3 Developing the Open-Source Mesh Processing Code meshDGP

Research on computational discrete Global Geometric Structures is based on code implementation, rather than abstract algorithms. Since 2007, the author has independently developed the mesh processing software meshDGP. This software, consisting of hundreds of thousands of lines of code developed in the C# language, integrates dozens to hundreds of various mesh processing algorithms, covering functions such as 3D model simplification, refinement, deformation, and parameterization, as shown in the figure below.

meshDGP is oriented toward learning, scientific research, and teaching, rather than industrial applications. The C# language was chosen for development to lower the learning threshold—it not only allows beginners to get started quickly but also enables experts to focus more on algorithms themselves rather than programming techniques. The software also integrates calling interfaces for mainstream C++ model processing libraries (such as Libigl and Tetgen) and tools like MATLAB, and is equipped with a supporting interactive interface and menu system, further enhancing learning convenience. Meanwhile, its advanced software architecture and design patterns make it more suitable for students to learn algorithm principles and expand new functions.

According to rough statistics, approximately 40,000 to 50,000 science and engineering teachers, students, and engineers across the country are currently using this software for learning, scientific research, and teaching activities. Over the past decade, it has been widely applied in teaching, scientific research, and engineering practice across multiple disciplinary fields, including computer graphics, mechanical analysis, mechanical engineering, differential geometry, virtual reality, biomaterials, CAD/CAE, finite element analysis, and industrial software.



(Figure 41: The demonstrations of many modling algorithms, generated by meshDGP.)

According to statistics from real-name registrations, tens of thousands of teachers, students, and engineers from interdisciplinary fields and various industries are currently learning and using the meshDGP code. Their research directions include: Mechanical Engineering, Architectural Design,

Numerical Analysis, Aeronautical and Astronautical Science and Technology, Applied Mathematics, GIS and BIM, Structural Engineering, Civil Engineering, Design and Media Engineering, Architecture and Civil Engineering, Architecture, Computational Mathematics, Hydraulic and Hydropower Engineering, Intelligent Construction, Solar Physics, Computer Vision, Computer Science and Technology, Mesh Generation and Application, Electromagnetic Computing, Pure Mathematics, Numerical Solutions of Partial Differential Equations, Differential Geometry, Applied Mathematics, Measurement and Control, Conformal Geometry, Graphics, Control Science and Engineering, Engineering Mechanics, Partial Differential Equations, Mesh Processing, Isogeometric Analysis Engineering Application, Stochastic Analysis, Optical Engineering, High-Performance Computing, Radio Physics, Intelligent Computing, Software Engineering, Automatic Control, Computational Fluid Dynamics, Finite Elements, Signal and Information Processing, Geological Modeling, Physics, Aeronautical and Astronautical Engineering, Software Engineering, Mechanics, Biology, High-Performance Computing for Multiphase Flow-Structure Interaction, Circuits and Systems, Medical Image Computing, Artificial Intelligence, Applied Statistics, Control Theory and Control Engineering, Structural Design, Vehicle Engineering, Applied Physics, Innovative Mechanical Structural Design, Biomechanics, Medical Image Analysis, Quantitative Research on Architectural Form, Engineering Thermophysics, Nuclear Science and Engineering, Wind Turbine Aerodynamics, Mechanical Design and Theory, Materials Science, Geometry Processing, Generalized Finite Elements, Aircraft Design, Quasicrystals, Fractal Geometry, Graph Embedding, Structural Topology Optimization, Communications, Electronic Science and Technology, Fluid Mechanics, Electromagnetic Fields, Computer Security, and Department of Aerospace Engineering and Mechanics.

Related to the open-source meshDGP code, there are five textbooks authored by Professor Zhao Hui. Through the cross-reference of these five textbooks and the code, numerous teachers, students, and engineers have been able to gain interdisciplinary access to mesh processing-related algorithms in the shortest possible time.

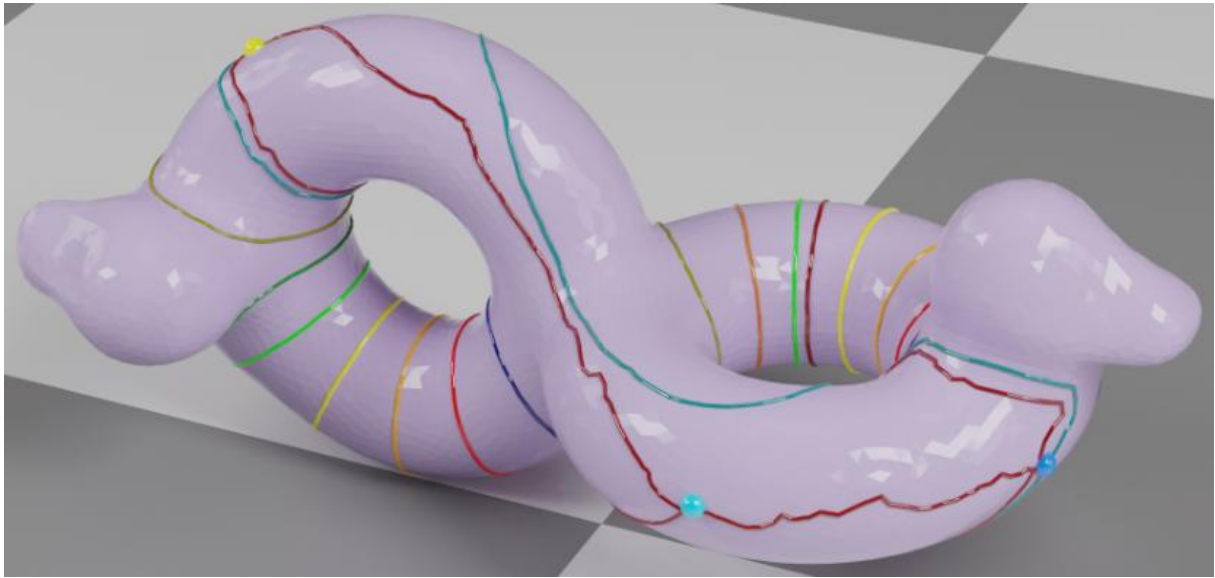
- (1) Computer Graphics: An Introduction to 3D Model Processing Algorithms (C# Version), by Zhao Hui et al., Ocean Press, 2014.
- (2) Computer Graphics: OpenGL 3D Rendering (C# Version), by Zhao Hui et al., Ocean Press, 2016.
- (3) 3D Model Parameterization Algorithms: Theory and Practice (C# Version), by Zhao Hui et al., Electronic Industry Press, 2017.
- (4) 3D Model Deformation Algorithms: Theory and Practice (C# Version), by Zhao Hui et al., Electronic Industry Press, 2017.
- (5) Fundamentals and Examples of GLSL Rendering Programming (C# Version), by Zhao Hui et al., Electronic Industry Press, 2017.



According to real-name registration information, teachers, students, and engineers who use meshDGP come from various universities, colleges, and enterprises. Some of these institutions are listed as follows:

(1) University of Shanghai for Science and Technology, (2) Northwestern Polytechnical University, (3) Yantai University, (4) Xi'an University of Architecture and Technology, (5) Kunming University of Science and Technology, (6) Beijing University of Technology, (7) Tongji University, (8) University of Chinese Academy of Sciences, (9) Xi'an Jiaotong University, (10) Beijing Normal University, (11) Zhejiang University, (12) Tsinghua University, (13) Xiangtan University, (14) Zhongyuan University of Technology, (15) Taiyuan University of Technology, (16) Beihang University, (17) Chongqing University, (18) Hangzhou Normal University, (19) Dalian University of Technology, (20) Sichuan University, (21) Jilin University, (22) Jiangxi University of Science and Technology, (23) Guangdong Pharmaceutical University, (24) Henan Normal University, (25) South-Central Minzu University, (26) National University of Defense Technology, (27) Beijing Union University, (28) Wuhan University of Technology, (29) South China Normal University, (30) Tianjin Polytechnic University, (31) Guangxi University, (32) Tianjin Chengjian University, (33) Changzhou Institute of Technology, (34) Central South University, (35) Zhengzhou University, (36) Nanjing Normal University, (37) Nanjing University, (38) Henan University, (39) China Institute of Water Resources and Hydropower Research, (40) Beijing Normal University, (41) Hunan University, (42) Beijing University of Posts and Telecommunications, (43) Xihua University, (44) University of South China, (45) Tianjin University of Technology and Education, (46) Harbin Institute of Technology, (47) Silhu University, (48) Kashgar University, (49) Politecnico di Milano, (50) Iwate University, (51) Université catholique de Louvain, (52) Stony Brook University, (53) University of Liverpool, UK, (54) The University of Hong Kong, (55) City University of Hong Kong, (56) Technion – Israel Institute of Technology, (57) École Polytechnique, (58) China Academy of Engineering Physics, (59) National Supercomputing Center in Zhengzhou, (60) Hong Kong Cardiovascular Health Research Center, (61) Hefei Institutes of Physical Science, Chinese Academy of Sciences, (62) China Academy of Building Research, (63) Hong Kong Cardiovascular and Cerebrovascular Disease Research Center, (64) Jizhi Research Institute, (65) Sichuan Tianshang Youjia Network Technology Co., Ltd., (66) Tengzhou Urban Construction Development Group, (67) Shenzhen Airport, (68) Liqing Automobile Technology (Shanghai)

Co., Ltd., (69) Guangzhou Da' an Gene Co., Ltd., (70) Hubei Jiutongfang Microelectronics Co., Ltd., (71) Shenzhen Conformal Technology Co., Ltd., (72) Huawei Technologies Co., Ltd., (73) Mihoyo Co., Ltd., (74) Aibo Lingdong (Beijing) Technology Co., Ltd., (75) Beijing Star Barrier Technology Co., Ltd., (76) Tencent Technology Co., Ltd., (77) Jiangsu Jizhi Graphene Research Institute Co., Ltd., (78) Yunnan Ciwei Biotechnology Co., Ltd., (79) Zhengzhou Aviation (Note: Full name not specified in original text), (80) Shanghai Feijiegou Digital Technology Co., Ltd., (81) Hubei Jiutongfang Electronics Co., Ltd., (82) Beijing Kela Ming Technology Co., Ltd., (83) Shanghai Huanshen Technology Co., Ltd., (84) Zhejiang Zhihe Technology Co., Ltd., (85) PIX MOVING, (86) Zhongwang Software Co., Ltd., (87) Glodon Technology Co., Ltd., (88) MediaTek Inc., (89) Lenovo Group, (90) Guangxian Cloud (Hangzhou) Technology Co., Ltd., (91) China National Offshore Oil Corporation, (92) ByteDance, etc.



(Figure 42: A harmonic foliation, generated by meshDGP.)

14.4 Differential Geometry Teaching Based on meshDGP

The theoretical foundation of computational discrete Global Geometric Structures is differential geometric topology, with "algorithms" as its tools and applications in disciplines such as mechanics, physics, and materials science. Therefore, it is an interdisciplinary research direction. This necessitates the widespread popularization of differential differential geometric topology theories. Currently, there are still relatively few courses on differential geometric topology offered in China, especially those designed for teachers and students from interdisciplinary majors.

To address this, with the support of leaders from multiple universities, the computational discrete Global Geometric Structures Laboratory, together with teachers from the School of Science of Zhongyuan University of Technology, Henan University, Kunming University of Science and Technology, Beijing University of Posts and Telecommunications, Yantai University, Hohai University, University of Chinese Academy of Sciences and other institutions, initiated the "Innovative Teaching Reform Seminar on Differential Geometry Visualization Teaching Based on meshDGP". The aim is to enhance students' interest in learning differential geometry and help them gain a deeper understanding of the connotation of the discipline. Relying on the meshDGP platform, by presenting concepts and theorems in differential geometry in a visual way, it helps students quickly understand abstract geometric concepts, theorems and formulas.

2023 创新教改



meshDGP实验平台/《微分几何》

2023 第二届《meshDGP+微分几何》创新教改之技术入门和教改入门

会议简介

零基础开始学习meshDGP，技术交流普及

会议内容

1. 赵海川 北京师范大学 博士生 meshDGP技术入门详解一
2. 郭晋源 中山大学 本科生 meshDGP技术入门详解二
3. 冷雁 烟台大学 创新教改成果展示
4. 郝建兵 滕州城建开发集团 学习经验分享
5. 周瑞芳 中原工学院 创新教改经验分享
6. 自由交流、提问、专家答疑
7. 张家玲 昆明理工大学 创新教改会议预告

主持人 河南大学数学与统计学院 李娜

会议时间：2023.05.09 下午3:00

腾讯会议：401-4449-8649

密码：0509

微信群 腾讯会议

教改联系人：李娜

邮箱：mengyifei001@163.com



《MeshDGP+微分几何》 教学改革创新改革 第六届研讨交流会 之渤海方案



会议简介

《微分几何》课程理论程度高，推导繁琐，不容易透过概念理解几何性质。赵海川老师开发的 meshDGP 有 20 万 Csharp 代码，数千个函数，对微分几何的很多几何拓扑概念、定理、几何结构进行了可视化，有菜单方便交互操作，并且有五本配套教材，资料详细全面。从而可以方便学生快速理解相关抽象的几何概念、定理和公式。

一方面为学生进一步学习深入的几何拓扑知识打下基础；另一方面为计算机视觉、三维重建、CAD/CAE、几何建模、网格处理、虚拟现实、影视特效、医学成像、计算机图形处理和渲染、虚拟人、生物数据分析、机器人运动学、拓扑材料、力学分析、土木建筑、计算共形几何、可计算离散整体几何结构、等等高精尖工程技术应用打下扎实的微分几何理论基础。

会议内容

主持人 开场白

1. 曾珂 西安建筑科技大学 微分几何结合 meshDGP 在土木的应用
2. 郝建兵 滕州城建开发集团 meshDGP 带来的学习微分几何的兴趣滕州经验
3. 周瑞芳 中原工学院 meshDGP 微分几何教学在中原工学院的实践
4. 冷雁 烟台大学 meshDGP 微分几何教学在烟台大学的实践
5. 赵海川 北京师范大学 博士生 meshDGP 之 Calabi-Yau 图形编程实践

主持人 总结展望

主持人 烟台大学数学学院副院长 孙丰云

会议时间：2023.05.19 晚上 19:00

腾讯会议：726-507-884

教改联系人：冷雁

Email: lengyan3511@163.com

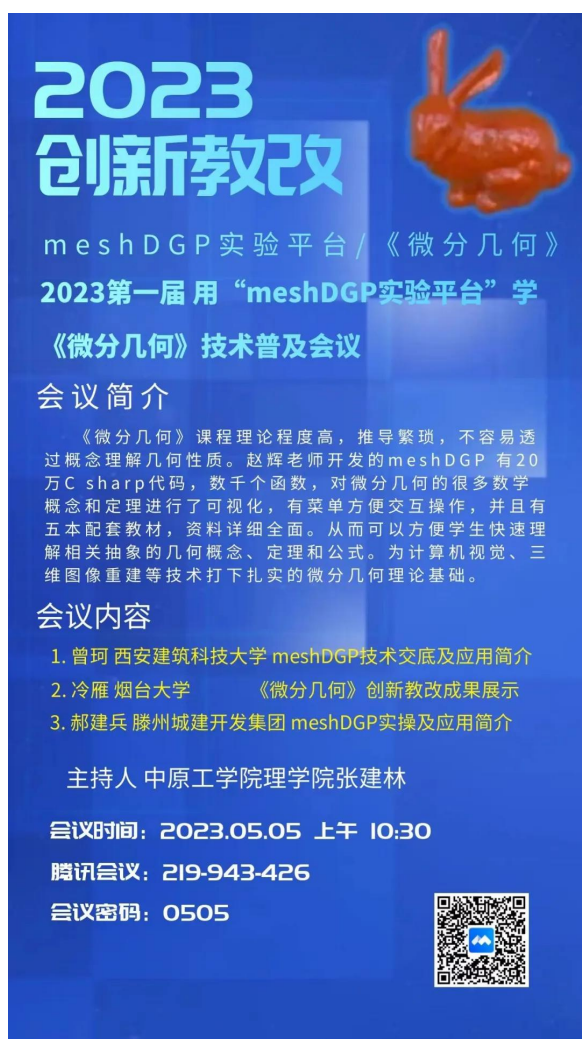


(Figure 43: Poster of the Differential Geometry Seminar.)

By merely learning concepts and theorems, students often feel confused about the practical value of differential geometry. However, with the help of the meshDGP platform, students can intuitively see the connection between differential geometry and real-world applications — for example, using parameterization to realize surface texture remapping. This builds a bridge between mathematical language and computer language, enabling students to deeply understand the practical value of the discipline. This not only attracts students to engage in differential geometry-related research but also helps reserve talents for this field. Modern differential geometry provides key mathematical tools for many engineering technologies, yet its high learning difficulty leads to a shortage of practitioners; meanwhile, the development of industrial technology cannot do without this tool. Therefore, the meshDGP platform can serve as a "stepping stone" to stimulate students' interest and promote more applications of differential geometry in the engineering field.

Currently, researchers in disciplines such as physics, chemistry, architecture, biology, computer science, and mechanics often use mathematical tools like Fourier transform and neural networks to solve engineering problems, but there are still a large number of difficult problems that cannot be solved with these tools. If more scholars can learn differential geometry through algorithm platforms, it is expected to provide new ideas for solving complex problems. For non-mathematics majors, their mathematical foundation is relatively weak, making it difficult for them to master the knowledge system and framework of contemporary differential geometry through online self-study alone—unlike postgraduates who have a solid grasp of basic knowledge such as Fourier transform and neural networks. The meshDGP platform, however, can help them quickly understand this mathematical tool and develop learning and practical abilities at an early stage.

The meshDGP algorithm platform contains hundreds of thousands of lines of algorithm code, allowing students to directly access a complete algorithm framework. Therefore, while imparting theoretical knowledge, our teaching model simultaneously trains students' coding skills. Through the learning path of "mutual complementarity between theory and algorithms," it broadens their employment channels. At present, high-precision and cutting-edge engineering fields—such as computer vision, 3D reconstruction, CAD/CAE, geometric modeling, mesh processing, virtual reality, film and television special effects, medical imaging, computer graphics processing and rendering, virtual humans, biological data analysis, robot kinematics, topological materials, mechanical analysis, civil architecture, and computational conformal geometry (for detailed information, see the articles on the WeChat subscription account of computational discrete Global Geometric Structures)—have an extremely urgent demand for differential geometry. These fields not only need talents who master theoretical knowledge but also require practical talents who can transform theory into algorithms. The teaching model proposed in the seminar is precisely committed to cultivating interdisciplinary talents with both theoretical foundation and hands-on ability.



2023 创新教改

meshDGP实验平台 / 《微分几何》

2023第一届用“meshDGP实验平台”学《微分几何》技术普及会议

会议简介

《微分几何》课程理论程度高，推导繁琐，不容易透过概念理解几何性质。赵辉老师开发的meshDGP有20万Csharp代码，数千个函数，对微分几何的很多数学概念和定理进行了可视化，有菜单方便交互操作，并且有五本配套教材，资料详细全面。从而可以方便学生快速理解相关抽象的几何概念、定理和公式。为计算机视觉、三维图像重建等技术打下扎实的微分几何理论基础。

会议内容

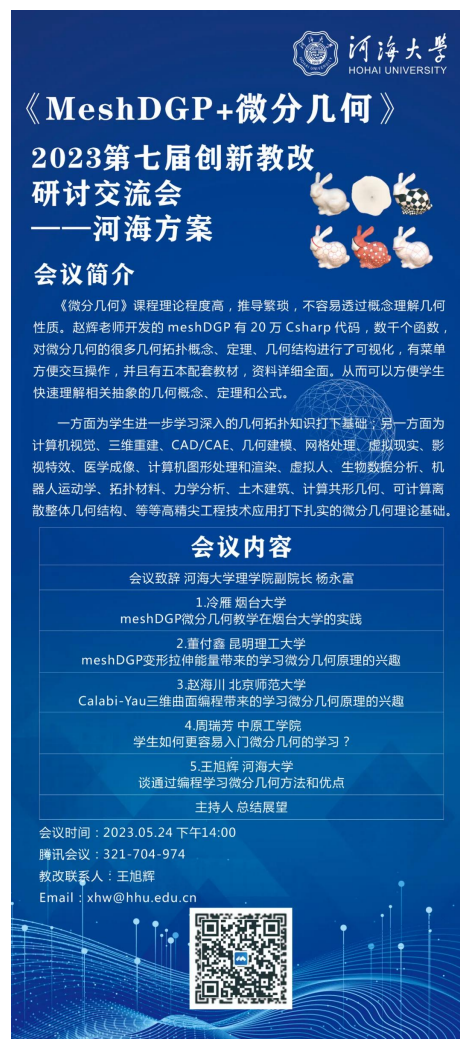
1. 曾珂 西安建筑科技大学 meshDGP技术交底及应用简介
2. 冷雁 烟台大学 《微分几何》创新教改成果展示
3. 郝建兵 滕州城建开发集团 meshDGP实操及应用简介

主持人 中原工学院理学院张建林

会议时间：2023.05.05 上午 10:30

腾讯会议：219-943-426

会议密码：0505

河海大学
HOHAI UNIVERSITY

《MeshDGP+微分几何》

2023第七届创新教改研讨交流会——河海方案

会议简介

《微分几何》课程理论程度高，推导繁琐，不容易透过概念理解几何性质。赵辉老师开发的meshDGP有20万Csharp代码，数千个函数，对微分几何的很多几何拓扑概念、定理、几何结构进行了可视化，有菜单方便交互操作，并且有五本配套教材，资料详细全面。从而可以方便学生快速理解相关抽象的几何概念、定理和公式。

一方面为学生进一步学习深入的几何拓扑知识打下基础；另一方面为计算机视觉、三维重建、CAD/CAE、几何建模、网格处理、虚拟现实、影视特效、医学成像、计算机图形处理和渲染、虚拟人、生物数据分析、机器人运动学、拓扑材料、力学分析、土木建筑、计算共形几何、可计算离散整体几何结构、等等高精尖工程技术应用打下扎实的微分几何理论基础。

会议内容

会议致辞 河海大学理学院副院长 杨永富
1. 冷雁 烟台大学 meshDGP微分几何教学在烟台大学的实践
2. 董付鑫 昆明理工大学 meshDGP变形拉伸能量带来的学习微分几何原理的兴趣
3. 赵海川 北京师范大学 Calabi-Yau三维曲面编程带来的学习微分几何原理的兴趣
4. 周瑞芳 中原工学院 学生如何更容易入门微分几何的学习？
5. 王旭辉 河海大学 谈通过编程学习微分几何方法和优点
主持人 总结展望

会议时间：2023.05.24 下午14:00

腾讯会议：321-704-974

教改联系人：王旭辉

Email：xhw@hhu.edu.cn



(Figure 44: Poster of the Differential Geometry Seminar.)

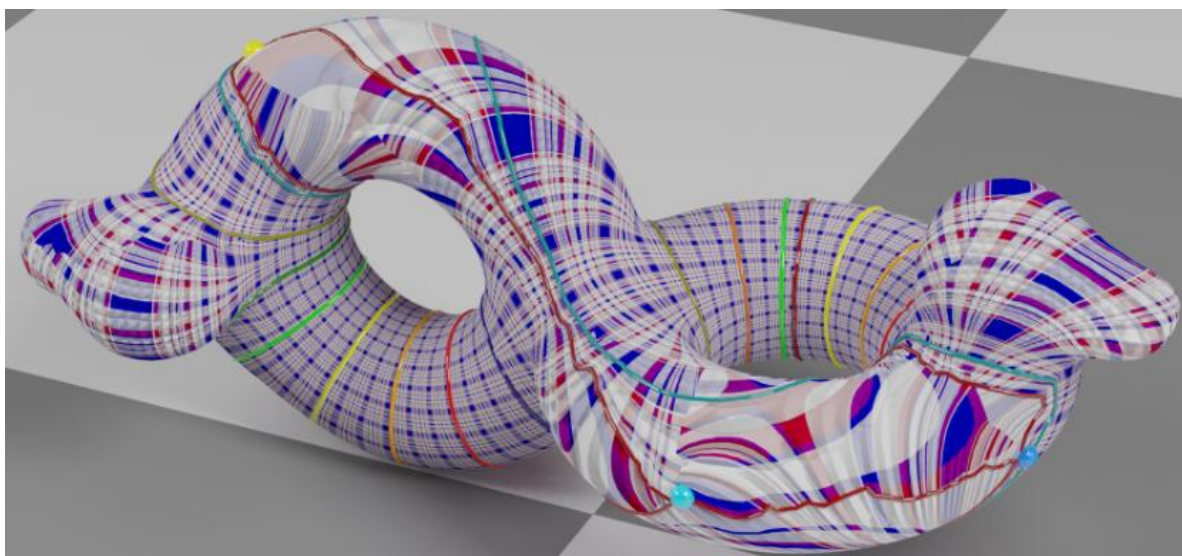
Through the seminar on differential geometry teaching, a new approach to differential geometry teaching has been successfully pioneered domestically and internationally, especially for non-mathematics majors. After this seminar, many schools have started relevant courses based on meshDGP, incorporating some knowledge of differential geometry. Teachers from various disciplines in these universities include not only those from computer graphics but also many interdisciplinary fields such as finite element computation, CAD/CAE, differential geometry, etc. Examples of such courses are as follows:

- (1) Kashgar University: Computer Graphics
- (2) Zhongyuan University of Technology: General Education Course - Introduction to Practical Differential Geometry
- (3) Beijing Yanqi Lake Institute of Applied Mathematics: computational discrete Global Geometric Structures (Autumn 2022)
- (4) Yantai University: Differential Geometry
- (5) Xi'an University of Architecture and Technology: Engineering Modeling, Computer-Aided Design
- (6) School of Mechanical Engineering, Shandong University of Technology: Mechanical CAE and Theory Course

(7) Computer Graphics (Modernized) Course

The research on computational discrete Global Geometric Structures is inseparable from the widespread popularization of differential geometric topology theories. To study a global geometric structure, one needs to accurately understand the mathematical concept of that geometric structure, which in turn relies on the support of 10 subsequent concepts, each of which then requires 10 new concepts for support. This quickly leads to hundreds of new mathematical concepts that teachers and students in general science and engineering disciplines do not encounter. Therefore, this requires significant learning investment. For teachers and students across disciplines, it is necessary to find suitable methods tailored to themselves, and they cannot simply copy the learning methods of mathematics majors, as they do not have the corresponding learning environment, atmosphere, and incentive conditions of mathematics departments.

The innovative seminar on differential geometry teaching we organized aims to promote the dissemination of differential geometry in other disciplines through means such as meshDGP code and visualization, thereby laying a knowledge foundation for the subsequent research and development of computational discrete Global Geometric Structures. So far, good results have been achieved. Teachers and students from many domestic schools have already understood this approach, and it is expected that this new way of learning differential geometric topology will gradually expand in the future.



(Figure 45: A harmonic foliation and the Ribbon graph, generated by Geometric .)

14.5 National Touring Art Exhibition of computational discrete Global Geometric Structures

The research direction of computational discrete Global Geometric Structures is not limited to studies in mathematics; it mainly targets research on applied geometry in computer science and other science and engineering fields. However, frontiers of differential geometric topology theories related to "global geometric structures" are rarely accessible to teachers and students of non-mathematics majors. Talents who understand and master these theories while being able to apply them in algorithm design are extremely scarce.

Although the geometric topology theories developed by mathematicians over the past decades have been gradually penetrating into engineering fields and applied in research practices, this interdisciplinary integration process is still in a stage of continuous development. Nevertheless, due to the high abstraction of these theories, teachers and students of non-mathematics majors often need a strong learning motivation to conduct in-depth exploration. The art form, however, can effectively stimulate the interest in learning and research among this group.

Based on the author's achievements in algorithm design in the application field of geometric topology theories, and with the help of rendering technology in computer graphics, these abstract geometric topology concepts and theories have been transformed into visual presentations. On this basis, in 2024, the author launched the [Global Touring Art Exhibition of computational discrete Global Geometric Structures](#), which has so far been held at more than ten universities and research institutions across the country, including:

- (1) [Zhongyuan University of Technology](#),
- (2) [Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences](#),
- (3) [North China Electric Power University](#),
- (4) [Henan University](#),
- (5) [Xihua University](#),
- (6) [Renmin University of China](#),
- (7) [South-Central Minzu University](#),
- (8) [Tianjin Chengjian University](#),
- (9) [Taiyuan University of Technology](#),
- (10) [High Magnetic Field Science Center, Hefei Institutes of Physical Science, Chinese Academy of Sciences](#),
- (11) [Jiangxi University of Science and Technology](#),
- (12) [Dali Li Zhengdao Science and Art Center](#).

The national touring art exhibition is not yet concluded, and dozens of more universities will host the exhibition in the follow-up.

The images exhibited cover geometric topology concepts such as Ribbon-graph, Square-Tiling, harmonic foliations, holomorphic quadratic differentials, meromorphic quadratic differentials, differential 1-forms, discrete Calabi flow, and normal flow.

During the exhibition, teachers and students from many universities feedback that this art exhibition has pioneered a new paradigm of "intrinsic geometric structure art" and serves as an innovative bridge connecting cutting-edge mathematical theories with public aesthetics. For the first time, the exhibition has systematically presented the intrinsic structures of surfaces that are difficult to perceive intuitively, transforming abstract theories such as holomorphic differentials and harmonic foliations into digital paintings with vivid colors, thus turning them from theoretical symbols into tangible visual entities.

Through this exhibition, the audience not only developed a strong sense of curiosity and desire for exploration but also truly felt the infinite charm generated by the interdisciplinary integration of mathematics and computer science. After visiting the exhibition, many teachers and students went to the library to look up relevant materials and learn about the mathematical concepts illustrated in the exhibition. Unconsciously, they developed an interest in these cutting-edge geometric topology theories.

Following the art exhibition, institutions including the [School of Science of Tianjin Chengjian University](#) and the [School of Science of Jiangxi University of Science and Technology](#) were still eager for more; they made full sets of reproductions of all over 50 artworks in the exhibition for collection and have launched long-term exhibitions on their campuses.

Compared with common exhibitions themed "Beauty of Mathematics" or "Geometric Art", this National Touring Art Exhibition has significant differences in both the core geometric and mathematical theories it presents and the external forms of presentation. Its core content lies in the "global geometric structures" generated through computer graphics technology and algorithms, and these geometric artworks are closely associated with applications in "chokepoint" fields such as industrial software, with the potential to solve relevant technical problems—which is exactly the unique value and innovative feature that previous lectures and exhibitions themed "Beauty of Mathematics" lack.

Although American mathematicians such as Thurston have popularized cutting-edge geometric theories and concepts through art forms, they were limited by the rendering technology available at that time and could hardly achieve the visual aesthetics supported by modern computer graphics. In addition, the fact that this exhibition is held as a national tour with the spontaneous participation of university teachers is an innovative attempt in itself.

Many teachers and students who came into contact with these differential geometric topology theories for the first time have summarized a large number of accurate insights and views on global geometric structures and computable research based on their own understanding, which has achieved the purpose of holding the exhibition.

Some teachers engaged in the research of these geometric topology theories also stated that the visual presentation is very helpful for their research work, as it can bring new perspectives and insights.

A number of schools have published graphic reports on the exhibition, including:

- (1) [Henan University](#),
- (2) [Renmin University of China](#),
- (3) [Tianjin Chengjian University](#),
- (4) [Taiyuan University of Technology](#),
- (5) [High Magnetic Field Science Center, Hefei Institutes of Physical Science, Chinese Academy of Sciences](#),
- (6) [Jiangxi University of Science and Technology](#).

Some schools have integrated the exhibition with [other mathematical activities](#), achieving even better results. For example:

[The 8th exchange and discussion activity of "RUC Mathematics Time I"](#) was held;

[The 8th Session of "RUC Mathematics Time I"](#) — Geometric Structures: Theory, Algorithms and Visualization.

Another example of integration with topological materials:

[The first science-art integrated art exhibition of Dali Li Zhengdao Science and Art Center opened at Taibao Home Dali Community](#).

“可计算离散整体几何结构”全国巡回艺术展

从 2023 年一月份开始,“可计算离散整体几何结构”实验室根据在图形学、应用微分几何拓扑、网格处理和生成等方面 30 年的教学、研究、工程工作,在国内首次提出通过各种艺术表现方法,例如图片、视频、雕塑、绘画等,以及用各种计算机图形学技术,例如虚拟现实、可视化等,来体现“当代微分几何拓扑”的数学理论和概念,尤其是“整体几何结构”的理论和概念。为了方便交流,我们把这种艺术称之为“当代几何拓扑”艺术流派。美国在上个世纪就有一些的体现几何拓扑理论的计算机动画艺术,如 *Outside in*, *Not knot* 等,以及一些艺术雕塑。

“当代几何拓扑”艺术主要目的有两个,第一个是艺术层面,通过视觉信息来直观感受当代学者们开拓的美轮美奂的前沿的微分几何拓扑之美。第二个是工程科技层面,当前的卡脖子高科技需要对很多前沿的微分几何拓扑理论进行熟练掌握和深入应用,不仅仅是应用几百年前、一百年前的数学理论。而这些基础数学理论的掌握不是一蹴而就的,很多时候无法通过一个环节就能应用到工业技术上,需要一环扣一环,一个理论接一个理论的环环相扣的接力,最终走到工业技术产品层面。在中间阶段,没有工业产品激励的情况下,通过可视化、艺术展现带来的视觉美感可以对每一环理论的辛苦学习和掌握提供巨大的激励。

当前对基础科学数学理论和应用科学技术之间的关系有很多争议,主要原因就是有很多基础科学数学理论没有直接在第一个环节就被应用到工业技术上,所以很多朋友难以理解基础科学数学理论的重要性。“可计算离散整体几何结构”实验室提出艺术-科普-图示-算法-代码-理论一条龙的思路,从而可以促进大家通过形象展示对某些基础理论有感性认识,进而理解基础理论如何通过一环一环的链条最终对工业技术发挥作用。

我们这次艺术展挑选了一些对调和一形式、调和叶状结构、全纯二次微分、微分形式、法向流等若干“整体几何结构”进行可视化的、由我们研究或开发的计算机算法生成的图片。这些“整体几何结构”的理论,及其在 CADCAE 工业软件、拓扑材料、斯格明子物理材料、生物膜等交叉学科上应用的更多详细信息可以参阅微信订阅号“可计算离散整体几何结构”。

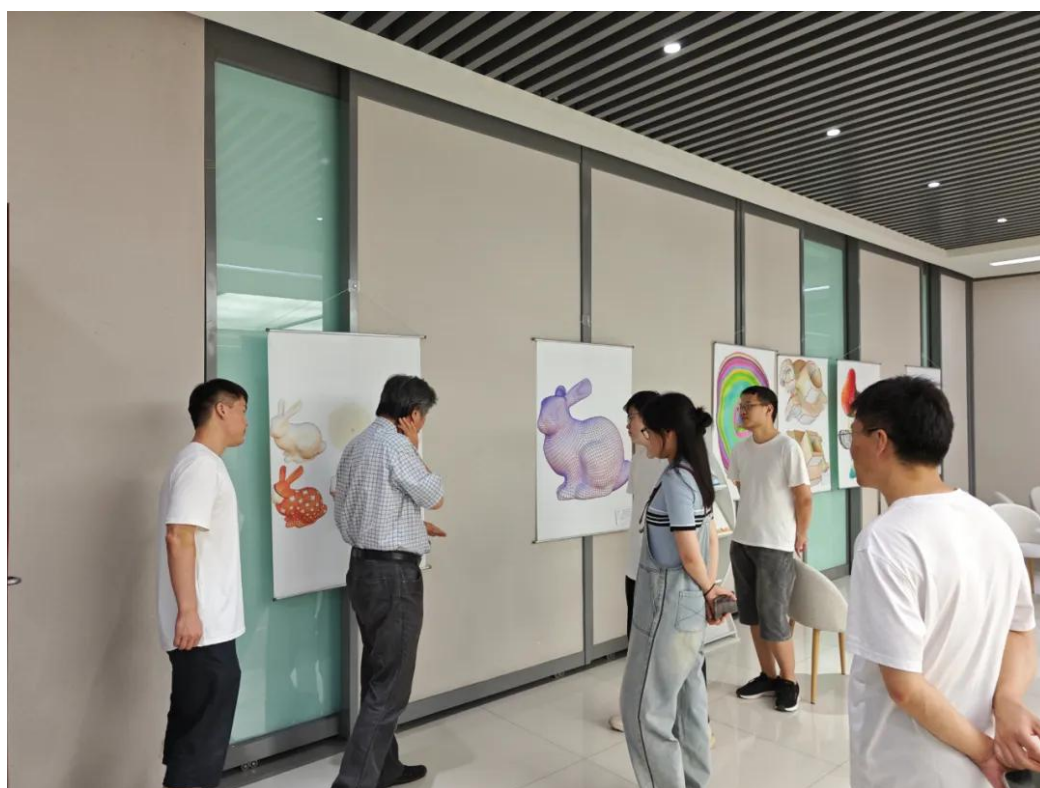


赵辉

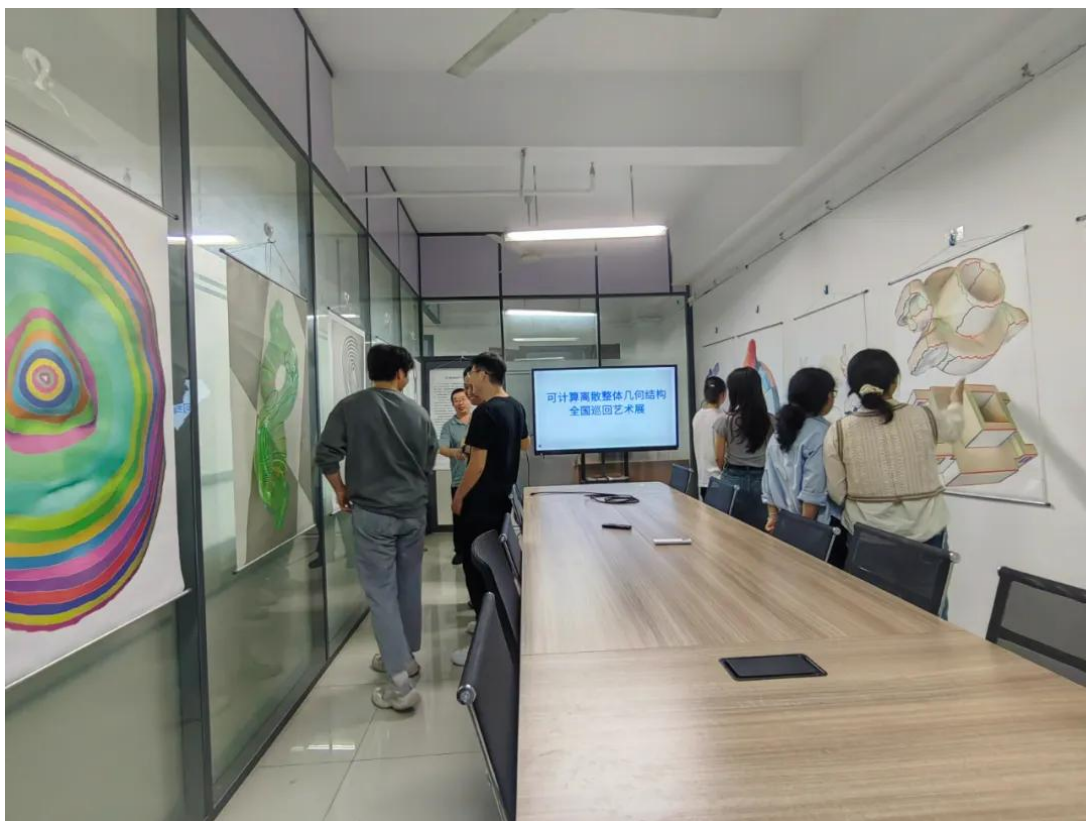
2024 年 6 月 15 日



(Figure 46: Exhibition of computational discrete Global Geometric Structures at Zhongyuan University of Technology.)



(Figure 47: Exhibition of computational discrete Global Geometric Structures at the Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences.)



(Figure 48: Exhibition of computational discrete Global Geometric Structures at North China Electric Power University.)



(Figure 49: Exhibition of computational discrete Global Geometric Structures at Xihua University.)



(Figure 50: Exhibition of computational discrete Global Geometric Structures at Henan University.)



(Figure 51: Exhibition of computational discrete Global Geometric Structures at Renmin University of China.)



(Figure 52: Exhibition of computational discrete Global Geometric Structures at South-Central Minzu University.)



(Figure 53: Exhibition of computational discrete Global Geometric Structures at Tianjin Chengjian University.)



(Figure 54: Exhibition of computational discrete Global Geometric Structures at Taiyuan University of Technology.)



(Figure 55: Exhibition of computational discrete Global Geometric Structures at Jiangxi University of Science and Technology.)



(Figure 56: Exhibition of computational discrete Global Geometric Structures at the Hefei Institutes of Physical Science, Chinese Academy of Sciences.)



(Figure 57: Exhibition of computational discrete Global Geometric Structures at Dali Li Zhengdao Science and Art Center.)

可计算离散整体几何结构全国巡回艺术展第九站：太原理工大学 新闻发布会

主持人	时间	内容	发言人	单位
嘉宾	16:00-16:10	领导讲话	武东升	山西发改委
	16:10-16:15	院领导发言	贺衍	太原理工大学数学学院
	16:15-16:30	太原理工大学全国巡回艺术展介绍	雷敏	太原理工大学数学学院
嘉宾	16:30-16:35	全国巡回艺术展第八站心得、体会、收获、经验	张东	天津城建大学理学院
嘉宾	16:35-16:40	全国巡回艺术展第七站心得、体会、收获、经验	朱剑林	中南民族大学计算机科学学院
嘉宾	16:40-16:45	全国巡回艺术展第四站心得、体会、收获、经验	韩喆	河南大学数学与统计学院
嘉宾	16:45-16:50	全国巡回艺术展第一站心得、体会、收获、经验	周瑞芳	中原工学院数学与信息科学学院
嘉宾	16:50-16:55	全国巡回艺术展第五站心得、体会、收获、经验	陈宏	西华大学机械学院
嘉宾	16:55-17:00	几何+艺术	孙志斌	中国科学院大学
嘉宾	17:00-17:05	几何+艺术	杨火根	江西理工大学理学院
嘉宾	17:05-17:10	科幻+艺术	陆贝珂	三体电视剧导演
嘉宾	17:10-17:15	数学+艺术	陈鹏	安徽师范大学
嘉宾	17:15-17:20	几何+艺术	刘绍辉	哈尔滨工业大学计算机科学与技术学院
嘉宾	17:20-17:25	内蕴整体几何结构 + 艺术	赵辉	可计算离散整体几何结构实验室
		现场师生交流+合影	师生	太原理工大学数学学院

14.6 Series of Mesh SquareTable Meetings

The algorithm research on computational discrete Global Geometric Structures focuses on meshes, investigating the corresponding global geometric structures on discrete meshes and smooth surfaces. Mesh research is not a purely abstract theoretical exploration; instead, it is a comprehensive study integrating the application of geometric topology theories, numerical computation, code implementation, visual rendering, and other dimensions. To achieve the goal of advancing the development of this field, with the support of teachers and deans from many universities across the country, the author established and launched the [Series of Mesh SquareTable Meetings](#).

It is worth noting that the reason this series of meetings is named "Square Table" rather than the common "Round Table" lies in the fact that square tiling is a core challenge in mesh partitioning. Consequently, the author proposed the concept of "super-structured quadrilateral meshes." "Super-structured quadrilateral meshes" refer to the research on dividing quadrilateral grids on surfaces using cutting-edge differential geometric topology theories such as various intrinsic global geometric structures. The shape of quadrilaterals highly aligns with the imagery of "square," so the name "Square Table" more accurately reflects the core of the research.

Up to now, eight sessions of the Mesh Square Table Meeting series have been held, with each session focusing on a different theme related to mesh research — such as code programming, industrial development, visual rendering, differential geometry teaching, optimization problems, solvers, interdisciplinary integration with computer graphics, paper writing styles, and mesh art. Each seminar attracts participants including teachers, doctoral students from different majors at various universities across the country, and engineers from enterprises. This series of meetings is still ongoing, and more mesh-related discussion topics will be covered in the future.

- (1) [The 1st Mesh Square Table Meeting](#): Code Programming Level
Hosted by the [School of Science, Tianjin Chengjian University](#)
Opening Speech: [Dean Jia Guozhi](#)
Organizer: Prof. Zhang Dong
- (2) [The 2nd Mesh Square Table Meeting](#): CAD/CAE Small Software Development Level
Hosted by the [School of Physics and Electronic Information, Anhui Normal University](#)
Opening Speech: [Dean Zhang Aiqing](#)
Organizer: Prof. [Chen Peng](#)
- (3) [The 3rd Mesh Square Table Meeting](#): Visual Rendering Level
Hosted by the [School of Science, Jiangxi University of Science and Technology](#)
Opening Speech: [Dean Xu Zhonghui](#)
Organizer: Prof. Yang Huogen
- (4) [The 4th Mesh Square Table Meeting](#): Code-Oriented Visual Learning and Teaching of Differential Geometry
Hosted by the [School of Mathematics and Statistics, Henan University](#)
Opening Speech: [Dean Tang Hengcai](#)
Organizer: Prof. [Han Zhe](#)

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- (5) [The 5th Mesh Square Table Meeting](#): Optimization Problems, Linear Systems, and Solvers
Hosted by the [School of Mathematics, Statistics and Mechanics, Beijing University of Technology](#)
Opening Speech: Dean Huang Qiumei
Organizer: Prof. Zhuge Changjing
- (6) [The 6th Mesh Square Table Meeting](#): Interdisciplinary Integration of Computer Graphics + Other Disciplines
Hosted by the [School of Mathematics and Information Science, Zhongyuan University of Technology](#)
Opening Speech: Dean Yan Zhenya
Organizer: Prof. Zhou Ruifang
- (7) [The 7th Mesh Square Table Meeting](#): Evaluation of Paper Writing Styles in the Field of Mesh + Computable Global Geometric Structures
Hosted by the [Dean of the School of Science, Changzhou Institute of Technology](#)
Opening Speech: [Dean Chen Rongjun](#)
Organizer: Prof. Liu Yongcai
- (8) [The 8th Mesh Square Table Meeting](#): Mesh + Global Geometric Structures + Art
Hosted by the [High Magnetic Field Science Center, Hefei Institutes of Physical Science, Chinese Academy of Sciences](#)
Opening Speech: [Director Du Haifeng](#)
Organizer: [Director Du Haifeng](#)

The online live broadcast of each Mesh Square Table Meeting attracts 3,000 to 6,000 viewers online. This wide participation has effectively promoted the popularization of mesh research and laid a foundation for the algorithm research on "global geometric structures."

At present, the Mesh Square Table Meeting series is mainly held online. To further deepen research and exchanges in the mesh field (especially in the direction of super-structured quadrilateral meshes), Teacher Han from the School of Mathematics and Statistics, Henan University has applied for a conference project at the Tianyuan International Center for Mathematical Exchange, and Professor Yang Huogen from the School of Science, Jiangxi University of Science and Technology has also applied for a conference project at the Xiangshan Science and Technology Conference Center. Both conferences are planned to be held offline, which will create more direct and in-depth communication opportunities for participating experts and scholars, and facilitate the collision and advancement of research.

The purpose of the computational discrete Global Geometric Structures Laboratory in establishing the Series of Mesh Square Table Meetings is to address the second step of the assertion that "abstract mathematics and basic sciences are the foundation and engine of scientific and technological development," namely: "How to unleash the enormous potential contained in abstract basic mathematics, especially cutting-edge geometric topology theories, into technology, and promote the development of industrial technology."

Currently, the academic community, engineering sector, business community, and all sectors of society have reached a consensus on the first part of the assertion and recognized its importance. However, what everyone is eagerly looking forward to is the second step. Only through the second step can the first

assertion be truly interpreted, transforming it from a viewpoint into reality.

This is not a conceptual but a practical purpose. Abstract mathematics can be pursued with just paper, pens, and blackboards, but industrial technology requires "experimental tools." For example, the renowned physicist Samuel C. C. Ting needs to master every detail of the experiment, such as the voltage of each instrument; physical laboratories are so expensive that ordinary people cannot afford them.

In contrast, the research on "computational discrete Global Geometric Structures" on meshes proposed by us only requires a computer, which is affordable for tens of millions of science and engineering professionals. Our experimental technology is "programming." Nevertheless, even such experiments require mastering a large number of detailed experimental programming code details; otherwise, it would be mere armchair theorizing. One of the purposes of the Series of Mesh Square Table Meetings is to solve various practical problems such as experimental details, rather than just conducting exchanges on algorithm theories.



(Figure 58: The 1st Mesh Square Table Meeting)



(Figure 59: The 2st Mesh Square Table Meeting.)



(Figure 60: The 3st Mesh Square Table Meeting)



(Figure 61: The 4st Mesh Square Table Meeting)



(Figure 62: The 5st Mesh Square Table Meeting)



(Figure 63: The 6st Mesh Square Table Meeting)



(Figure 64: The 7st Mesh Square Table Meeting)



(Figure 65: The 8st Mesh Square Table Meeting)

14.7 Organizing meshDGP Study Sessions

The research platform for computational discrete Global Geometric Structures is built on the original mesh code platforms such as meshDGP and Geometric. Since the release of meshDGP, tens of thousands of teachers, students, and engineers across the country have learned and used it. To address the question of "how to quickly get started with meshDGP using spare fragmented time", we have successively held more than 80 real-name themed study sessions. Through these study sessions, over 80 teachers, students, and engineers from various universities across the country and with different research directions have efficiently mastered the key points for getting started in their spare time, laying a foundation for in-depth research. Some of the study topics and participants are as follows.

题目	学习人
MeshDGP里的Morse函数	潘安 兰州
MeshDGP里的特征向量计算展示	瞿悦呈 北航计算机学院博士生
MeshDGP里面的QEM简化算法展示和代码流程	冯骊骅 老师 重庆科技大学计算机学院
MeshDGP里的向量场展示和代码流程【1/3】	赵晴 中原工学院数学学院 本科生
MeshDGP里的Calabi-Yau曲面编程	叶然然 本科生 中原工学院数学与信息科学学院数学与应用数学
MeshDGP里面的参数化效果图展示、代码概述：频谱，ABF算法篇章【1/2】	殷思麒 湖南大学机械学院 博士生
MeshDGP里的OpenGL展示和代码流程【1/2】	康亚卓 北京 中机认检
MeshDGP里的拉伸能量算法展示、代码概述【1/4】	丁弘晖 人大统计学博士生
MeshDGP里的OpenGL展示和代码流程II	潘安 兰州
MeshDGP里的GLSL效果展示和代码流程	祁桐 北京构力科技有限公司
四边形六面体网格生成之Polycube Shap Space	瞿悦呈 北航计算机学院博士生
基于Stretching Energy的变形实验展示	潘安 兰州
meshDGP剪辑等操作的初步认识和后续学习目标	杜志楠 中国航空制造技术研究院
meshDGP里的GLSL渲染二刷	胡志林 博士 清华精密仪器系毕业
meshdgp里面的高斯曲率，平均曲率，欧拉示性数，亏格，点，边，面展示	朱施恩 威斯康星-麦迪逊大学 UW madison 统计专业 研究生
线性代数之——MeshDGP里的OpenGL之矩阵变换展示和代码流程	孙竞争 广东省科学院智能制造研究所
meshDGP软件架构和设计模式	康亚卓 潘安 彭贵军 黄锦龙 祁桐 vs 曾珂 西安建筑大学
meshDGP里如何实现样条曲面【1/2】	宋洋 博士 犹他大学 师从样条前辈教授
meshDGP里的骨骼动画和蒙皮展示与代码流程	肖智文 博士 清华航院
meshDGP里如何实现细分subdivision曲面【1/2】	杨火根 教授 江西理工大学
meshDGP里的贝塞尔曲线展示和代码流程【1/3】	银润龙 老师 忻州师范学院数学系
meshDGP里的简化光滑等操作和代码流程【1/3】	陆云桥 工程师 上海正雅齿科
meshDGP里的梯度，拉普拉斯与双调和操作和代码【1/2】（PPT代替）	刘亚雄 湖南
meshDGP里的三角形、四边形网格生成展示和代码流程【1/2】	王少阳 工程师 制造行业
meshDGP里的B-Spline曲线展示和代码流程【1/3】	黄锦龙 工程师 深圳 激光行业
meshDGP里的三维模型点边面的增加删除操作展示和代码流程【1/2】	陈鹏 老师 安徽师范大学物理与电子信息化学院
meshDGP里的三维模型补洞、边界操作展示和代码流程【1/2】	王卫 中学老师 世青国际学校
meshDGP里的刚体转动 通过OpenGL【1/3】	姬振华 老师 郑州航院力学
MeshDGP里面的5,6,7度数操作展示和代码流程	李林 老师 太原科技大学机械工程学院
meshDGP里的matlab调用	刘宇辉 中南大学 老师 人工智能方向
meshDGP里的网格切割操作展示和代码流程	杨佩桥 北航电子信息工程 研究生
三维模型参数化扭曲度量及和谐参数化理论代码展示（无视频 使用PPT）	李鹏 北航
meshDGP里的补洞挖洞操作和代码流程	詹国峰 工程师 三维技术相关行业
MeshDGP里的ArcBall里面的（线性变换）原理和选择功能展示和代码流程	段育鹏 哈尔滨工程大学 博士生
meshDGP里的人机交互和UI	刘婧媛 东京大学五十岚 健夫Takeo Igarashi教授实验室博士后 香港科技大学visgraph lab博士
meshDGP里的线性代数50个应用之一	诸葛昌靖 北京工业大学数学学院 线性代数老师
MeshDGP里的三维模型特征线条抽取算法功能展示和代码流程	刘永才 老师 常州工学院理学院
meshDGP里的posion重建效果展示和对照代码	曾珂 教授 西安建筑大学
meshDGP里的法流操作展示和代码对照	古鹏举 博士 贵州大学
meshDGP的posion重建操作展示和对照代码	王川川 工程师 宁波
meshDGP里的polycube变形	陈晨曦 博士 南洋理工大学
meshDGP里的GLSL渲染	孟楠 教授 香港大学医学院
meshDGP概览和研究生指导辅助工具	孙志斌 教授 中国科学院大学 中国科学院国家空间科学中心
meshDGP里的三维模型分段算法展示	邵阳毅 博士 航发商发上海
meshDGP里的贴图和featureLine学习	伍世聪 工程师 北京+苏州

The Study Sessions are an innovative interdisciplinary learning and exchange model established by the computational discrete Global Geometric Structures Laboratory, with several prominent features:

- (1) Utilizing spare time;
- (2) Making use of fragmented time;
- (3) Enabling quick access to basics (quick getting-started);
- (4) No restrictions on academic discipline or major;
- (5) No requirements for prior background knowledge;
- (6) Learning with reference to code;
- (7) Visualized, menu-based, and interactive learning;
- (8) Participants who have already 入门 (mastered the basics) guiding new beginners;
- (9) No time or energy burden on either party—if either side feels burdened, the Study Session is deemed unsuccessful;
- (10) A completely different model from specialized in-school learning;
- (11) Particularly suitable for teachers and students with clear learning objectives;
- (12) Suitable for teachers and students with strong self-directed learning abilities;
- (13) No requirement for light workload in one's regular job—nearly 100 real-name registered teachers, students, and engineers who got started through the Study Sessions all did so in their spare time amid busy work;
- (14) Relatively suitable for teachers and students who are good at communication, as those who help with learning also do so in their spare time (on a voluntary basis), rather than being a teacher's regular job in school which involves guided instruction.

With its visualized interface, concise and clear-format code, convenient menu interaction, five supporting textbooks, and code-level application of a large number of mathematical theories, meshDGP is highly favored by a wide range of science and engineering teachers, students, and engineers across the country, and has especially gained recognition from many senior engineers. Judging from the enthusiastic participation of senior engineers from various industries in the Study Sessions, we have put forward the goal of "equipping 50 million science and engineering engineers with computational discrete Global Geometric Structures".

Currently, many cutting-edge computational science and technology are still in the stage of development and diffusion, which is a historical evolutionary process. Due to the limitations of the times and conditions back then, many senior engineers failed to access cutting-edge computing technologies in a timely manner. However, through the meshDGP Study Sessions, they can make up for this technological generation gap and achieve "looking back and catching up" in the course of technological development.

Over the past 50 years, many people among several generations in China have aspired to become scientists and engineers. However, with the changes in social development, this trend has weakened. In the future, Chinese teenagers may be more inclined to engage in fields such as finance. This is similar to the current situation in the United States where many engineering and technology talents come from other countries, or like the gradual reduction of the peasant population in China decades ago, with fewer people willing to engage in agriculture. Therefore, the core of our proposed goal of "equipping 50 million science and engineering engineers with computational discrete Global Geometric Structures" is to focus on the older generation of science and engineering engineers trained in China over the past 50 years, rather than only on current teenagers. When senior engineers master cutting-edge computing technologies, they will naturally form a demonstration and inheritance effect, laying a solid foundation for the growth and development of teenagers, and achieving a "natural and smooth" cultivation of technical talents.

At present, nearly 100 teachers, students, and engineers are conducting real-name online learning through the Study Sessions, accounting for one-fifth of all 500 real-name registered teachers and students, and 1% of the tens of thousands of teachers and students who use meshDGP in total. According to feedback from the teachers and students participating in the Study Sessions, basically, only three half-days of preparation in spare time are needed to get started. This proves that the mechanism of our Study Sessions is practical and feasible, conforming to actual conditions.

14.8 Joint Research Center for Super-Structured Quadrilateral Meshes

The research content of computational discrete Global Geometric Structures covers professional fields such as differential geometric topology theory, computer algorithm design, code implementation, and mechanical engineering. It is not a single algorithm; therefore, it requires interdisciplinary and cross-disciplinary collaboration, and tackles key challenges through an "organized scientific research" model.

Since the Laboratory of computational discrete Global Geometric Structures proposed the research direction of "super-structured quadrilateral meshes" to solve technical problems related to industrial software, it has attracted the attention of many professional teachers, scholars, and experts across the country, and relevant research has gradually unfolded. To advance "organized scientific research", the author has jointly established the "Joint Research Center for Super-Structured Quadrilateral Meshes" with multiple universities.

The joint mechanism of colleges from multiple universities fundamentally avoids the "five-only" tendencies (such as "only focusing on papers, only focusing on projects") that hinder scientific research development. The core value of the Joint Research Center lies in: taking "super-structured quadrilateral meshes" as the common research core, integrating the expert resources from multiple disciplines of various universities, and promoting the formation of a closed loop among multiple elements including "education, scientific research, science popularization, code, programming, programs, interfaces, interaction, visualization, art, geometric topology, mathematics, computation, algorithms, engineering, software, and academic exchanges". This realizes a positive cycle and iterative development of mutual promotion. Through the value demonstrated by the achievements, methods, and models themselves, it completely breaks the current "five-only" dilemma in the scientific research field, accelerates the transformation of the innovative achievements of human thinking contained in abstract mathematical concepts and theories into the field of science and technology, and ultimately contributes significantly to the development of high technology at home and abroad.

The specific objectives of the Joint Research Center include:

- (1) Taking the specific application of super-structured quadrilateral meshes in industrial software as an example, intuitively demonstrating the promoting role of "cutting-edge basic mathematics" in "high technology";
- (2) Demonstrating how to unleash the huge productivity contained in cutting-edge mathematical theories through specific application scenarios;
- (3) Establishing a communication channel for teachers and students from different schools through a specific platform;
- (4) Conducting cooperation in academic papers, code programming, industrial applications, and other aspects;
- (5) Organizing relevant academic and science popularization activities;
- (6) Studying the mathematical theories involved in super-structured quadrilateral meshes;
- (7) Studying code implementation;

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- (8) Inviting experts in algorithms required for super-structured quadrilateral meshes to give exchange reports;
 - (9) Understanding the research ideas of computational discrete Global Geometric Structures through the case of super-structured quadrilateral meshes;
 - (10) Taking super-structured quadrilateral meshes as the core, laying a technical foundation for the next leapfrog development of domestic industrial software such as CAD and CAE.

At present, the School of Science of Jiangxi University of Science and Technology, the School of Science of Tianjin Chengjian University, and the School of Mathematics and Statistics of Henan University have successively passed the review of the college affairs meeting and obtained the signature confirmation of the dean, officially becoming the initiating units and core members of the Joint Research Center. Each of these three institutions has selected several teachers and graduate students to participate in the research work; up to now, the total number of teachers and students in the Joint Research Center has reached more than 20.

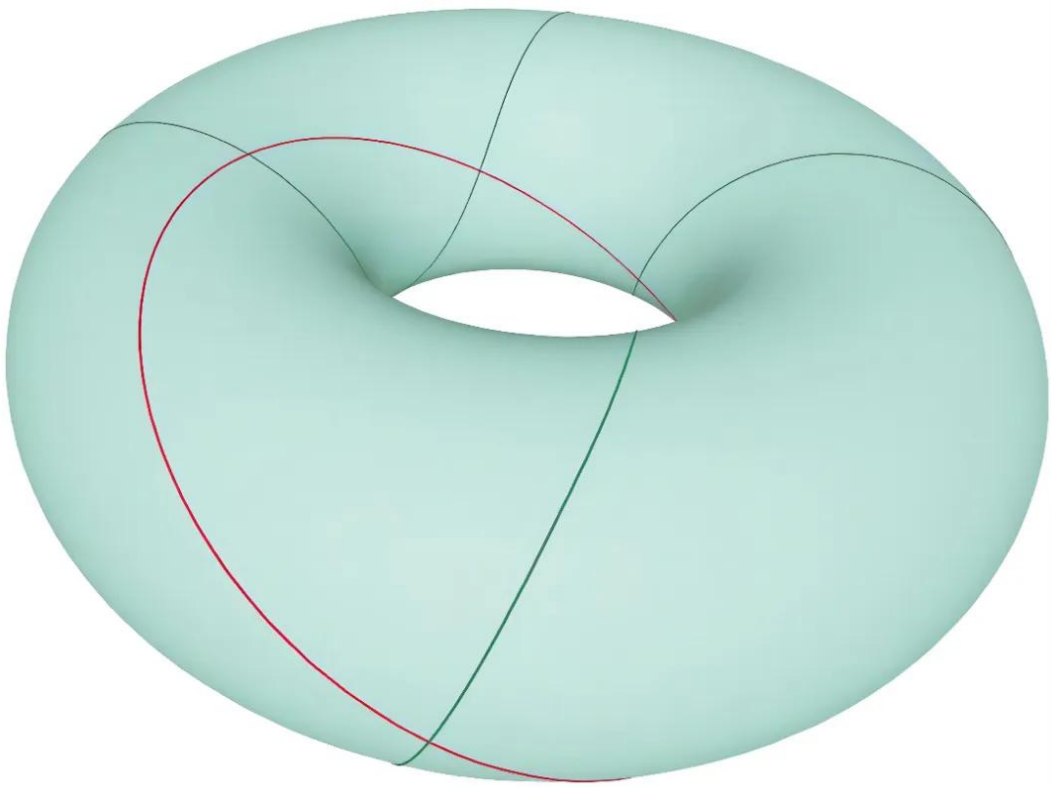
The research goal of computational discrete Global Geometric Structures is to release the productivity contained in the global geometric structure theory into various industrial technologies through algorithms. Compared with local mathematical structures, which have been relatively widely and maturely applied in industry, the productivity of global geometric structures is still in the embryonic stage and has broad prospects. This cannot be achieved overnight or by one person alone. Through large-scale organized scientific research, organizing teachers, students, engineers, scholars, and experts from various disciplines and professional backgrounds, it is expected to achieve mutual assistance, mutual exchange, and mutual improvement, thereby promoting the development of this new research field.

Super-structured quadrilateral meshes are a type of global geometric structure within computational discrete Global Geometric Structures. Due to their industrial application value, a platform such as the "Joint Research Center" was established first to provide further support for interested teachers and students. There are also various other global geometric structures, and in the future, organized scientific research may be carried out for them in a similar manner.

14.9 Super-Structured Quadrilateral Mesh Programming Competition

Computational discrete Global Geometric Structures is not merely a research on abstract mathematical theories, but a scientific and technological research that takes programming code debugging as an experimental tool. This means the research process requires in-depth and extensive support from code programming technologies and tools, especially the specialized programming capabilities in the mesh field—just as physics research cannot do without various physical equipment and experimental operations.

However, code programming, as a research tool, is often overlooked in this field: people tend to think it can be mastered without adequate preparation, and it also does not receive sufficient attention in the research process. Based on our practical experience, mesh-related code programming must be supported by adequate tool reserves, and researchers need to be proficient in various mesh programming technologies. For this reason, to cultivate this key technical capability, we have launched the "[Super-Structured Quadrilateral Mesh Programming Competition](#)" and set specific programming tasks. By completing these tasks, participants can systematically master most mesh-related programming technologies, laying a tool foundation for in-depth research.



(Figure 66: Schematic Diagram 1 of the Mesh Programming Competition Questions.)

The specific problem description is as follows:

Input a 3D model (closed) of a torus composed of triangles (which can be generated from mathematical formulas using the methods in the five textbooks);

First generate a quadrilateral torus, then generate a triangular torus;

Input an $N \times N$ grid on the torus (generated according to the torus formula);

The grid has N horizontal lines and N vertical lines (no need for the complex non-horizontal lines

shown in the above figure; simple horizontal and vertical lines are used);

Cut the triangles on the torus into $N \times N$ small models along the grid;

Each small model is composed of triangles;

Render these small models in different colors;

Output these small models as files in OBJ or OFF format;

The diagram of the cutting process for the entire problem is shown in the duck diagram below.

Notes on the programming problem and its difficulties:

10. The programming difficulty lies in the fact that cutting the grid requires proficient mastery of the operations of the half-edge data structure for meshes (refer to the five textbooks for a large number of basic operations);

11. This programming task does not require algorithms or theories; it mainly tests the ability to write code;

12. A large number of "if-else" statements are needed to handle various cases of grid intersections;

13. Cutting along the horizontal and vertical lines of the grid requires passing through triangles, and each of these triangles will be cut into several smaller triangles;

14. Since all elements are generated by formulas, the intersection points between the grid and each triangle on the torus are already known during input;

15. Triangles that are not passed through do not need to be modified;

16. There are a large number of "if" cases that cannot be considered in the first place, and they need to be addressed while writing the program;

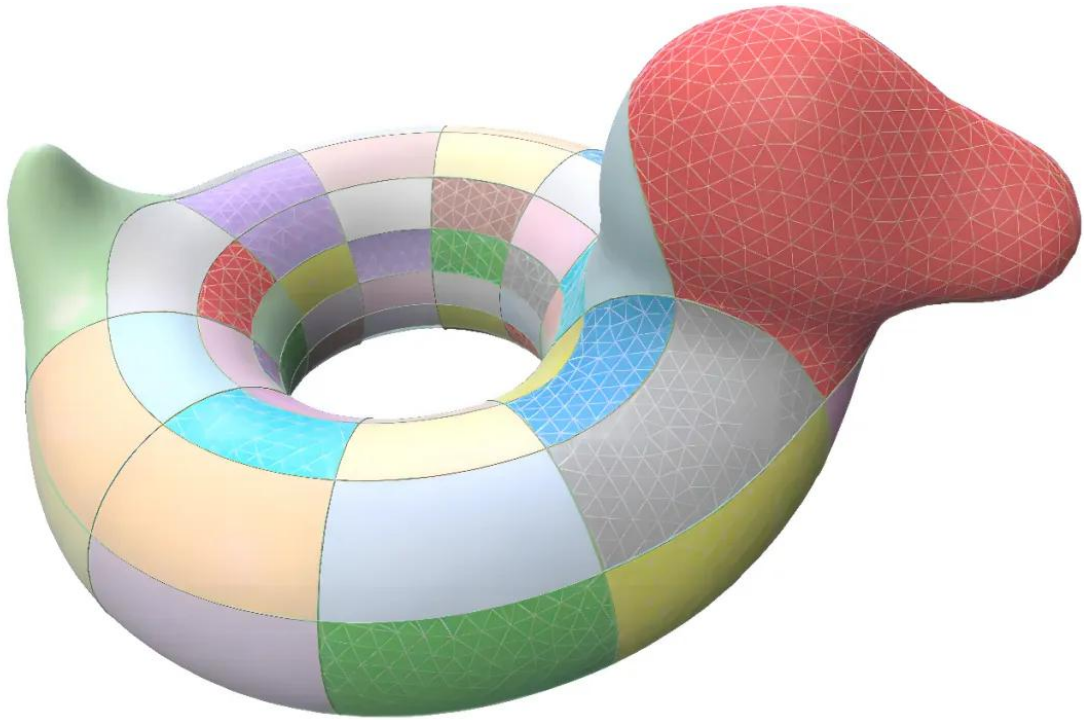
17. Numerical errors need to be handled through programming. For example, a horizontal line may be very close to the vertex of a triangle, so robust coding skills are required to determine how to cut through the triangle;

18. The key challenge is that the coding for the intersection of horizontal and vertical lines is extremely complex (it is estimated that dozens of "if-else" cases need to be considered), and the process requires great caution. Only after the cutting is successfully implemented will one feel a sense of relief (like a breeze);

19. Similar to the old oil peddler in the story, proficiency comes from practice;

20. Based on existing mesh programming experience, completing this section of code may require 3 to 4 weeks of immersive, full-time programming;

21. A large number of detailed UIs and rendering techniques are needed for visual assistance during the cutting process, which makes it easier to identify bugs visually (refer to the UI of meshDGP and the OpenGL and GLSL content in the five textbooks).



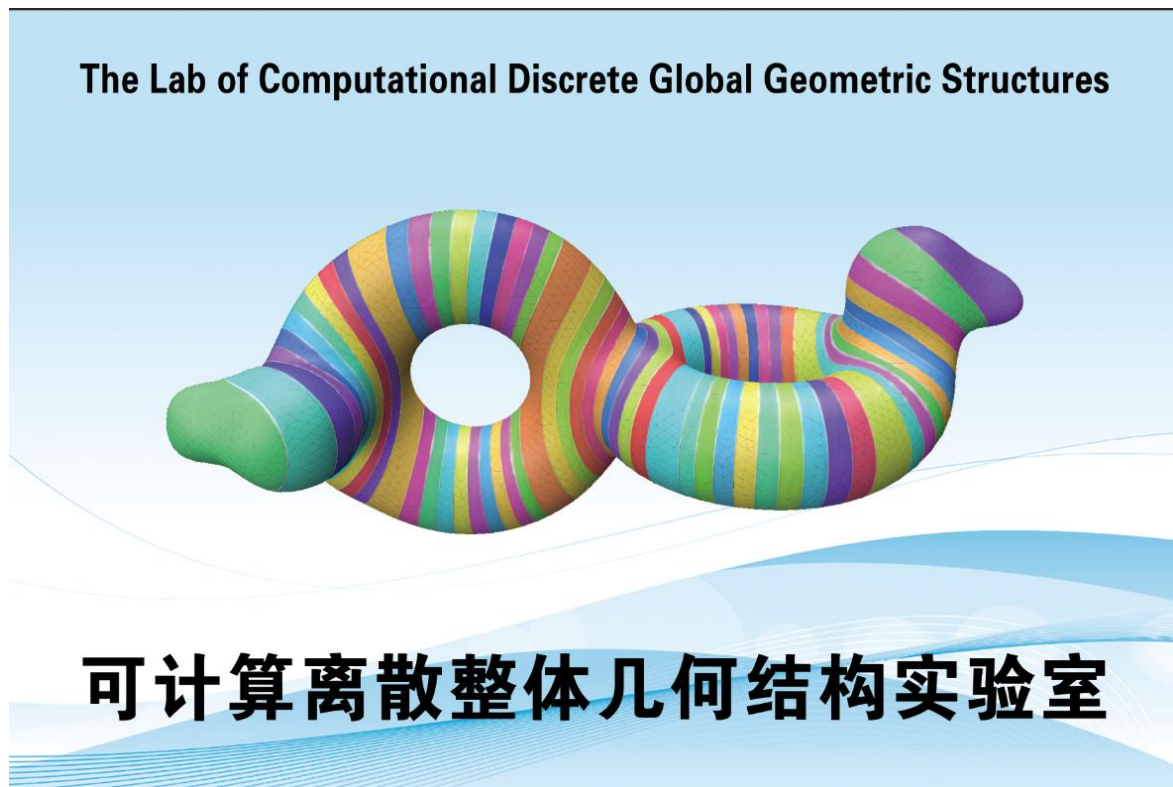
(Figure 67: Schematic Diagram 2 of the Mesh Programming Competition Questions.)

14.10 Establish a Laboratory

To advance the work in the field of computational discrete Global Geometric Structures, the author established the "Laboratory of computational discrete Global Geometric Structures". A series of the aforementioned related work has been carried out through this platform; the figure below shows the nameplate of the laboratory.

Since its establishment, the laboratory has held nearly 100 study sessions, 8 Mesh Square Table Meetings, 8 Differential Geometry Teaching Seminars, 12 national touring art exhibition events, seminars on modernized courses of Computer Graphics, and seminars on the application of Linear Algebra. It has also established a Joint Research Center, among other initiatives.

These activities have attracted extensive participation from teachers, students of many domestic universities and colleges, as well as enterprise engineers, and have accumulated nearly 100,000 views on live streaming platforms. Through these numerous activities, the dissemination of the research concept of computational discrete Global Geometric Structures has been effectively promoted, enabling more teachers and students to understand the innovative aspects of this research concept.



(Figure 68: Laboratory Logo.)

14.11 Establish an award

To advance the research on computational discrete Global Geometric Structures, especially at the current stage, so as to further clarify the innovation of this new research direction, content, and field, the Laboratory of computational discrete Global Geometric Structures proposes the establishment of the computational discrete Global Geometric Structures Achievement Award.

1. Purpose of Establishing the Award

- (1) The purpose of this award is not to conduct research on abstract mathematics or prove geometric theorems.
- (2) The purpose of establishing this award is to promote the algorithm design, code programming, and visual computing of global geometric structures, as well as the interdisciplinary applications of these computing results in industrial software, physical experiments, topological materials, and other fields.
- (3) The purpose of this award is to inspire and encourage future researchers to engage in research and investment in this field through the successful global geometric structure algorithm designs by current scholars.
- (4) At present, there are already some scattered global geometric structures for which algorithms have been successfully designed and implemented with code. However, when developing these algorithms, previous scholars did not have a macroscopic systematic concept of global geometric structures, but rather focused on researching specific individual problems. By establishing this award, a research field dedicated to the algorithm design of global geometric structures can be built conceptually and ideologically, thereby accelerating research in this area and further promoting the emergence of various new technologies.
- (5) To encourage scholars to design more global geometric structure algorithms and apply these algorithms in various disciplines.
- (6) To transform the research on global geometric structure algorithms from scattered efforts to systematic development.
- (7) Different research directions and disciplines have distinct values. Through this award, unique values for the research on global geometric structure "algorithms" will be gradually established—such as relying on "visualization" and using "code" as an experimental tool.
- (8) To promote in-depth exchanges among researchers engaged in studies related to global geometric structure algorithms, and facilitate the establishment and expansion of academic communities in this field.
- (9) To encourage teachers and students from various professional disciplines to engage with global geometric structures and use the algorithmic tools of global geometric structures to explore solutions to scientific and technological problems in their respective fields.

2. Eligibility Criteria for the Award

- (1) We firmly believe that the achievements of scientific research cannot be separated from the efforts of a team. However, to more clearly encourage researchers to "deeply understand" the mathematical concepts and theories of global geometric structures, this award targets key scholars rather than the achievements themselves.

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- (2) The awardee must be able to "mathematically" understand the global geometric structure of the algorithm they have contributed to, and consciously design algorithms to implement this structure. Their contributions should not be limited to participating in a single paper, but should promote the research on global geometric structure algorithms through various aspects of their work.
 - (3) In other words, this award is not based solely on papers; it is an award for individuals, not for papers.
 - (4) The successful algorithm of the awardee must be verified by code and presented through visualization.
 - (5) The awardee's algorithm must truly maintain the "global" characteristics of a specific global geometric structure, rather than undermining global constraints through local computations.
 - (6) Alternatively, the awardee's work may involve clear visualization that illustrates a specific global geometric structure.
 - (7) The awardee's work does not need to explicitly target "global geometric structures" as its goal. For example, previous research by scholars on vector fields and sequence counting was not explicitly aimed at developing global geometric structures, but rather focused on other applications.
 - (8) This award mainly targets the construction of global geometric structures through "algorithms". However, global geometric structures constructed through other methods (such as physical materials) may also be considered.

3. Selection Process

- (1) This award does not require applications. Instead, the selection committee will 不定期 select global geometric structure algorithms from around the world, and notify the awardee (s) after discussion and evaluation.
- (2) There is no limit on the number of awardees. Any scholar who has successfully designed, visualized, and applied an algorithm for a specific global geometric structure is eligible for the award.
- (3) The selection and award presentation dates are not fixed. The selection committee will promptly identify new successful algorithms and issue the award to the key scientists behind the algorithm in a timely manner.
- (4) Unlike general awards such as the Nobel Prize, the Fields Medal, and the Future Science Prize, the algorithms selected for this award are not compared or chosen against each other. Any key scholar who has successfully designed and implemented an algorithm for a global geometric structure is eligible to receive the award. Therefore, the selection committee can also be regarded as a "mineral exploration team" (tasked with identifying valuable algorithmic achievements).

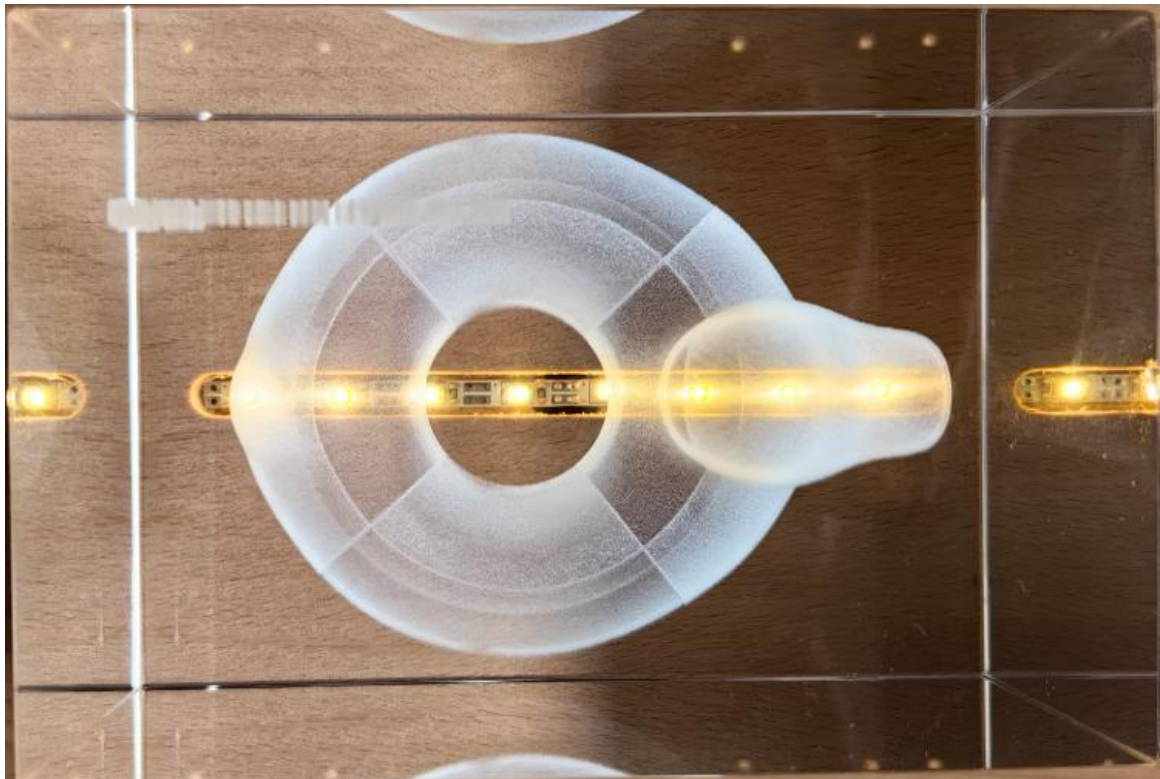
4. Role of the Award

- (1) We anticipate that through this award, the academic community, industrial sector, and all sectors of society will recognize and understand that the "computability of global geometric structures" will play a key role in the next stage of technological development, and how to release the enormous productivity contained in the abstract mathematics of global geometric structures into applied technology through "algorithms".
- (2) We also anticipate that over the next 30 years, this award will catalyze the application of global geometric structures in interdisciplinary research fields such as topological materials, quantum

computers, industrial software, and artificial intelligence, leading to the development of various new materials and technologies. Ultimately, this will accelerate the rapid advancement of science and technology and realize technologies that are currently unimaginable.

5. Award Medal

The preliminary design of the award medal sculpture is as follows: the sculpture will embody global geometric structures through a visualized form.



(Figure 69: Top Side of the Preliminary Award Sculpture.)



(Figure 70: Front Side of the Preliminary Award Sculpture.)

4. Review Committee

- (1) It is not merely an academic review; the review process is a link where participants from various sectors engage to build consensus.
- (2) It will be composed of mathematicians, computer scientists, physicists, and other professionals related to global geometric structures;
- (3) It will include teachers and students who are interested in global geometric structure algorithms;
- (4) It will involve representatives from the industrial sector who plan to explore the industrial and technological applications of global geometric structures.

14.12 Journal

A proposal is put forward to establish a journal with an entirely new mechanism, titled computational discrete Global Geometric Structures (with the English name "Journal of Computational Discrete Global Geometric Structures"). This journal aims for innovation: leveraging internet and artificial intelligence technologies, it will operate in a fully new form, covering both its mechanism and publication methods.

(1) The idea for this journal is to make it an electronic version; therefore, it will not be published periodically. Instead, each article will be published online as soon as it is ready.

(2) The current mechanism of academic journals was established by European and American institutions 200 years ago. It was suitable for the social development context of that time and promoted the advancement of science and technology. For instance, the phenomenon of "over-reliance on papers" (the "only-papers" tendency) did not exist hundreds of years ago, nor was it feasible back then. However, after hundreds of years, societal technologies and article dissemination channels have completely changed, and the old mechanism is no longer fully appropriate. Problems such as the "only-papers" tendency— which hinder scientific research— have emerged. Thus, establishing a new driving mechanism for academic journals can keep pace with the times and further promote the development of current science and technology.

(3) In addition to the conventional first author, corresponding author, etc., the key innovation of this journal lies in: each article will additionally set up "sponsor authors".

(4) The sponsor authors of each article will consist of current sponsor authors and former sponsor authors.

(5) A sponsor author will be included in the author list by sponsoring a certain amount of money, such as 900,000 RMB.

(6) There is no limit on the number of current sponsor authors.

(7) The sponsorship amount for each article is not fixed; it is determined by the sponsor themselves or the sponsor's scientific advisory team based on their assessment of the article's value and significance.

(8) A current sponsor author may resell their "sponsor author" position to a new sponsor author, after which the original current sponsor author will be included in the list of former sponsor authors of the article.

(9) The amount of money that a new sponsor intends to contribute will be equally shared by the current sponsor author (who is reselling the position) and the other authors excluding the sponsor authors (there may be more than one current sponsor author). For example, suppose X and Y are current sponsor authors, and A, B, C, D are the first author, second author, third author, and corresponding author respectively. If Z sponsors 1,000,000 RMB to this article to purchase X's sponsor position, then A, B, C, and D will share this 1,000,000 RMB, with each receiving 200,000 RMB. After this transaction, the author list of the article will change from [A, B, C, D, X, Y] to [A, B, C, D, Y, Z], and X will be added to the list of former sponsor authors in the article's appendix. In other words, although X is no longer a current sponsor author, they will still retain a certain honor through being listed in the former sponsor authors appendix.

(10) This mechanism is similar to the buying, selling, and collecting of antiques or calligraphy and paintings; the only difference is that sponsors are collecting the honor associated with academically valuable articles. For example, if a sponsor had collected the article on neural network backpropagation in 1990, they could now resell it to another collector at a high price, gaining both fame and fortune.

(11) A scientific advisory team will provide professional opinions to sponsors. Sponsors may organize their own advisory teams to assess whether the academic value of a given article is long-term or short-term, thereby deciding whether to hold the sponsor position for the long term or short term.

-
- (12) A sponsor author only holds the right to the honor of the article and the right to resell the sponsor position; they do not possess the intellectual property rights to the content of the article.
- (13) This mechanism encourages researchers to conduct valuable research and publish high-quality articles, as they can obtain continuous and multiple substantial incomes from this process—providing them with sufficient motivation. The greater the academic value of an article, the more new incomes will be generated each time a new sponsor author joins over time. In contrast, a low-quality "filler article" will likely attract no sponsor authors.
- (14) This mechanism ensures that the income researchers obtain from sponsorships is higher than the benefits they would gain from catering to the "five-only" tendencies. As a result, researchers will have the motivation to engage in valuable and long-term scientific research.
- (15) The appendix of each article in this journal will state: "This article is not intended for use in professional title evaluation, project applications, or similar purposes; it is merely a by-product of scientific research achievements."
- (16) Whenever there is a change in the sponsor authors, the author list in the PDF version of the article on the journal's website will be updated promptly.
- (17) Being a sponsor author of an article is a great honor. For example, if the articles on Einstein's theory of relativity had included the design of sponsor authors back then, many wealthy individuals like Carnegie would have likely competed to bid for the sponsor author positions.
- (18) This journal imposes no restrictions on research fields or directions; it covers fields such as physics, chemistry, computer science, mathematics, and more.
- (19) Each article will be reviewed by no fewer than 5 reviewers.
- (20) Reviewers will remain anonymous during the review process; after the article is accepted, the reviewers will be publicly listed in the appendix of the article. This creates a positive cycle: as the value of valuable articles becomes more evident over time, the discerning judgment of the reviewers will also be recognized and validated.
- (21) The content of submissions to the journal can include popular science articles, such as those published in Quanta Magazine.
- (22) A distinctive feature of the journal's submissions is the inclusion of illustrations. Articles with illustrations allow sponsor authors to form a more intuitive perception, which may encourage them to provide higher sponsorships.
- (23) This journal is directly linked to financial interests—it is not just about writing articles in a rigid, dull manner. If an article is well-written, it will attract sponsor authors; researchers who publish three to five high-quality articles may soon achieve financial freedom. Therefore, the motivation for contributing to this journal is extremely strong.
- (24) The journal will have a pre-release "-1 preview version". After the -1 version is made public, sponsor authors can provide sponsorship within a specified period (e.g., three months), following which the official Version 1 will be published.

Journal of Computational Global Discrete Geometric Structures

赵辉、王阳明、李白、牛顿、比尔·盖茨、马云

November 5, 2024

Abstract

《可计算离散整体几何结构》实验室提出一种全新的科研学术期刊杂志运转机制。



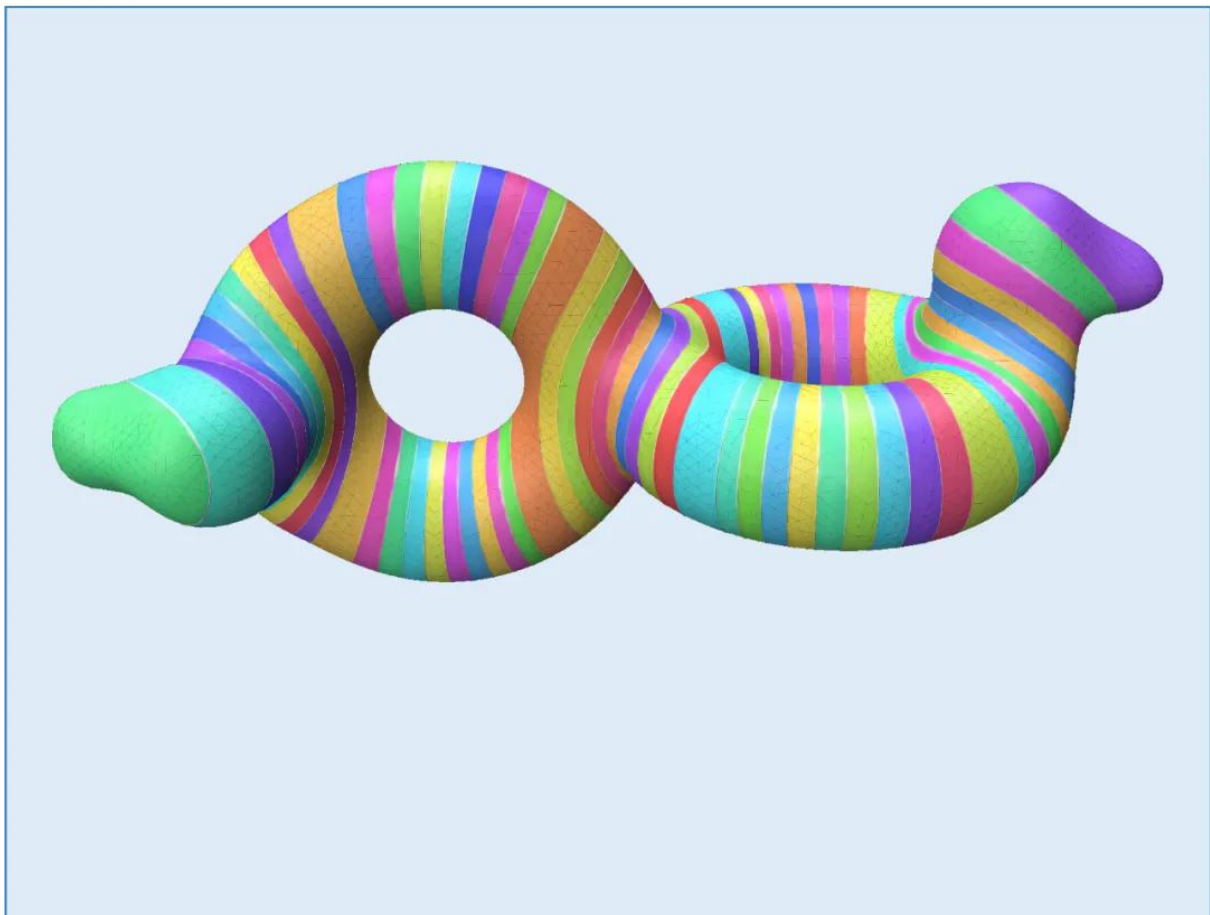
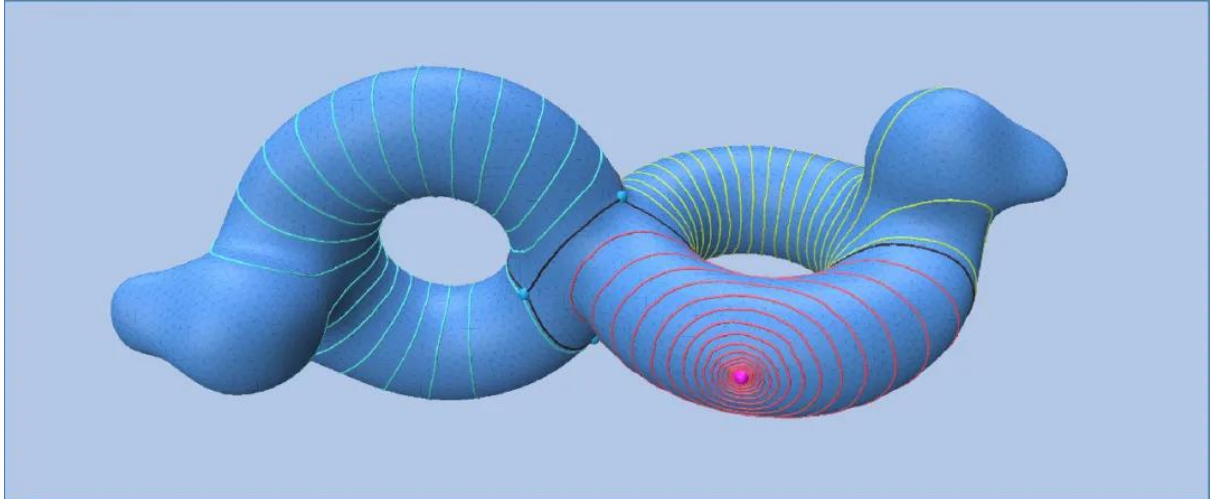
Figure 1: 调和叶状结构图示。

1 作者说明

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2 版本说明

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